

MA643 - Numerical Analysis of Partial
Differential Equations
Lecture Notes

Abhinav Jha
Indian Institute of Technology, Gandhinagar
Semester 1 2026

Preface

These notes are designed to provide a structured and comprehensive understanding of the course content. They will cover key topics, concepts, and computational techniques that are fundamental to numerical analysis of partial differential equations. Please note that this is the Version 1.0.1 of the notes and hence there is a chance that some of the content is incorrect. If you find some flaws, please email me at `abhinav.jha@iitgn.ac.in`.

Differences from Version 1.0.0

1. Chapter 1
 - (a) Added examples for Elliptic, Parabolic, and Hyperbolic PDEs.
2. Chapter 2
 - (a) Added Remark 2.19.
 - (b) Added Example 2.2 and Example 2.3. I would like to acknowledge Abhijit Venkat (B.Tech, Mechanical 2026, IITGN) and Hrriday Ruparel (B.Tech, Computer Science 2026, IITGN) for providing the codes.
3. Chapter 3
 - (a) Added proof for Lemma 3.6.
 - (b) Added proof for Theorem 3.18.

Just because someone stumbles and loses their path, doesn't mean they're lost forever.

—Prof. Charles Xavier

Contents

1	Partial Differential Equations	7
1.1	Heat Equation	8
1.1.1	Boundary Conditions	10
1.1.2	Poisson Equation	11
1.1.3	Non-Dimensional Form	12
1.2	Classification of PDEs	13
1.3	Analytical Solution to Elliptic PDEs	16
2	Finite Difference Methods	19
2.1	Dirichlet Problem in Square Domain	19
2.1.1	Implementation of Five-Point Stencil	22
2.1.2	Convergence Theory	27
2.1.3	Discrete Maximum Principle	33
2.2	Advanced Topics	35
2.2.1	Neumann Problem in Square Domain	35
2.2.2	Non-Rectangular Domains	42
2.2.3	Higher-Order Discretisation	47
2.3	Summary	51
3	Introduction to Functional Analysis	53
3.1	Function Spaces	54
3.1.1	Banach Spaces	54
3.1.2	Space of Measurable Functions	60
3.1.3	Dual Space	68
3.1.4	Hilbert Space	71
3.2	Sobolev Spaces	75
3.2.1	Sobolev Embedding Theorem	83
3.2.2	Trace	85
3.2.3	Gauss and Green Formulas	87
3.2.4	Domains	88
3.3	Fixed Points	90

Chapter 1

Partial Differential Equations

In real life, many physical phenomena are governed by differential equations. From the steam rising from your morning coffee to the spread of pollution in the air, these processes can often be modeled using partial differential equations (PDEs).



Figure 1.1: Smoke coming out of an industrial chimney.

Let us consider a simple example. Imagine smoke rising from an industrial chimney (see Fig. 1.1). What do you observe? First, the smoke is carried in the direction of the wind. Then, it begins to disperse into the surrounding air. Finally, although not directly visible, chemical reactions take place between the smoke and components of the atmosphere. The equation that models such behavior is known as the convection–diffusion–reaction equation—one among many PDEs used to describe natural phenomena.

The origin of differential calculus can be traced back to the foundational work of Isaac Newton and Gottfried Wilhelm Leibniz. The study of PDEs, however, is typically credited to Jean le Rond d’Alembert, who investigated the wave equation in the 18th century. Subsequent major contributions came from Leonhard Euler, who formulated what are now called the Euler equations for fluid dynamics.

This chapter provides a brief introduction to partial differential equations: how they are derived, how they are classified, and what challenges arise when attempting to solve them analytically. For a deeper understanding, the reader is encouraged to consult textbooks dedicated to this topic. Some recommended references include [6, 10].



Figure 1.2: From left to right: Isaac Newton: 4 January 1643 – 31 March 1727, Gottfried Wilhelm Leibniz: 1 July 1646 – 14 November 1716, Jean le Rond d'Alembert : 16 November 1717 – 29 October 1783, and Leonhard Euler : 15 April 1707 – 18 September 1783.

1.1 Heat Equation

Engineers and physicists are often interested in properties of materials that vary continuously in space and time. Let $\mathbf{x} := (x, y, z)$ denote the spatial coordinates and t denote time. One fundamental concept in this context is that of *conservation*.

So, what is a *conservation law*? At its core, a conservation law is concerned with tracking changes in a material property within a control volume V , which is enclosed by a surface S . It relates the change of that property inside V to its production within the volume and its flow across the boundary.

There are three key components involved in formulating a conservation law:

1. The quantity Q of the property per unit volume,
2. The rate F at which the property is produced or destroyed (i.e., net production) per unit volume, and
3. The net *flux* of the property through the surface S , i.e., the flow of the property into or out of the volume.

If $\mathbf{q}$ denotes the flux vector (i.e., flow rate per unit area) at a point inside V , then the flux across the boundary at a point on S is given by $\mathbf{q} \cdot \mathbf{n}$, where $\mathbf{n}$ is the outward unit normal vector (see Fig. 1.3).

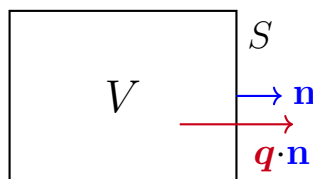


Figure 1.3: Flux vector $\mathbf{q}$ flowing through the boundary S enclosing volume V .

A conservation law can be succinctly stated as:

The rate of change of Q inside the volume is equal to the net production within the volume minus the net outward flux through its boundary.

As Q is the quantity per unit volume. Hence, Q over V is given by $\int_V Q dV$ and similarly

for F and $\mathbf{q}$. Mathematically, the conservation principle can be expressed as

$$\frac{d}{dt} \int_V Q dV = \int_V F dV - \int_S \mathbf{q} \cdot \mathbf{n} dS. \quad (1.1)$$

Using the *divergence theorem* (also known as Gauss's theorem), the surface integral can be converted into a volume integral:

$$\int_S \mathbf{q} \cdot \mathbf{n} dS = \int_V \nabla \cdot \mathbf{q} dV,$$

where $\nabla \cdot \mathbf{q}$ denotes the divergence of the flux vector $\mathbf{q}$.

Furthermore, if the control volume V is fixed (i.e., independent of time), we can interchange the order of time differentiation and volume integration in the first term of Eq. (1.1), yielding

$$\int_V \left(\frac{dQ}{dt} + \nabla \cdot \mathbf{q} - F \right) dV = 0.$$

Since V is arbitrary, the integrand must vanish pointwise (see [12, Proposition 6.3.3]). This gives the local form of the conservation law:

$$\frac{dQ}{dt} + \nabla \cdot \mathbf{q} = F. \quad (1.2)$$

The quantities Q , $\mathbf{q}$, and F generally depend on the unknown variable u (e.g., temperature or concentration), the spatial position $\mathbf{x}$, time t , and physical properties of the medium such as conductivity or density.

We now turn to a classical example: the heat equation, where the quantity of interest is temperature, denoted by $u(t, \mathbf{x})$. Joseph Fourier is credited with the derivation and solution of the heat equation, which he introduced in his 1822 work *Théorie analytique de la chaleur*.



Figure 1.4: Joseph Fourier: 21 March 1768 – 16 May 1830

Let $C(u)$ be the specific heat capacity, i.e., the amount of heat energy required to raise the temperature of a unit mass by one degree. This has the units of Joules/Kelvin. If u_0 is a reference (or base) temperature, then the thermal energy per unit mass is

$$E = \int_{u_0}^u C(v) dv.$$

The thermal energy per unit volume is then ρE , where ρ is the material density. Thus, the conserved quantity is

$$Q = \rho E = \rho \int_{u_0}^u C(v) dv.$$

If ρ and C are constant, this simplifies to

$$Q = \rho C(u - u_0).$$

The heat flux $\mathbf{q}$ is given by *Fourier's law*:

$$\mathbf{q} = -\lambda \nabla u,$$

where λ is the thermal conductivity—a material property that quantifies the ability to conduct heat. The negative sign indicates that heat flows from regions of high temperature to low temperature.

Assuming no other modes of heat transfer and letting $F(u, \mathbf{x}, t)$ represent the internal heat sources (or sinks) per unit volume, the energy conservation law (1.2) becomes

$$\rho C u_t - \nabla \cdot (\lambda \nabla u) = F.$$

If λ is constant, this further simplifies to the classical heat equation:

$$u_t - \varepsilon \Delta u = f, \tag{1.3}$$

where the thermal diffusivity is $\varepsilon = \frac{\lambda}{\rho C}$ and the source term is $f = \frac{F}{\rho C}$.

We assume that Eq. (1.3) is posed over a spatial domain Ω with boundary Γ , and the time variable t ranges over the interval $[0, T]$. It is also important to note that we have assumed an idealistic situation where the material has uniform density and conductivity.

The heat equation is not limited to thermal processes. Equations of the form (1.3) arise in many other diffusive phenomena, such as chemical diffusion, pollutant dispersion, and image smoothing.

1.1.1 Boundary Conditions

Another important aspect of a partial differential equation is the specification of appropriate *boundary conditions*. There are three major types of boundary conditions commonly encountered in physical problems:

1. **Dirichlet Boundary Condition** Named after Peter Gustav Lejeune Dirichlet, this condition prescribes the value of the solution itself on a portion of the boundary. In the context of heat transfer, it specifies the temperature $u(t, \mathbf{x})$ on a part of the boundary:

$$u = g, \quad \text{on } (0, T) \times \Gamma_D,$$

where $\Gamma_D \subset \Gamma$. These are also referred to as *essential boundary conditions*.

2. **Neumann Boundary Condition** Named after Carl Neumann, this condition prescribes the heat flux across the boundary. It specifies the normal derivative of the solution on a portion of the boundary:

$$-\frac{\partial u}{\partial \mathbf{n}} = g, \quad \text{on } (0, T) \times \Gamma_N,$$

where $\Gamma_N \subset \Gamma$. These are also called *natural boundary conditions*.

3. **Robin Boundary Condition** Named after Victor Gustav Robin, this condition models convective heat exchange at the boundary according to Newton's law of cooling. It is a combination of Dirichlet and Neumann conditions:

$$\frac{\partial u}{\partial \mathbf{n}} + h(u - u_{\text{env}}) = 0, \quad \text{on } (0, T) \times \Gamma_R,$$

where $\Gamma_R \subset \Gamma$, h is the heat transfer coefficient, and u_{env} is the ambient temperature.

In addition to boundary conditions, an initial condition is required to fully determine the solution. This specifies the state of the system at the initial time $t = 0$:

$$u(0, \mathbf{x}) = u_0(\mathbf{x}), \quad \text{for } \mathbf{x} \in \Omega.$$



Figure 1.5: Peter Gustav Lejeune Dirichlet : 13 February 1805 – 5 May 1859 (left) and Carl Neumann : 7 May 1832 – 27 March 1925 (right).

1.1.2 Poisson Equation

If the temperature remains constant in time, we obtain a special case of the heat equation:

$$-\varepsilon \Delta u = f, \quad \text{in } \Omega, \tag{1.4}$$

which is known as the *Poisson equation*. It is named after Siméon Denis Poisson, who published it in 1823. When the source term vanishes, i.e., $f(\mathbf{x}) = 0$, the equation reduces to the *Laplace equation*, named after Pierre-Simon Laplace, who studied it in 1786.

Solutions to the Laplace equation are known as *harmonic functions*. These functions play a central role in an area of mathematics known as *harmonic analysis*, which connects PDEs with Fourier analysis, potential theory, and other fields.

The focus of this course will be on developing numerical methods for solving the Poisson equation.

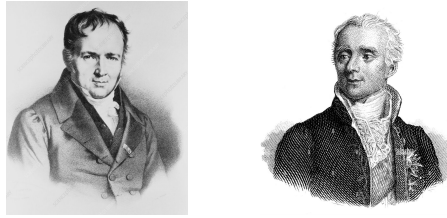


Figure 1.6: Siméon Denis Poisson : 21 June 1781 – 25 April 1840 (left) and Pierre-Simon Laplace : 23 March 1749 – 5 March 1827 (right).

1.1.3 Non-Dimensional Form

Until now, we have discussed PDEs from the perspective of engineers and physicists, deriving equations based on physical laws. However, one important aspect we have not yet addressed is the role of *units* or *dimensions* in these equations.

In the natural sciences, units play a fundamental role. The most widely used and standardized unit system is the *International System of Units (SI)*. For example, the SI unit of energy is the joule. More fundamentally, the SI system is built upon seven *base quantities*, each associated with a specific unit (see Table 1.1). Other units can be expressed in terms of these base units. For instance, one joule is equivalent to $\text{kg m}^2 \text{s}^{-2}$ in base units.

Symbol	Name	Base Quantity
s	second	time
m	metre	length
kg	kilogram	mass
A	ampere	electric current
K	kelvin	thermodynamic temperature
mol	mole	amount of substance
cd	candela	luminous intensity

Table 1.1: SI base units corresponding to the seven fundamental physical quantities.

We now examine whether the heat equation (1.3) is *dimensionally consistent*, i.e., whether all terms in the equation have the same physical units. From this point onward, we restrict ourselves to the SI base unit system.

Recall that $u(t, \mathbf{x})$ represents temperature, which is measured in kelvins (K). The thermal diffusivity $\varepsilon = \lambda/(\rho C)$ is a material parameter composed of:

- Thermal conductivity λ with units $\text{kg m s}^{-3} \text{K}^{-1}$,
- Density ρ with units kg m^{-3} ,
- Specific heat capacity C with units $\text{m}^2 \text{s}^{-2} \text{K}^{-1}$.

Thus, the thermal diffusivity ε has units:

$$\varepsilon = \frac{\lambda}{\rho C} \Rightarrow \text{m}^2 \text{s}^{-1}.$$

The source term is given by $f = F/(\rho C)$, where F has units of heat generated per unit volume per unit time: $\text{kg m}^{-1} \text{s}^{-3}$. Consequently,

$$f = \frac{F}{\rho C} \Rightarrow \text{K s}^{-1}.$$

On the left-hand side of the heat equation, the time derivative u_t has units K s^{-1} , while the Laplacian Δu has units K m^{-2} , and multiplying it by ε yields K s^{-1} . Therefore, all three terms in the heat equation have consistent SI base units, confirming that the equation is dimensionally valid.

In numerical analysis, it is often beneficial to work with a *dimensionless form* of the equation. This allows for a formulation that is independent of measurement systems—for example, temperature could be measured in either Kelvin or Celsius.

Let L , U , and T^* represent the characteristic length, temperature, and time scales of the problem, respectively. We define the dimensionless variables (denoted by primes) as:

$$\mathbf{x}' = \frac{\mathbf{x}}{L}, \quad u' = \frac{u}{U}, \quad t' = \frac{t}{T^*}.$$

Using the chain rule, the derivatives transform as:

$$u_t = \frac{U}{T^*} u'_{t'}, \quad \Delta u = \frac{U}{L^2} \Delta' u'.$$

Substituting these into the heat equation gives:

$$\begin{aligned} \frac{U}{T^*} u'_{t'} - \varepsilon \frac{U}{L^2} \Delta' u' &= f, \quad \text{in } \left(0, \frac{T}{T^*}\right) \times \Omega', \\ u'_{t'} - \varepsilon \frac{T^*}{L^2} \Delta' u' &= \frac{T^*}{U} f, \quad \text{in } \left(0, \frac{T}{T^*}\right) \times \Omega', \end{aligned}$$

where Ω' is the dimensionless spatial domain. Note that both $\varepsilon \frac{T^*}{L^2}$ and $\frac{T^*}{U} f$ are dimensionless, so all terms in the rescaled equation are nondimensional.

By abuse of notation, we drop the primes and obtain the *dimensionless heat equation*:

$$u_t - \varepsilon \Delta u = f, \quad \text{in } (0, T) \times \Omega,$$

where ε and f are now interpreted as nondimensional parameters. A similar procedure can be used to obtain the dimensionless form of the Poisson equation.

1.2 Classification of PDEs

After exploring the modeling and physical derivation of partial differential equations, we now shift to a more abstract and mathematical perspective. In this course, we are primarily interested in a specific class of PDEs known as *second-order partial differential equations*. To make this precise, we begin by defining the notion of the *order* of a PDE.

Definition 1.1 (Order of a PDE). The *order* of a partial differential equation is the order of the highest derivative of the unknown function appearing in the equation. For example, if the highest derivative is a second derivative, such as $\partial^2 u / \partial x^2$, then the equation is said to be of *second order*.

Second-order PDEs play a central role in modeling a wide range of physical phenomena. These equations are broadly classified into three types: *elliptic*, *parabolic*, and *hyperbolic*.

Let $\Omega \subset \mathbb{R}^d$, where $d \in \mathbb{N}$. A general linear second-order PDE can be written in the form

$$\sum_{j,k=1}^d a_{jk}(\mathbf{x}) \partial_j \partial_k u(\mathbf{x}) + F(\mathbf{x}, u, \partial_1 u, \dots, \partial_d u) = 0,$$

or equivalently,

$$\nabla \cdot (\mathbf{A}(\mathbf{x}) \nabla u) + F(\mathbf{x}, u, \partial_1 u, \dots, \partial_d u) = 0,$$

where $\mathbf{A}(\mathbf{x}) = [a_{jk}(\mathbf{x})]$ is a matrix-valued function defined on Ω .

If $u(\mathbf{x})$ is sufficiently smooth, then by Schwarz's theorem, the mixed second partial derivatives commute: $\partial_j \partial_k u = \partial_k \partial_j u$. This allows us to assume that the coefficient matrix $\mathbf{A}(\mathbf{x})$ is symmetric. Even if $\mathbf{A}(\mathbf{x})$ is not symmetric initially, we can rewrite the equation in symmetric form:

$$\begin{aligned} \sum_{j,k=1}^d a_{jk}(\mathbf{x}) \partial_j \partial_k u(\mathbf{x}) &= \sum_{j < k} a_{jk}(\mathbf{x}) \partial_j \partial_k u + \sum_{j > k} a_{jk}(\mathbf{x}) \partial_j \partial_k u + \sum_{j=1}^d a_{jj}(\mathbf{x}) \partial_{jj} u \\ &= \sum_{j < k} a_{jk}(\mathbf{x}) \partial_j \partial_k u + \sum_{j > k} a_{jk}(\mathbf{x}) \partial_k \partial_j u + \sum_{j=1}^d a_{jj}(\mathbf{x}) \partial_{jj} u \\ &= \sum_{j < k} (a_{jk}(\mathbf{x}) + a_{kj}(\mathbf{x})) \partial_j \partial_k u + \sum_{j=1}^d a_{jj}(\mathbf{x}) \partial_{jj} u \\ &= \sum_{j,k=1}^d \left(\frac{a_{jk}(\mathbf{x}) + a_{kj}(\mathbf{x})}{2} \right) \partial_j \partial_k u(\mathbf{x}). \end{aligned}$$

Hence, without loss of generality, we may take $\mathbf{A}(\mathbf{x})$ to be symmetric. As a result, all eigenvalues of $\mathbf{A}(\mathbf{x})$ are real.

The classification of a second-order PDE depends on the signs of the eigenvalues of the matrix $\mathbf{A}(\mathbf{x})$. Let:

- α be the number of positive eigenvalues,
- β the number of negative eigenvalues,
- γ the number of zero eigenvalues.

We say the PDE is of type (α, β, γ) , and classify it as:

- **Elliptic PDE** : Type $(d, 0, 0)$ or $(0, d, 0)$.
- **Parabolic PDE** : Type $(d - 1, 0, 1)$ or $(0, d - 1, 1)$.
- **Hyperbolic PDE** : Type $(d - 1, 1, 0)$ or $(1, d - 1, 0)$.

This matrix-based definition may be unfamiliar to those who have previously studied PDEs only in two variables. To reconcile the two perspectives, consider the classical form of a second-order PDE in two dimensions:

$$\mathbf{A}u_{xx} + \mathbf{B}u_{xy} + \mathbf{C}u_{yy} + \mathbf{D}u_x + \mathbf{E}u_y + \mathbf{F}u = \mathbf{G}.$$

Focusing only on the second-order part, the associated coefficient matrix is:

$$\begin{bmatrix} \mathbf{A} & \frac{\mathbf{B}}{2} \\ \frac{\mathbf{B}}{2} & \mathbf{C} \end{bmatrix},$$

with determinant

$$\mathbf{AC} - \frac{\mathbf{B}^2}{4}.$$

From linear algebra, the determinant of a 2×2 symmetric matrix equals the product of its eigenvalues (see [16]). Therefore, the sign of the discriminant $\mathbf{B}^2 - 4\mathbf{AC}$ determines the PDE type:

- Elliptic if $\mathbf{B}^2 - 4\mathbf{AC} < 0$,
- Parabolic if $\mathbf{B}^2 - 4\mathbf{AC} = 0$,
- Hyperbolic if $\mathbf{B}^2 - 4\mathbf{AC} > 0$.

This classical definition coincides with the eigenvalue-based classification and extends naturally to higher dimensions.

Examples:

- The **Poisson equation**

$$\begin{aligned} -\Delta u &= f \quad \text{in } \Omega, \\ u &= g \quad \text{on } \Gamma_D, \end{aligned}$$

is elliptic.

- The **Heat equation**

$$\begin{aligned} u_t - \Delta u &= f \quad \text{in } (0, T) \times \Omega, \\ u &= g \quad \text{on } (0, T) \times \Gamma_D, \\ u(0, \mathbf{x}) &= u_0(\mathbf{x}) \quad \text{on } \Omega, \end{aligned}$$

is parabolic.

- The **Wave equation**

$$\begin{aligned} u_{tt} - \Delta u &= 0 & \text{in } (0, T) \times \Omega, \\ u &= 0 & \text{on } (0, T) \times \Gamma_D, \\ u(0, \mathbf{x}) &= f(\mathbf{x}) & \text{on } \Omega, \\ u_t(0, \mathbf{x}) &= g(\mathbf{x}) & \text{on } \Omega, \end{aligned}$$

is hyperbolic.

1.3 Analytical Solution to Elliptic PDEs

Until now, we have derived partial differential equations (PDEs), classified them, and discussed various boundary conditions. We now turn to the important question of the *existence and uniqueness* of solutions, followed by how such solutions can be computed.

Before presenting the main result, we introduce some useful notations and definitions.

Definition 1.2 (Multi-Index). A d -dimensional multi-index is a tuple $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d)$, where each $\alpha_i \in \mathbb{N} \cup \{0\}$. The corresponding partial derivative is defined as

$$\partial^\alpha := \partial_1^{\alpha_1} \partial_2^{\alpha_2} \dots \partial_d^{\alpha_d}, \quad \text{where } \partial_i^{\alpha_i} := \frac{\partial^{\alpha_i}}{\partial x_i^{\alpha_i}}.$$

We denote the order by $|\alpha| = \sum_{i=1}^d \alpha_i$.

Example 1.1. Let $|\alpha| = 2$. If $d = 2$ then $\alpha = (1, 1), (2, 0)$, and $(0, 2)$ and if $d = 3$ then $\alpha = (1, 1, 0), (0, 1, 1), (1, 0, 1), (2, 0, 0), (0, 2, 0)$, and $(0, 0, 2)$.

Definition 1.3 (Hölder Continuous Functions). Let Ω be an open subset of $\mathbb{R}^d$, $0 < \alpha \leq 1$, and $k \geq 0$ an integer. The space of Hölder continuous functions is defined as

$$\mathcal{C}^{k,\alpha}(\overline{\Omega}) := \left\{ f \in \mathcal{C}^k(\overline{\Omega}) : \sup_{\mathbf{x} \neq \mathbf{y} \in \overline{\Omega}} \frac{|\partial^\beta f(\mathbf{x}) - \partial^\beta f(\mathbf{y})|}{|\mathbf{x} - \mathbf{y}|^\alpha} < \infty \text{ for all multi-index } \beta \text{ with } |\beta| = k \right\}.$$

Remark 1.1. If $\alpha = 1$ and $k = 0$, then $\mathcal{C}^{0,1}(\overline{\Omega})$ corresponds to the space of Lipschitz continuous functions. For $\alpha = 0$ and $k = 0$, we recover the space of bounded continuous functions.

Remark 1.2. A domain $\Omega \subset \mathbb{R}^d$ is said to have a $\mathcal{C}^{k,\alpha}$ boundary if, near every boundary point, the domain can be locally represented as the graph of a $\mathcal{C}^{k,\alpha}$ function. Intuitively, when you zoom in near any boundary point, the boundary looks smooth up to order k with Hölder-continuous derivatives of order k , and the domain lies entirely on one side of this surface. For example:

- The set $\Omega = \{(x, y) \in \mathbb{R}^2 : y^2 > x\}$ has a $\mathcal{C}^{1,1}$ boundary, since near each boundary point the curve can be written as $y = \pm\sqrt{x}$, whose first derivatives are Lipschitz continuous.

- In contrast, $\Omega = \{(x, y) \in \mathbb{R}^2 : y > |x|\}$ does *not* have a $\mathcal{C}^1$ boundary, because the boundary $y = |x|$ has a corner at the origin where the derivative does not exist.

Theorem 1.3 (Existence and Uniqueness of Solution [9, Theorem 6.14]). *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with $\mathcal{C}^{2,\alpha}$ boundary. Consider the elliptic boundary value problem:*

$$\begin{aligned} -\varepsilon \Delta u &= f & \text{in } \Omega, \\ u &= g & \text{on } \Gamma. \end{aligned}$$

If $f \in \mathcal{C}^{0,\alpha}(\overline{\Omega})$ and $g \in \mathcal{C}^{2,\alpha}(\overline{\Omega})$, then there exists a unique solution $u \in \mathcal{C}^{2,\alpha}(\overline{\Omega})$.

Notice that the theorem provides only sufficient conditions for existence of a classical solution. These assumptions are not necessary, as the following examples demonstrate. Let us now consider some examples of the Poisson equation and their solutions:

1. On $\Omega = \{(x, y) : x^2 + y^2 < 1\}$ (the unit disk), with $g = 0$ and

$$f(x, y) = 4,$$

the solution (see Fig. 1.7b) in polar coordinates (r, θ) is

$$u(r, \theta) = 1 - r^2.$$

2. On $\Omega = [0, 1] \times [0, 1]$, with $g = 0$ and

$$f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y),$$

the exact solution (see Fig. 1.7a) is

$$u(x, y) = \sin(\pi x) \sin(\pi y).$$

Although Ω has corners and therefore does not possess a $\mathcal{C}^{2,\alpha}$ boundary, the problem still admits a classical solution since the data satisfy the necessary compatibility conditions at the corners. For further reading on elliptic problems in polygonal domains, we refer to [11].

3. On

$$\Omega = [-1, 1]^2 \setminus [0, 1] \times [-1, 0],$$

with $g = 0$ and

$$f(x, y) = 1,$$

the problem does not, in general, admit a classical solution. This is due to the re-entrant corner at $(0, 0)$, which introduces an intrinsic corner singularity. Unlike the previous example, compatibility conditions alone cannot remove this singularity. Consequently, higher derivatives of the solution fail to remain bounded near the re-entrant corner. We refer the interested reader to [11] for a detailed treatment of elliptic problems on polygonal domains.

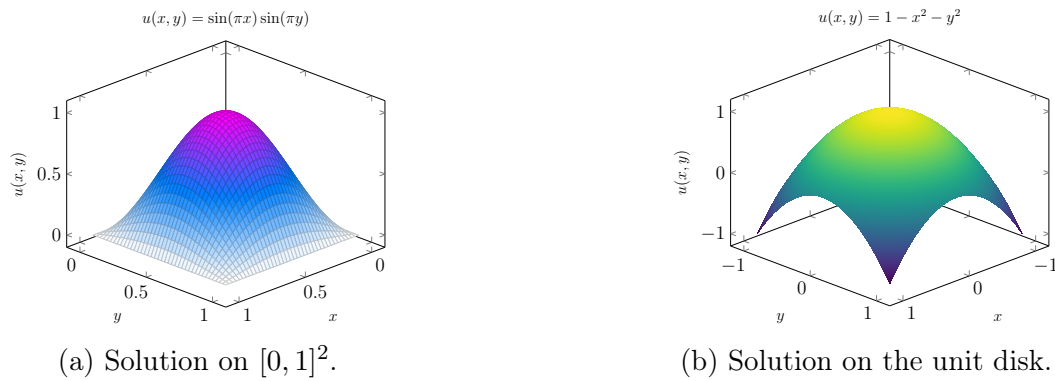


Figure 1.7: Solutions to the Poisson equation in two simple domains.

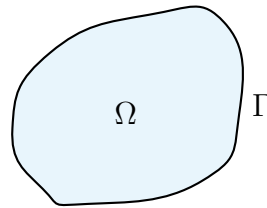


Figure 1.8: A domain Ω with smooth, curved boundary Γ .

The first two examples can be solved analytically using the *method of separation of variables*. In general, if a PDE admits a unique classical solution and the domain is regular (e.g., a rectangle or a disk), then separation of variables can often be applied. However, this approach fails for irregular domains such as that shown in Figure 1.8.

We observe that even for relatively simple domains, classical solutions may fail to exist. For more complex domains, even when existence and uniqueness are guaranteed theoretically, computing the solution explicitly can become intractable.

This leads us to two major motivations:

1. To reconsider and generalize the notion of a “solution” (e.g., to weak solutions).
2. To employ numerical methods to approximate solutions.

The next chapter will focus on Point 2: numerical methods. Point 1 will be revisited later when we discuss the variational and weak formulations of PDEs.

Chapter 2

Finite Difference Methods

In the previous chapter, we provided a general overview of partial differential equations (PDEs), including their physical motivation, classification, boundary conditions, and the question of existence and uniqueness of classical solutions. We saw that analytical solutions are often limited to highly idealized cases, and that even simple domains can present significant challenges. This motivates the use of numerical methods, which will be the focus of this chapter.

We now turn our attention to the numerical solution of PDEs, specifically the Poisson equation, which serves as a prototype for elliptic problems. This chapter will build on ideas from the course **MA637: Numerical Analysis and Computing**, where we studied boundary value problems in one dimension. In particular, we considered equations of the form

$$-y''(x) = f(x, y(x), y'(x), y''(x)), \quad \text{for } a \leq x \leq b, \quad (2.1)$$

with Dirichlet boundary conditions $y(a) = \alpha$ and $y(b) = \beta$ (see [7, Chapter 11]).

The methods developed for such one-dimensional problems extend naturally to two-dimensional settings. By the end of this chapter, we aim to understand how to discretize elliptic PDEs using finite difference methods, how to incorporate various types of boundary conditions, and how to analyze the resulting schemes in terms of consistency, stability, and convergence. We will also address practical aspects of solving the resulting linear systems efficiently.

2.1 Dirichlet Problem in Square Domain

We begin our study of numerical methods with the Poisson equation subject to Dirichlet boundary conditions. The goal is to find $u \in \mathcal{C}^4(\Omega)$ such that

$$-\Delta u = f \quad \text{in } \Omega, \quad (2.2)$$

$$u = g \quad \text{on } \Gamma, \quad (2.3)$$

where $\Omega \subset \mathbb{R}^2$ is a domain with boundary $\Gamma := \partial\Omega$, and $f \in \mathcal{C}(\Omega)$, $g \in \mathcal{C}(\Gamma)$. We begin by considering the case $\Omega = (0, 1) \times (0, 1)$, which allows us to adapt ideas from finite difference methods (FDM) for boundary value problems in one dimension, similar to Eq. (2.1).

The first step in solving the Poisson equation is to discretize the domain Ω using a uniform grid of size $h \times h$ where $h = 1/M$ for some integer M . The internal grid points are defined by

$$\Omega_h = \{(x_i, y_j) : x_i = ih, y_j = jh, i, j = 1, 2, \dots, M-1\}.$$

We denote the boundary grid points by $\partial\Omega_h$ and the complete set of grid points by $\bar{\Omega}_h = \Omega_h \cup \partial\Omega_h$ (see Fig. 2.1).

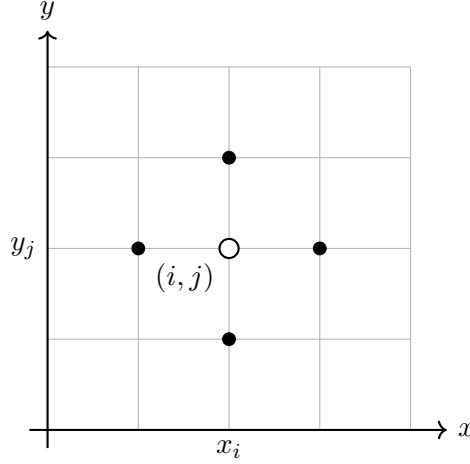


Figure 2.1: A uniform grid on the domain $\Omega = (0, 1) \times (0, 1)$. The white point represents an interior grid node (i, j) , and the black points are its four nearest neighbors used in the five-point stencil.

Definition 2.1 (Grid Function). A vector $\mathbf{u} \in \mathbb{R}^{(M+1) \times (M+1)}$ that assigns a function value to each grid point is called a *grid function*.

For a grid point $(x_i, y_j) \in \Omega_h$, where $x_i = ih$ and $y_j = jh$, we denote the numerical approximation of the solution by $\mathbf{u}_{i,j}$, and the exact analytical solution by $u_{i,j} := u(x_i, y_j)$. While the boundary values of u are known from the prescribed Dirichlet condition, the values at interior points are computed by solving a system of equations.

Remark 2.1. Whenever we use bold notation such as $\mathbf{u}$ or $\mathbf{f}$, it refers to the vector representation over the grid. The subscripted form $\mathbf{u}_{i,j}$ refers to the scalar value at the grid point (i, j) .

Returning to Eq. (2.2), we recall that the Laplacian is given by

$$\Delta u = - \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right).$$

We use a Taylor expansion to approximate the second derivatives. At the grid point (x_i, y_j) , we have

$$\frac{\partial^2 u}{\partial x^2} \Big|_{i,j} = \frac{1}{h^2} (u_{i-1,j} - 2u_{i,j} + u_{i+1,j}) + \mathcal{O}(h^2).$$

Letting $\delta_x^2 u_{i,j} := u_{i-1,j} - 2u_{i,j} + u_{i+1,j}$, we get

$$\frac{\partial^2 u}{\partial x^2} \Big|_{i,j} = \frac{\delta_x^2 u_{i,j}}{h^2} + \mathcal{O}(h^2).$$

Similarly, in the y -direction:

$$\left. \frac{\partial^2 u}{\partial y^2} \right|_{i,j} = \frac{\delta_y^2 u_{i,j}}{h^2} + \mathcal{O}(h^2), \quad \text{where } \delta_y^2 u_{i,j} := u_{i,j-1} - 2u_{i,j} + u_{i,j+1}.$$

This gives the approximation:

$$-\Delta u|_{i,j} \approx -\frac{1}{h^2} (\delta_x^2 u_{i,j} + \delta_y^2 u_{i,j}) + \mathcal{O}(h^2).$$

The finite difference equations that approximate the PDE are obtained by ignoring the truncation error and replacing $u_{i,j}$ with the grid function value $\mathbf{u}_{i,j}$. This leads to the algebraic equation

$$-\frac{1}{h^2} (\delta_x^2 \mathbf{u}_{i,j} + \delta_y^2 \mathbf{u}_{i,j}) = f_{i,j},$$

where $f_{i,j} := f(x_i, y_j)$. Substituting the definitions of δ_x^2 and δ_y^2 , we get:

$$4\mathbf{u}_{i,j} - \mathbf{u}_{i+1,j} - \mathbf{u}_{i-1,j} - \mathbf{u}_{i,j+1} - \mathbf{u}_{i,j-1} = h^2 f_{i,j}, \quad (2.4)$$

for each $(x_i, y_j) \in \Omega_h$.

Since the values of $\mathbf{u}$ at the boundary grid points are known from the Dirichlet condition, we obtain a system of $(M-1)^2$ linear equations for the unknowns at the interior points. Each equation involves the grid point and its four immediate neighbors, hence this is referred to as the *five-point stencil*.

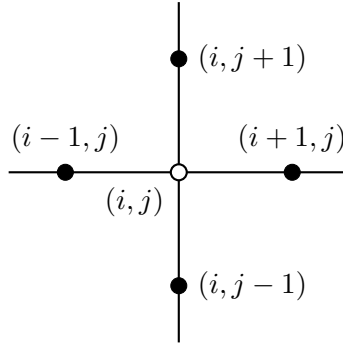


Figure 2.2: Five-point stencil: the central white node (i, j) is surrounded by its four nearest neighbors $(i \pm 1, j)$ and $(i, j \pm 1)$.

In the special case where $f \equiv 0$ (i.e., Laplace's equation), Eq. (2.4) simplifies to

$$\mathbf{u}_{i,j} = \frac{1}{4} (\mathbf{u}_{i+1,j} + \mathbf{u}_{i-1,j} + \mathbf{u}_{i,j+1} + \mathbf{u}_{i,j-1}).$$

That is, the value at an interior grid point is the arithmetic mean of its four immediate neighbors (see Fig. 2.2). This is reminiscent of the mean value property satisfied by harmonic functions in complex analysis. Remarkably, our discrete solution satisfies a similar property, highlighting the consistency of the method with the underlying theory.

2.1.1 Implementation of Five-Point Stencil

We now examine the organization of the unknowns $\mathbf{u}_{i,j}$, which are indexed by two indices. To convert them into a single column vector suitable for matrix-based computations, we arrange the interior grid values into a matrix and then stack the columns. This yields

$$\mathbf{u} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_{M-1}]^\top,$$

where each $\mathbf{u}_i \in \mathbb{R}^{M-1}$ represents a column vector of values along the vertical grid line x_i , i.e.,

$$\mathbf{u}_i = [\mathbf{u}_{i,1} \quad \mathbf{u}_{i,2} \quad \cdots \quad \mathbf{u}_{i,M-1}]^\top.$$

Next, consider the matrix structure corresponding to the finite difference equation (2.4). Each row in the matrix corresponds to a grid point $(i, j) \in \Omega_h$ and involves at most five non-zero entries due to the five-point stencil. At boundary-adjacent interior nodes, some terms involve known values from the Dirichlet boundary condition and are moved to the right-hand side.

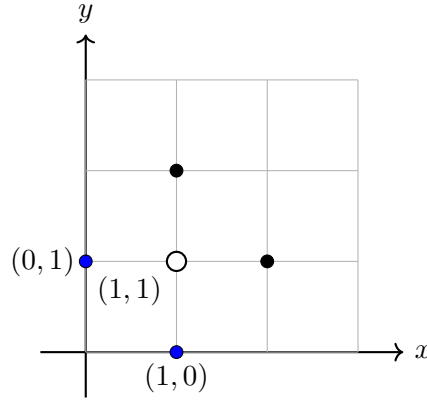


Figure 2.3: Stencil at the bottom-left corner of the grid. The interior node $(1,1)$ (white) depends on neighboring nodes. The Dirichlet boundary values at $(0,1)$ and $(1,0)$ (blue) are known from boundary conditions, while the others are unknown.

For instance, consider $(i, j) = (1, 1)$. The finite difference equation is

$$\begin{aligned} 4\mathbf{u}_{1,1} - \mathbf{u}_{2,1} - \mathbf{u}_{1,2} - \mathbf{u}_{0,1} - \mathbf{u}_{1,0} &= h^2 f_{1,1}, \\ \Rightarrow 4\mathbf{u}_{1,1} - \mathbf{u}_{2,1} - \mathbf{u}_{1,2} &= h^2 f_{1,1} + g_{0,1} + g_{1,0}, \end{aligned}$$

where $\mathbf{u}_{0,1} = g_{0,1}$ and $\mathbf{u}_{1,0} = g_{1,0}$ are known from the Dirichlet data. Hence, only three unknowns remain on the left-hand side.

We now focus on the matrix representation column by column. Fix a vertical index i and consider the values $\mathbf{u}_{i,j}$ for $j = 1, 2, \dots, M-1$. Each $\mathbf{u}_i$ is influenced by its neighbors $\mathbf{u}_{i-1}$ and $\mathbf{u}_{i+1}$, as shown below.

For $j = 1$:

$$\frac{1}{h^2} (4\mathbf{u}_{i,1} - \mathbf{u}_{i-1,1} - \mathbf{u}_{i+1,1} - \mathbf{u}_{i,2} - \mathbf{u}_{i,0}) = f_{i,1}.$$

Since $\mathbf{u}_{i,0} = g_{i,0}$ is known, this becomes:

$$4\mathbf{u}_{i,1} - \mathbf{u}_{i-1,1} - \mathbf{u}_{i+1,1} - \mathbf{u}_{i,2} = h^2 f_{i,1} + g_{i,0}.$$

For $j = 2$:

$$4\mathbf{u}_{i,2} - \mathbf{u}_{i-1,2} - \mathbf{u}_{i+1,2} - \mathbf{u}_{i,1} - \mathbf{u}_{i,3} = h^2 f_{i,2}.$$

For $j = M - 1$:

$$4\mathbf{u}_{i,M-1} - \mathbf{u}_{i-1,M-1} - \mathbf{u}_{i+1,M-1} - \mathbf{u}_{i,M-2} = h^2 f_{i,M-1} + g_{i,M},$$

where $\mathbf{u}_{i,M} = g_{i,M}$ is again known from the boundary.

Writing these equations in matrix form for fixed i , we observe a tridiagonal structure:

$$\begin{aligned} [-1] [\mathbf{u}_{i-1,1}] + [4 \quad -1] \begin{bmatrix} \mathbf{u}_{i,1} \\ \mathbf{u}_{i,2} \end{bmatrix} + [-1] [\mathbf{u}_{i+1,1}] &= h^2 f_{i,1} + g_{i,0}, \\ [-1] [\mathbf{u}_{i-1,2}] + [-1 \quad 4 \quad -1] \begin{bmatrix} \mathbf{u}_{i,1} \\ \mathbf{u}_{i,2} \\ \mathbf{u}_{i,3} \end{bmatrix} + [-1] [\mathbf{u}_{i+1,2}] &= h^2 f_{i,2}, \\ &\vdots \\ [-1] [\mathbf{u}_{i-1,M-1}] + [-1 \quad 4] \begin{bmatrix} \mathbf{u}_{i,M-2} \\ \mathbf{u}_{i,M-1} \end{bmatrix} + [-1] [\mathbf{u}_{i+1,M-1}] &= h^2 f_{i,M-1} + g_{i,M}. \end{aligned}$$

Each column vector $\mathbf{u}_i$ satisfies a linear system involving a tridiagonal matrix $\mathbf{D} \in \mathbb{R}^{(M-1) \times (M-1)}$:

$$\mathbf{D} = \begin{bmatrix} 4 & -1 & 0 & \cdots & 0 \\ -1 & 4 & -1 & \cdots & 0 \\ 0 & -1 & 4 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & -1 \\ 0 & 0 & \cdots & -1 & 4 \end{bmatrix}. \quad (2.5)$$

Each $\mathbf{u}_i$ is also coupled with neighboring columns $\mathbf{u}_{i-1}$ and $\mathbf{u}_{i+1}$ through identity matrices (denoted by $\mathbb{I}$). The full system for $i = 1, 2, \dots, M - 1$ becomes:

$$-\mathbb{I}\mathbf{u}_{i-1} + \mathbf{D}\mathbf{u}_i - \mathbb{I}\mathbf{u}_{i+1} = h^2 \mathbf{f}_i + \mathbf{g}_i,$$

where

$$\mathbf{f}_i = [f_{i,1} \quad f_{i,2} \quad \cdots \quad f_{i,M-1}]^\top, \quad \text{and} \quad \mathbf{g}_i = [g_{i,0} \quad 0 \quad \cdots \quad 0 \quad g_{i,M}]^\top.$$

The values $\mathbf{u}_0 = \mathbf{g}_0$ and $\mathbf{u}_M = \mathbf{g}_M$ correspond to the Dirichlet boundary and are moved to the right-hand side.

Stacking the equations for all $i = 1, \dots, M - 1$, we obtain a linear system of the form

$$\mathbf{A}\mathbf{u} = \mathbf{f},$$

where $\mathbf{A} \in \mathbb{R}^{(M-1)^2 \times (M-1)^2}$, and both $\mathbf{u}$ and $\mathbf{f}$ are vectors in $\mathbb{R}^{(M-1)^2}$. The matrix $\mathbf{A}$ has a block tridiagonal structure given by

$$\mathbf{A} = \frac{1}{h^2} \begin{bmatrix} \mathbf{D} & -\mathbb{I} & & \\ -\mathbb{I} & \mathbf{D} & \ddots & \\ & \ddots & \ddots & -\mathbb{I} \\ & & -\mathbb{I} & \mathbf{D} \end{bmatrix}.$$

The right-hand side takes the form

$$\mathbf{f} = \begin{bmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \\ \vdots \\ \mathbf{f}_{M-1} \end{bmatrix} + \frac{1}{h^2} \begin{bmatrix} \mathbf{g}_0 + \mathbf{g}_1 \\ \mathbf{g}_2 \\ \vdots \\ \mathbf{g}_{M-1} + \mathbf{g}_M \end{bmatrix},$$

which accounts for the contributions from the boundary conditions.

Let us now examine the structure of the matrix $\mathbf{A}$ in more detail. Recall that $\mathbf{A} \in \mathbb{R}^{(M-1)^2 \times (M-1)^2}$, so it contains approximately M^4 entries in total. However, if we inspect the structure of $\mathbf{A}$, particularly the second block row, we find that each block row contains roughly $5M$ non-zero entries—due to contributions from the main diagonal block and the two neighboring identity blocks. Consequently, the total number of non-zero entries in $\mathbf{A}$ is approximately $5M^2$.

This indicates that the matrix $\mathbf{A}$ is *sparse*, meaning that the majority of its entries are zero. Fig. 2.4 illustrates the sparsity pattern of $\mathbf{A}$ for $M = 10$, with non-zero entries shown in black.

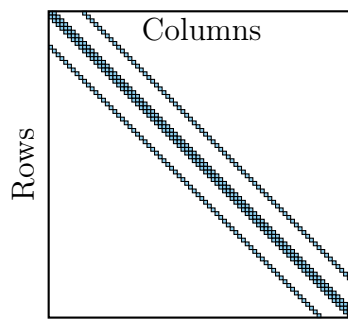


Figure 2.4: Sparsity pattern for matrix $\mathbf{A}$ for $M = 10$.

Definition 2.2 (Sparse Matrix). A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is called a *sparse matrix* if a significant fraction of its entries are zero. The *sparsity* of a matrix is typically quantified as the ratio

$$\text{sparsity}(\mathbf{A}) = 1 - \frac{\text{number of non-zero entries in } \mathbf{A}}{n^2}.$$

Solving linear systems involving sparse matrices requires methods that exploit their structure to reduce computational cost and memory usage. Among the most widely used direct methods is LU decomposition, which factorizes $\mathbf{A}$ into a product of a lower triangular matrix $\mathbf{L}$ and an upper triangular matrix $\mathbf{U}$.

Remark 2.2. We might revisit sparse matrix storage schemes and efficient solution techniques—both direct and iterative—later in the course when we focus on linear solvers.

Let us now look at the implementation of the matrix $\mathbf{A}$ and the right-hand side vector $\mathbf{f}$ for the finite difference method (FDM).

Definition 2.3 (Kronecker Product). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$. The *Kronecker product* $\mathbf{A} \otimes \mathbf{B}$ is defined as the block matrix:

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix} \in \mathbb{R}^{mp \times nq}.$$

In component form, this means that each scalar entry a_{ij} of matrix $\mathbf{A}$ is multiplied by the entire matrix $\mathbf{B}$. For example, if $\mathbf{A}$ is 2×2 and $\mathbf{B}$ is 2×2 , then:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix},$$

then

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{bmatrix}.$$

For the 2D finite difference discretization of the Poisson equation, the global stiffness matrix $\mathbf{A}$ can be written as

$$\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2,$$

where

$$\mathbf{A}_1 = \mathbb{I} \otimes \mathbf{D}, \quad \text{and} \quad \mathbf{A}_2 = \bar{\mathbb{I}} \otimes \mathbb{I}.$$

Here $\bar{\mathbb{I}}$ is the tridiagonal matrix with zero diagonal and -1 on its first sub- and super-diagonals:

$$\bar{\mathbb{I}} = \begin{bmatrix} 0 & -1 & 0 & \cdots & 0 \\ -1 & 0 & -1 & \cdots & 0 \\ 0 & -1 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 0 \end{bmatrix}.$$

Thus, the global matrix becomes:

$$\mathbf{A} = \mathbb{I} \otimes \mathbf{D} + \bar{\mathbb{I}} \otimes \mathbb{I}.$$

In Python, the Kronecker products can be constructed using the function `np.kron(A, B)` from the NumPy library. Algorithm 1 presents a basic implementation of this method for the 2D Poisson equation with Dirichlet boundary conditions.

Algorithm 1 Finite Difference Method for 2D Poisson Equation with Dirichlet BCs

Given: Domain $\Omega = (0,1)^2$, Number of nodes per dimension: M , right-hand side: $f(x, y)$, boundary data: $g(x, y)$

Find: Approximate solution $\mathbf{u}$ on interior grid

Step 1: Meshing

Initialize $h = \frac{1}{M}$

Define interior grid points $x_i = ih$, $y_j = jh$, for $i, j = 1, 2, \dots, M - 1$

Total number of interior nodes $N = M - 1$

Step 2: Assemble matrix $\mathbf{A}$

Define matrix $\mathbf{D}, \bar{\mathbb{I}} \in \mathbb{R}^{N \times N}$ and by

for $i = 0$ **to** $N - 1$ **do**

$\mathbf{D}_{i,i} = 4$, $\bar{\mathbb{I}}_{i,i} = 0$

if $i > 0$ **then**

$\mathbf{D}_{i,i-1} = -1$, $\bar{\mathbb{I}}_{i,i-1} = -1$

end if

if $i < N - 1$ **then**

$\mathbf{D}_{i,i+1} = -1$, $\bar{\mathbb{I}}_{i,i+1} = -1$

end if

end for

Define identity matrix $\mathbb{I} \in \mathbb{R}^{N \times N}$

Define $\mathbf{A} = (\mathbb{I} \otimes \mathbf{D} + \bar{\mathbb{I}} \otimes \mathbb{I}) / h^2$

Step 3: Assemble the RHS

Initialize load vector $\mathbf{F} \in \mathbb{R}^{N^2}$

for $i = 0$ **to** $N - 1$ **do**

for $j = 0$ **to** $N - 1$ **do**

 Compute grid point: $x = (i + 1)h$, $y = (j + 1)h$

 Compute linear index: $\ell = i \cdot N + j$

$\mathbf{F}_\ell = f(x, y)$

Apply Dirichlet BCs:

if $i = 0$ **then**

$\mathbf{F}_{\ell+} = g(0, y) / h^2$

end if

if $i = N - 1$ **then**

$\mathbf{F}_{\ell+} = g(1, y) / h^2$

end if

if $j = 0$ **then**

$\mathbf{F}_{\ell+} = g(x, 0) / h^2$

end if

if $j = N - 1$ **then**

$\mathbf{F}_{\ell+} = g(x, 1) / h^2$

end if

end for

end for

Step 4: Solve the linear system

Solve $\mathbf{A}\mathbf{u} = \mathbf{F}$ for $\mathbf{u} \in \mathbb{R}^{N^2}$

26

return $\mathbf{u}$

2.1.2 Convergence Theory

Implementation of a numerical scheme is one aspect that primarily concerns engineers. Another important aspect, which concerns mathematicians, is the study of its convergence properties. We say a numerical solution $\mathbf{u}$ is a good approximation of the analytical solution u if the error between them reduces as the discretization parameter h tends to zero. One important question is how to measure the closeness of $\mathbf{u}$ and u , as $\mathbf{u}$ consists of discrete values, whereas u is a continuous function.

We re-write Eq. (2.2) as $\mathcal{L}u = \mathcal{F}$, where

$$\mathcal{L}u = \begin{cases} -\Delta u & \text{in } \Omega, \\ u & \text{on } \Gamma, \end{cases} \quad \text{and} \quad \mathcal{F} = \begin{cases} f & \text{in } \Omega, \\ g & \text{on } \Gamma. \end{cases} \quad (2.6)$$

We define the five-point stencil operator by $\mathcal{L}_h \mathbf{u} = \mathcal{F}_h$, where

$$\mathcal{L}_h \mathbf{u} = \begin{cases} \mathfrak{L}_h \mathbf{u} & \text{in } \Omega_h, \\ \mathfrak{B}_h \mathbf{u} & \text{on } \Gamma_h, \end{cases} \quad \text{and} \quad \mathcal{F}_h = \begin{cases} \mathbf{f} & \text{in } \Omega_h, \\ \mathbf{g} & \text{on } \Gamma_h. \end{cases} \quad (2.7)$$

Here, $\mathfrak{L}_h \mathbf{u}_{i,j} := -h^{-2} (\delta_x^2 + \delta_y^2) \mathbf{u}_{i,j}$ and $\mathfrak{B}_h \mathbf{u}_{i,j} := \mathbf{u}_{i,j}$, i.e., the identity operator.

The operator $\mathcal{L}_h$ approximates the continuous operator $\mathcal{L}$. To quantify how well it does so, we introduce the notion of *local truncation error*.

Definition 2.4. (Local Truncation Error) The local truncation error, denoted by $\mathfrak{R}_h$, is defined as the residual when the exact solution u of Eq. (2.6) is substituted into the discrete equation Eq. (2.7), i.e.,

$$\mathfrak{R}_h := \mathcal{L}_h u - \mathcal{F}_h.$$

Definition 2.5. (Consistency) The approximation $\mathcal{L}_h \mathbf{u} = \mathcal{F}_h$ is said to be *consistent* with $\mathcal{L}u = \mathcal{F}$ if $\mathfrak{R}_h \rightarrow 0$ as $h \rightarrow 0$. It is said to be consistent of order p if $\mathfrak{R}_h = \mathcal{O}(h^p)$ with $p > 0$.

The first step is to verify whether the FDM is consistent. Since Dirichlet boundary conditions are implemented exactly, we only need to analyze $\mathfrak{L}_h \mathbf{u}$. Thus, the local truncation error reduces to $\mathfrak{R}_h = \mathfrak{L}_h u - \mathbf{f}$. Using Taylor series expansion for $\delta_x^2 u_{i,j}$ and $\delta_y^2 u_{i,j}$ at the internal node (ih, jh) , we get

$$\begin{aligned} \mathfrak{R}_h|_{i,j} &= -h^{-2} (\delta_x^2 + \delta_y^2) u_{i,j} - \mathbf{f}_{i,j} \\ &= -(u_{xx}|_{i,j} + u_{yy}|_{i,j}) - \mathbf{f}_{i,j} - \frac{h^2}{12} (u_{xxxx} + u_{yyyy})|_{i,j} + \mathcal{O}(h^4) \\ &= -\frac{h^2}{12} (u_{xxxx} + u_{yyyy})|_{i,j} + \mathcal{O}(h^4), \end{aligned}$$

since $-\Delta u = f$. Hence, the method is consistent of order 2, provided $u \in \mathcal{C}^4(\Omega)$.

Consistency tells us how closely $\mathcal{L}_h$ approximates $\mathcal{L}$. The numerical solution $\mathbf{u}$ satisfies $\mathcal{L}_h \mathbf{u} = \mathcal{F}_h$, while the analytical solution u satisfies $\mathcal{L}_h u = \mathcal{F}_h + \mathfrak{R}_h$. To study their difference, let $\mathbf{e} = u - \mathbf{u}$, where $\mathbf{e}_{i,j} = u(x_i, y_j) - \mathbf{u}_{i,j}$. Then,

$$\mathcal{L}_h \mathbf{e} = \mathcal{L}_h \mathbf{u} - \mathcal{L}_h u = \mathfrak{R}_h.$$

We want the error $\mathbf{e}$ to go to zero as $h \rightarrow 0$. This happens if $\mathfrak{R}_h \rightarrow 0$ and $\mathcal{L}_h$ is invertible. This leads to the notion of *stability*.

Definition 2.6. (ℓ_∞ Stability) The discrete operator is said to be stable (with respect to the maximum norm $\|\cdot\|_{h,\infty}$) if there exists a constant $C > 0$, independent of h , such that the solution of the equation $\mathcal{L}_h \mathbf{u} = \mathcal{F}_h$ satisfies

$$\|\mathbf{u}\|_{h,\infty} \leq C \|\mathcal{F}_h\|_{h,\infty},$$

where C is known as the *stability constant*.

A few things to note: first, the stability of $\mathcal{L}_h$ makes no reference to the original operator $\mathcal{L}$; second, stability depends on the particular norm, in this case the maximum norm $\|\cdot\|_{h,\infty}$. This notion extends to other norms such as ℓ_p norms for $1 \leq p < \infty$. As a refresher, the ℓ_∞ norm is defined as

$$\|u - \mathbf{u}\|_{h,\infty} = \max_{i,j} |u(x_i, y_j) - \mathbf{u}_{i,j}|.$$

As mentioned in the introduction, this is how we measure the closeness of the numerical solution.

Remark 2.3. The notion of ℓ_∞ stability mirrors that of a well-posed problem. A boundary value problem with a unique solution is said to be well-posed if the solution varies continuously with respect to the input data. That is, if δf is a perturbation in the source or boundary terms and δu is the resulting change in the solution, then $\|\delta u\|_a \leq C \|\delta f\|_a$ for some norm $\|\cdot\|_a$.

But why do we need both consistency and stability? To answer that, we now formally define convergence.

Definition 2.7. (Convergence) A numerical method is said to *converge* if $\|u - \mathbf{u}\|_{h,\infty} \rightarrow 0$ as $h \rightarrow 0$. It is said to be convergent of order p if $\|u - \mathbf{u}\|_{h,\infty} = \mathcal{O}(h^p)$ for some $p > 0$.

We introduced consistency and stability first because there is a fundamental relationship among the three:

Theorem 2.4. (Convergence) Suppose that the discrete boundary value problem $\mathcal{L}_h \mathbf{u} = \mathcal{F}_h$ is a consistent approximation of $\mathcal{L}u = \mathcal{F}$ and that $\mathcal{L}_h$ is ℓ_∞ stable. Then $\|u - \mathbf{u}\|_{h,\infty} \rightarrow 0$ as $h \rightarrow 0$. Moreover, if the order of consistency is $p > 0$, then the order of convergence is also p .

Proof. Since $\mathcal{L}_h$ is ℓ_∞ stable and the error $\mathbf{e}$ satisfies

$$\mathcal{L}_h \mathbf{e} = \mathfrak{R}_h,$$

we have $\|\mathbf{e}\|_{h,\infty} \leq C\|\mathfrak{R}_h\|_{h,\infty}$. Because $\mathcal{L}_h$ is consistent of order p , we conclude that $\|\mathbf{e}\|_{h,\infty} \rightarrow 0$ as $h \rightarrow 0$ with the same order, provided C is independent of h . $\square$

We have already seen that our method is consistent, and hence, if it is stable, then it will also be convergent due to Theorem 2.4. Before we prove stability, we need to introduce a few concepts. We first mention certain properties of the operator $\mathcal{L}$.

- $\mathcal{L}$ is said to be linear if $\mathcal{L}(u + \alpha v) = \mathcal{L}(u) + \alpha\mathcal{L}(v)$ for scalar α .
- $\mathcal{L}$ is said to be *inverse monotone* if $\mathcal{L}u \geq 0$ implies that $u \geq 0$.

Remark 2.5. For a linear operator $\mathcal{L}$, the definition of inverse monotonicity can equivalently be stated using the opposite inequality:

$$\mathcal{L}u \leq 0 \implies u \leq 0.$$

Indeed, if $\mathcal{L}u \leq 0$, then by linearity

$$\mathcal{L}(-u) = -\mathcal{L}u \geq 0,$$

and inverse monotonicity gives $-u \geq 0$, or equivalently $u \leq 0$.

To prove stability, we also require a bounded, non-negative function $\varphi(x)$, called a *comparison function*, such that $\mathcal{L}\varphi(x) \geq 1$ for all $x \in [0, 1]$.

Lemma 2.6. (Stability) *Suppose that the operator $\mathcal{L}_h$ is linear and inverse monotone, and that there exists a comparison function $\Phi > 0$ such that $\mathcal{L}_h\Phi \geq 1$. Then $\mathcal{L}_h$ is stable with stability constant $C = \max_{i,j} \Phi_{i,j}$, provided Φ is bounded independently of h .*

Proof. Let $\mathbf{u}$ be a solution of $\mathcal{L}_h\mathbf{u} = \mathcal{F}_h$. We need to show that $\|\mathbf{u}\|_{h,\infty} \leq C\|\mathcal{F}_h\|_{h,\infty}$. For a particular (i, j) -th entry:

$$\begin{aligned} \mathcal{L}_h\mathbf{u}_{i,j} = (\mathcal{F}_h)_{i,j} &\leq \|\mathcal{F}_h\|_{h,\infty} \\ &= \|\mathcal{F}_h\|_{h,\infty} \times 1 \\ &\leq \|\mathcal{F}_h\|_{h,\infty} \mathcal{L}_h\Phi_{i,j} \\ &= \mathcal{L}_h(\|\mathcal{F}_h\|_{h,\infty}\Phi)_{i,j}, \end{aligned}$$

which implies $0 \leq \mathcal{L}_h(\|\mathcal{F}_h\|_{h,\infty}\Phi - \mathbf{u})_{i,j}$. Hence, by inverse monotonicity, $\mathbf{u}_{i,j} \leq \|\mathcal{F}_h\|_{h,\infty}\Phi_{i,j}$. Similarly, we obtain $-\mathbf{u}_{i,j} \leq \|\mathcal{F}_h\|_{h,\infty}\Phi_{i,j}$. Since $-\mathbf{u}_{i,j}, \mathbf{u}_{i,j} \leq |\mathbf{u}_{i,j}|$, we get $|\mathbf{u}_{i,j}| \leq \|\mathcal{F}_h\|_{h,\infty}\Phi_{i,j}$.

Now, taking the maximum over all i, j :

$$\|\mathbf{u}\|_{h,\infty} = \max_{i,j} |\mathbf{u}_{i,j}| \leq C\|\mathcal{F}_h\|_{h,\infty}, \quad \text{where } C = \max_{i,j} \Phi_{i,j}.$$

$\square$

Remark 2.7. In Lemma 2.6, we consider a grid function $\mathbf{u}$. When we say $\mathbf{u} > 0$, it means $\mathbf{u}_{i,j} > 0$ for all $i, j = 1, 2, \dots, M-1$.

We now prove the stability of our FDM operator $\mathcal{L}_h$.

Theorem 2.8. *The difference operator $\mathcal{L}_h$ defined in Eq. (2.7) is stable.*

Proof. We need to show two things for $\mathcal{L}_h$ to be stable. First is inverse monotonicity, and second, we need to compute a comparison function Φ such that $\mathcal{L}_h \Phi \geq 1$.

Inverse Monotonicity: We need to show that $\mathcal{L}_h \mathbf{u} \geq 0$ implies $\mathbf{u} \geq 0$.

We proceed by contradiction. Suppose this is not true. Then there exists a grid function $\mathbf{u}$ such that $\mathcal{L}_h \mathbf{u} \geq 0$ but $\mathbf{u} < 0$, i.e., there exists a point (i, j) such that

$$\mathbf{u}_{i,j} < 0. \quad (2.8)$$

Let (i_0, j_0) be the point where the minimum occurs.

Now,

$$\mathcal{L}_h \mathbf{u}_{i_0, j_0} = \frac{1}{h^2} (4\mathbf{u}_{i_0, j_0} - \mathbf{u}_{i_0+1, j_0} - \mathbf{u}_{i_0-1, j_0} - \mathbf{u}_{i_0, j_0+1} - \mathbf{u}_{i_0, j_0-1}).$$

Since $\mathbf{u}_{i_0, j_0}$ is the minimum, we have $\mathbf{u}_{i_0, j_0} \leq \mathbf{u}_{i_0 \pm 1, j_0}$ and $\mathbf{u}_{i_0, j_0} \leq \mathbf{u}_{i_0, j_0 \pm 1}$, which implies

$$\mathbf{u}_{i_0, j_0} - \mathbf{u}_{i_0 \pm 1, j_0} \leq 0 \quad \text{and} \quad \mathbf{u}_{i_0, j_0} - \mathbf{u}_{i_0, j_0 \pm 1} \leq 0.$$

Hence,

$$\begin{aligned} \mathcal{L}_h \mathbf{u}_{i_0, j_0} &= \frac{1}{h^2} (\mathbf{u}_{i_0, j_0} - \mathbf{u}_{i_0+1, j_0} + \mathbf{u}_{i_0, j_0} - \mathbf{u}_{i_0-1, j_0} + \mathbf{u}_{i_0, j_0} - \mathbf{u}_{i_0, j_0+1} + \mathbf{u}_{i_0, j_0} - \mathbf{u}_{i_0, j_0-1}) \\ &\leq 0. \end{aligned}$$

But we also assumed that $\mathcal{L}_h \mathbf{u}_{i_0, j_0} \geq 0$. Therefore, we must have $\mathcal{L}_h \mathbf{u}_{i_0, j_0} = 0$, and all terms in the sum must be zero:

$$\mathbf{u}_{i_0, j_0} = \mathbf{u}_{i_0 \pm 1, j_0} = \mathbf{u}_{i_0, j_0 \pm 1}.$$

That is, the value at (i_0, j_0) equals the values at its four neighbors. Now, pick one of these neighbors, say $(i_0 + 1, j_0)$, and apply the same argument. Again, you will conclude that all its neighbors must have the same value. Repeating this process across the domain implies that $\mathbf{u}$ is constant on $\overline{\Omega}_h$.

Moreover, from (2.8), this constant is strictly negative: $\mathbf{u} < 0$ on $\overline{\Omega}_h$. But on the boundary $\partial\Omega_h$, we use Dirichlet conditions: $\mathcal{L}_h \mathbf{u} = \mathfrak{B}_h \mathbf{u} = \mathbf{u} = g$, and from $\mathcal{L}_h \mathbf{u} \geq 0$ this implies $g \geq 0$ on $\partial\Omega_h$. So we obtain a contradiction since $\mathbf{u} < 0$ on the boundary. Hence, our assumption is false, and we conclude that $\mathcal{L}_h$ is inverse monotone.

Comparison Function: Next, we need to find a comparison function $\Phi(x, y) > 0$ and show that $\mathcal{L}_h \Phi \geq 1$. Let us consider the function

$$\Phi(x, y) = 1 + \frac{1}{2}x(1 - x),$$

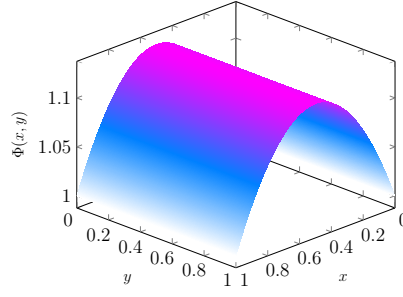


Figure 2.5: Surface plot of the comparison function $\Phi(x, y) = 1 + \frac{1}{2}x(1 - x)$, which is independent of y .

which is bounded and non-negative on $[0, 1]^2$ (see Figure 2.5).

Now, compute $\mathcal{L}_h \Phi_{i,j}$. Since Φ depends only on x , we can write:

$$\begin{aligned} \mathcal{L}_h \Phi_{i,j} &= -\frac{1}{h^2} (-4\Phi_{i,j} + \Phi_{i+1,j} + \Phi_{i-1,j} + \Phi_{i,j-1} + \Phi_{i,j+1}) \\ &= -\frac{1}{h^2} \left(-4 \left(1 + \frac{1}{2}x_i(1 - x_i) \right) + \left(1 + \frac{1}{2}x_{i+1}(1 - x_{i+1}) \right) \right. \\ &\quad \left. + \left(1 + \frac{1}{2}x_{i-1}(1 - x_{i-1}) \right) + \left(1 + \frac{1}{2}x_i(1 - x_i) \right) + \left(1 + \frac{1}{2}x_i(1 - x_i) \right) \right). \end{aligned}$$

Using $x_{i\pm 1} = x_i \pm h$, compute:

$$\begin{aligned} x_{i+1}(1 - x_{i+1}) &= (x_i + h)(1 - x_i - h) = x_i(1 - x_i) - h(1 - 2x_i) - h^2, \\ x_{i-1}(1 - x_{i-1}) &= (x_i - h)(1 - x_i + h) = x_i(1 - x_i) + h(1 - 2x_i) - h^2. \end{aligned}$$

Adding both:

$$x_{i-1}(1 - x_{i-1}) + x_{i+1}(1 - x_{i+1}) = 2x_i(1 - x_i) - 2h^2.$$

Substitute this into the expression for $\mathcal{L}_h \Phi_{i,j}$:

$$\begin{aligned} \mathcal{L}_h \Phi_{i,j} &= -\frac{1}{h^2} \left(-4 \left(1 + \frac{1}{2}x_i(1 - x_i) \right) + 2 \left(1 + \frac{1}{2}x_i(1 - x_i) \right) + \frac{1}{2}(2x_i(1 - x_i) - 2h^2) \right) \\ &= -\frac{1}{h^2} (-2 - x_i(1 - x_i) + x_i(1 - x_i) - h^2) = 1. \end{aligned}$$

Hence, $\mathcal{L}_h \Phi = 1$, and the comparison function condition is satisfied.

Therefore, both required properties are fulfilled, and $\mathcal{L}_h$ is ℓ_∞ stable. $\square$

We have the consistency and stability of our numerical method. Therefore we have the following result.

Corollary 2.9. *The five-point approximation to the Poisson equation with a Dirichlet boundary condition on the unit square, Ω , is second-order convergent if the fourth derivatives of the exact solution are bounded.*

Remark 2.10. (Experimental Order of Convergence (E.O.C.)) From theory, we expect second-order convergence for the five-point scheme. To verify this numerically, we define the order α of a method by

$$\|u - \mathbf{u}_h\|_{h,\infty} = \mathcal{O}(h^\alpha), \quad \text{for some } \alpha > 0, \quad (2.9)$$

where $\mathbf{u}_h$ denotes the solution obtained on a grid of width h . If we use a coarser grid of size $2h$, then

$$\|u - \mathbf{u}_{2h}\|_{2h,\infty} = \mathcal{O}(2^\alpha h^\alpha).$$

Therefore,

$$\frac{\|u - \mathbf{u}_h\|_{h,\infty}}{\|u - \mathbf{u}_{2h}\|_{2h,\infty}} \approx \frac{1}{2^\alpha}$$

which gives the estimate

$$\alpha \approx \frac{1}{\log(2)} \log \left(\frac{\|u - \mathbf{u}_{2h}\|_{2h,\infty}}{\|u - \mathbf{u}_h\|_{h,\infty}} \right).$$

This quantity can be computed directly from numerical results and is called the *experimental order of convergence* (E.O.C.).

Another way to estimate α is via a log-log plot. Taking the logarithm of Eq. (2.9) gives

$$\log(\|u - \mathbf{u}_h\|_{h,\infty}) \approx \alpha \log(h).$$

Defining $Y = \log(\|u - \mathbf{u}_h\|_{h,\infty})$ and $X = \log(h)$, we obtain a linear relation $Y \approx \alpha X$. The slope of the Y vs. X curve (for varying h) then provides an estimate of α .

Example 2.1. Consider Eq. (2.2) with

$$f = 2\pi^2 \sin(\pi x) \cos(\pi y), \quad g = 0.$$

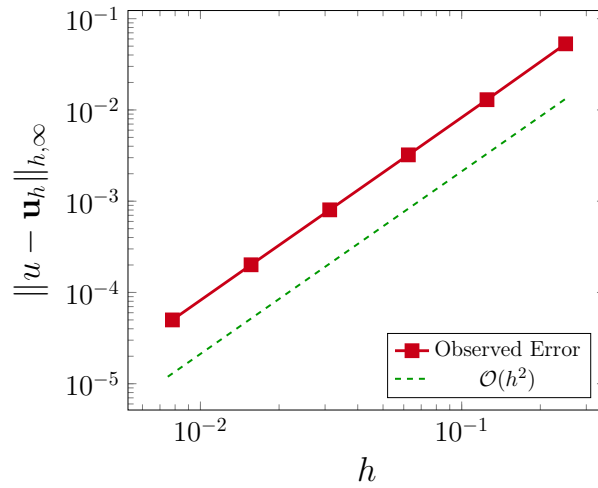
The analytical solution is

$$u(x, y) = \sin(\pi x) \cos(\pi y).$$

We compute the numerical solution using the five-point stencil and report the results in Table 2.1. We observe that the computed E.O.C. matches the theoretical prediction of second-order accuracy. We also plot the log-log plot in Fig. 2.6.

h	$\ u - \mathbf{u}\ _{h,\infty}$	E.O.C.
0.250000	0.053029	-
0.125000	0.012951	2.033754
0.062500	0.003219	2.008367
0.031250	0.000804	2.002087
0.015625	0.000201	2.000522
0.007812	0.000050	2.000130

Table 2.1: Example 2.1: Experimental order of convergence.

Figure 2.6: Log-log plot of error vs. mesh size h .

2.1.3 Discrete Maximum Principle

The analytical solution of Eq. (2.2) satisfies certain important properties. One such property is the *maximum principle*.

Theorem 2.11. (Maximum Principle for the Poisson Problem) Suppose that the function $u(x, y)$ satisfies the inequality

$$-u_{xx} - u_{yy} \leq 0,$$

for all $(x, y) \in \Omega$. Then $u(x, y)$ is either constant or attains its maximum value on $\partial\Omega$.

Remark 2.12. In some references (see [6]), maximum principles are distinguished as the *local maximum principle* and the *global maximum principle*. Here, we present a unified version encompassing both.

Remark 2.13. The maximum principle reflects the fundamental behavior of heat flow in steady-state thermal equilibrium. According to Fourier's law of heat conduction, thermal energy naturally flows from higher to lower temperatures. If a strict peak temperature existed at an interior point, heat would constantly drain outward from that point into its cooler surroundings in every direction. For that interior point to remain a peak without cooling down over time, an internal heat source would have to continuously generate energy at that exact location to replenish the loss. Without such a source, any interior hot spot inevitably flattens out, forcing the overall maximum temperature to be anchored on the boundary where external heat is supplied.

The above theorem states that the maximum of the solution can occur only on the boundary. If it is attained in the interior, then the solution must be constant.

An important aspect in designing numerical methods is ensuring that the numerical solution also satisfies a discrete analogue of the analytical properties. In this case it will be the *discrete maximum principle*.

Theorem 2.14. (Discrete Maximum Principle) *Suppose that the discrete operator $\mathcal{L}_h \mathbf{u}$ is defined as in Eq. (2.7) and satisfies $\mathcal{L}_h \mathbf{u} \leq 0$ at all grid points in Ω_h . Then, either $\mathbf{u}$ is constant in Ω_h or it attains its maximum on the boundary $\partial\Omega_h$.*

Proof. We proceed by contradiction. The proof follows the same structure as the proof of ℓ_∞ -stability of $\mathcal{L}_h$. Suppose $\mathbf{u}$ attains a non-constant maximum at an interior point (x_i, y_j) , where $0 < i, j < M$. Then, since $\mathcal{L}_h \mathbf{u}_{i,j} \leq 0$, we have

$$\mathbf{u}_{i,j} \leq \frac{1}{4} (\mathbf{u}_{i+1,j} + \mathbf{u}_{i-1,j} + \mathbf{u}_{i,j+1} + \mathbf{u}_{i,j-1}).$$

But since $\mathbf{u}_{i,j}$ is the maximum, it satisfies

$$\mathbf{u}_{i,j} \geq \mathbf{u}_{i\pm 1,j}, \quad \mathbf{u}_{i,j} \geq \mathbf{u}_{i,j\pm 1},$$

and hence

$$\mathbf{u}_{i,j} \geq \frac{1}{4} (\mathbf{u}_{i+1,j} + \mathbf{u}_{i-1,j} + \mathbf{u}_{i,j+1} + \mathbf{u}_{i,j-1}).$$

Combining both inequalities, we conclude

$$\mathbf{u}_{i,j} = \frac{1}{4} (\mathbf{u}_{i+1,j} + \mathbf{u}_{i-1,j} + \mathbf{u}_{i,j+1} + \mathbf{u}_{i,j-1}),$$

which can only happen if $\mathbf{u}_{i,j} = \mathbf{u}_{i\pm 1,j} = \mathbf{u}_{i,j\pm 1}$ as $\mathbf{u}_{i,j}$ is the maximum. That is, all five values are equal. Therefore, $\mathbf{u}$ attains the same maximum at the five-point stencil

$$\{(i, j), (i \pm 1, j), (i, j \pm 1)\}.$$

Repeating the same argument for the four neighboring points and propagating this reasoning through the domain, we find that either the solution is constant on $\overline{\Omega}_h$ or the maximum eventually appears on the boundary. This contradicts the assumption that the maximum occurs strictly in the interior. $\square$

Remark 2.15. The definition of inverse monotonicity is most naturally applied with homogeneous or nonnegative boundary data. If the prescribed boundary values g have mixed signs, the positivity argument can still be recovered by a simple shift. In particular, let

$$M = \|g\|_{\infty, \partial\Omega}.$$

Then

$$-M \leq g \leq M.$$

Consequently, the shifted functions

$$u + M \quad \text{and} \quad M - u$$

have nonnegative boundary values. Thus, provided the operator satisfies the corresponding discrete maximum principle, inverse monotonicity can be applied to these shifted functions to obtain bounds on u . For example, in the homogeneous case $\mathcal{L}u = 0$, one obtains

$$-M \leq u \leq M,$$

and hence

$$\|u\|_\infty \leq \|g\|_{\infty, \partial\Omega}.$$

Thus, inverse monotonicity does not require the boundary data themselves to be nonnegative when establishing stability estimates.

2.2 Advanced Topics

Until now, our focus has been on a simple problem, Eq. (2.2). A natural question that arises is: why solve Eq. (2.2) numerically when analytical solutions exist? One reason is to better understand the method, validate results computationally, and develop more advanced methods for problems where analytical solutions are not available. In this section, we explore such extensions.

2.2.1 Neumann Problem in Square Domain

Consider the Poisson problem with Neumann boundary conditions defined on the unit square $\Omega := (0, 1)^2$,

$$-\Delta u = f \quad \text{in } \Omega, \quad \partial_{\mathbf{n}} u = g \quad \text{on } \Gamma, \quad (2.10)$$

where $\partial_{\mathbf{n}} u := \nabla u \cdot \mathbf{n}$ denotes the derivative in the direction of the outward unit normal $\mathbf{n}$ to $\Gamma := \partial\Omega$. A fundamental property of the Poisson problem with pure Neumann boundary conditions is that the data f and g must satisfy the compatibility condition

$$\int_{\Omega} f, dx + \int_{\Gamma} g, ds = 0. \quad (2.11)$$

Indeed, integrating the differential equation over Ω and applying the divergence theorem gives

$$\int_{\Omega} f, dx = - \int_{\Omega} \Delta u, dx = - \int_{\Gamma} \nabla u \cdot \mathbf{n}, ds = - \int_{\Gamma} g, ds, \quad (2.12)$$

which proves that (2.11) is necessary for the existence of a solution. Conversely, under the standard assumptions on f and g , this compatibility condition is also sufficient for the existence of a weak solution. However, the solution is not unique: if u is a solution, then $u + C$ is also a solution for any constant $C \in \mathbb{R}$. Thus, the Neumann problem is well-posed only after fixing the additive constant, for example by imposing the normalization

$$\int_{\Omega} u, dx = 0. \quad (2.13)$$

To develop a finite difference method for Eq. (2.10), we follow the same procedure as in the previous section. We will consider two distinct cases: first, the case $i = 0$ and $0 < j < M$ (inner boundary grid points); and second, the case $i = 0$ with $j = 0$ or $j = M$ (corner boundary grid points).

Inner Boundary Grid Points. Consider the grid point $(0, jh)$ for $0 < j < M$, i.e., a point lying on the boundary $x = 0$ (see Fig. 2.7).

Here, the outward normal is $\mathbf{n} = (-1, 0)$. Hence, the Neumann boundary condition becomes

$$\partial_{\mathbf{n}} u = (-1, 0) \cdot (u_x, u_y) = -u_x = g(0, y), \quad \text{for } 0 < y < 1. \quad (2.14)$$

We recall the standard finite difference operators. Let Δ^+ denote the forward difference

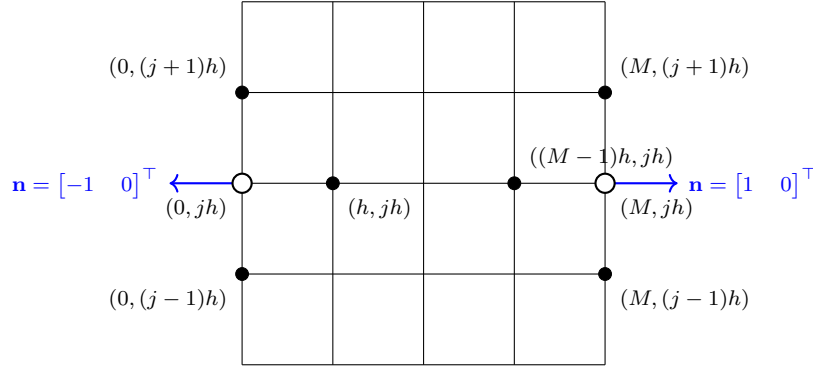


Figure 2.7: Stencil at the grid point $i = 0$ and $0 < j < M$. The interior node $(0, jh)$ (white) depends on neighboring nodes.

operator, and Δ^- the backward difference operator:

$$\begin{aligned}\Delta^+ u_i &= u_{i+1} - u_i = hu_x|_{ih} + \frac{h^2}{2}u_{xx}|_{ih} + \mathcal{O}(h^3), \\ \Delta^- u_i &= u_i - u_{i-1} = hu_x|_{ih} - \frac{h^2}{2}u_{xx}|_{ih} + \mathcal{O}(h^3).\end{aligned}$$

Let Δ_x^+ denote the forward difference operator in the x -direction. Applying it to $u_{0,j}$, we get

$$\Delta_x^+ u_{0,j} = u_{1,j} - u_{0,j} = hu_x|_{0,j} + \frac{h^2}{2}u_{xx}|_{0,j} + \mathcal{O}(h^3). \quad (2.15)$$

Rewriting,

$$-u_x(0, jh) = -\frac{1}{h}\Delta_x^+ u_{0,j} + \frac{h}{2}u_{xx}|_{0,j} + \mathcal{O}(h^2). \quad (2.16)$$

Now, since the point $(0, jh)$ lies on the boundary, we cannot directly apply a second-order centered difference for u_{xx} (as it would require $u_{-1,j}$). However, assuming u is smooth, we use the PDE itself to substitute:

$$u_{xx}|_{0,j} = -f_{0,j} - u_{yy}|_{0,j} = -f_{0,j} - \frac{1}{h^2}\delta_y^2 u_{0,j} + \mathcal{O}(h^2). \quad (2.17)$$

Substituting this into the previous expression:

$$-u_x(0, jh) = -\frac{1}{h}\Delta_x^+ u_{0,j} - \frac{h}{2}f_{0,j} - \frac{1}{2h}\delta_y^2 u_{0,j} + \mathcal{O}(h^2). \quad (2.18)$$

Replacing u_x with the Neumann condition and dropping higher-order terms gives the discrete boundary condition:

$$-\frac{1}{h}\Delta_x^+ \mathbf{u}_{0,j} - \frac{1}{2h}\delta_y^2 \mathbf{u}_{0,j} = \frac{h}{2}\mathbf{f}_{0,j} + \mathbf{g}_{0,j}. \quad (2.19)$$

Expanding the operators, this becomes:

$$-2(\mathbf{u}_{1,j} - \mathbf{u}_{0,j}) + (2\mathbf{u}_{0,j} - \mathbf{u}_{0,j+1} - \mathbf{u}_{0,j-1}) = h^2\mathbf{f}_{0,j} + 2h\mathbf{g}_{0,j}, \quad (2.20)$$

$$\Rightarrow 4\mathbf{u}_{0,j} - 2\mathbf{u}_{1,j} - \mathbf{u}_{0,j+1} - \mathbf{u}_{0,j-1} = h^2\mathbf{f}_{0,j} + 2h\mathbf{g}_{0,j}. \quad (2.21)$$

The corresponding stencil is shown in Fig. 2.7. A similar strategy can be applied at the opposite boundary ($i = M$), using a backward difference in the x -direction instead. The resulting system is slightly modified and is left as an exercise.

Let us now consider the consistency error. The local truncation error at the boundary point $(0, jh)$ is given by

$$\mathfrak{R}_h|_{0,j} := -\frac{1}{h}\Delta_x^+ \mathbf{u}_{0,j} - \frac{1}{2h}\delta_y^2 \mathbf{u}_{0,j} - \mathbf{g}_{0,j} - \frac{h}{2}\mathbf{f}_{0,j}. \quad (2.22)$$

Using Taylor expansions, we get

$$\begin{aligned} \mathfrak{R}_h|_{0,j} &= -\left(u_x + \frac{h}{2}u_{xx} + \mathcal{O}(h^2)\right)_{0,j} - \frac{h}{2}(u_{yy} + \mathcal{O}(h^2))_{0,j} - \mathbf{g}_{0,j} - \frac{h}{2}\mathbf{f}_{0,j} \\ &= (-u_x - \mathbf{g})_{0,j} + \frac{h}{2}(-u_{xx} - u_{yy} - \mathbf{f})_{0,j} + \mathcal{O}(h^2). \end{aligned}$$

Since $u_x = -g$ and $-\Delta u = f$ at $(0, jh)$, the truncation error vanishes up to $\mathcal{O}(h^2)$, confirming second-order consistency.

Corner Boundary Grid Point

Now, let us examine the corner case. Consider the point $(0, 1)$, corresponding to $i = 0$ and $j = M$ (see Fig. 2.8). Along the edge $x = 0$, the Neumann condition is $-u_x = g$, since the outward normal is $\mathbf{n} = [-1 \ 0]^\top$. Along the edge $y = 1$, the condition is $u_y = g$, since $\mathbf{n} = [0 \ 1]^\top$. At the corner, the outward normal is not uniquely defined. We therefore combine the two one-sided directional conditions associated with the adjacent edges. In particular, adding the two conditions gives

$$u_y - u_x = 2g. \quad (2.23)$$

This combination is particularly convenient for the finite-difference discretization, since the corresponding Taylor expansions involve the second derivatives u_{xx} and u_{yy} through their sum, which can be replaced using the governing equation $\Delta u = -f$.

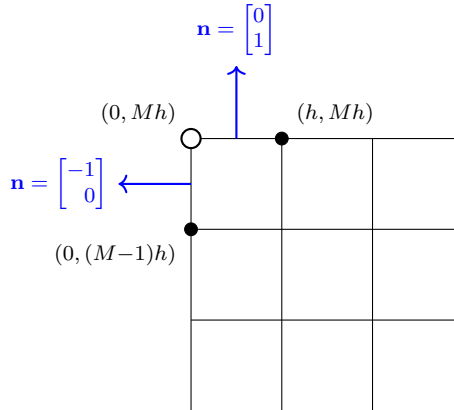


Figure 2.8: Stencil at the grid point $i = 0$ and $j = M$. The corner node $(0, Mh)$ (white) depends on neighboring nodes.

Let Δ_y^- denote the backward difference in the y -direction. Then,

$$\frac{1}{h}\Delta_y^- u_{0,M} = u_y|_{0,M} - \frac{h}{2}u_{yy}|_{0,M} + \mathcal{O}(h^2) \quad \Rightarrow \quad u_y|_{0,M} = \frac{1}{h}\Delta_y^- u_{0,M} + \frac{h}{2}u_{yy}|_{0,M} + \mathcal{O}(h^2). \quad (2.24)$$

Similarly, using the forward difference Δ_x^+ in the x -direction:

$$\frac{1}{h}\Delta_x^+ u_{0,M} = u_x|_{0,M} + \frac{h}{2}u_{xx}|_{0,M} + \mathcal{O}(h^2) \quad \Rightarrow \quad -u_x|_{0,M} = -\frac{1}{h}\Delta_x^+ u_{0,M} + \frac{h}{2}u_{xx}|_{0,M} + \mathcal{O}(h^2). \quad (2.25)$$

Adding both expressions gives:

$$2g_{0,M} = \frac{1}{h} (\Delta_y^- u_{0,M} - \Delta_x^+ u_{0,M}) + \frac{h}{2} (u_{xx} + u_{yy})_{0,M} + \mathcal{O}(h^2). \quad (2.26)$$

Replacing the second derivatives using the PDE $-\Delta u = f$ and dropping higher-order terms:

$$2\mathbf{g}_{0,M} = \frac{1}{h} (\mathbf{u}_{0,M} - \mathbf{u}_{0,M-1} - \mathbf{u}_{1,M} + \mathbf{u}_{0,M}) - \frac{h}{2}\mathbf{f}_{0,M}. \quad (2.27)$$

Simplifying,

$$2\mathbf{u}_{0,M} - \mathbf{u}_{0,M-1} - \mathbf{u}_{1,M} = \frac{h^2}{2}\mathbf{f}_{0,M} + 2h\mathbf{g}_{0,M}. \quad (2.28)$$

The local truncation error at the corner point is given by:

$$\begin{aligned} \mathfrak{R}_h|_{0,M} &= -\left(u_x + \frac{h}{2}u_{xx} + \mathcal{O}(h^2)\right)_{0,M} + \left(u_y - \frac{h}{2}u_{yy} + \mathcal{O}(h^2)\right)_{0,M} - \frac{h}{2}f_{0,M} - 2g_{0,M} \\ &= -\frac{h}{2}(u_{xx} + u_{yy} + f)_{0,M} + (u_y - g)_{0,M} - (u_x + g)_{0,M} + \mathcal{O}(h^2). \end{aligned}$$

Again, using the PDE and the Neumann boundary conditions at $(0, Mh)$, we conclude that the scheme is consistent with order $\mathcal{O}(h^2)$.

So far, we have not discussed the stability of the method. As seen in the previous section, the notion of stability is closely tied to the well-posedness of the underlying PDE. However, Eq. (2.10) does not have a unique solution: if u is a solution, then so is $u + c$ for any constant $c \in \mathbb{R}$.

Introducing a notion of stability without using function space theory is challenging. We may revisit this later from a more analytical perspective. In the meantime, it is worth pondering: with respect to which norm does the scheme exhibit stability?

Now, let us discuss the implementation of the method. One of the key differences from the Dirichlet case is that our number of unknowns increases, since the boundary values are also treated as unknowns. Thus, if

$$\mathbf{A}\mathbf{u} = \mathbf{f}$$

is the system of equations, then $\mathbf{A} \in \mathbb{R}^{(M+1)^2 \times (M+1)^2}$ and $\mathbf{u}, \mathbf{f} \in \mathbb{R}^{(M+1)^2}$.

For the interior grid points, the structure of the matrix remains the same. The additional rows and columns arise from the Neumann boundary conditions at $i = 0$, $i = M$, $j = 0$, and $j = M$. Looking at Figs. 2.7 and 2.8, we observe that:

- The unknowns at $i = 0$ depend on values at $i = 1$ (via forward difference).

- Similarly, values at $i = M$ depend on $i = M - 1$ (via backward difference).

Let the vector of unknowns at $i = 0$ be

$$\mathbf{u}_0 = [\mathbf{u}_{0,0} \quad \mathbf{u}_{0,1} \quad \dots \quad \mathbf{u}_{0,M}]^\top,$$

and similarly define the right-hand side vector $\mathbf{f}_0$. The Neumann boundary stencil gives, for $j = 1, \dots, M - 1$,

$$4\mathbf{u}_{0,j} - 2\mathbf{u}_{1,j} - \mathbf{u}_{0,j+1} - \mathbf{u}_{0,j-1} = h^2\mathbf{f}_{0,j} + 2h\mathbf{g}_{0,j}.$$

At the corners:

- For $(i, j) = (0, 0)$:

$$2\mathbf{u}_{0,0} - \mathbf{u}_{1,0} - \mathbf{u}_{0,1} = h^2/2\mathbf{f}_{0,0} + 2h\mathbf{g}_{0,0},$$

- For $(i, j) = (0, M)$:

$$2\mathbf{u}_{0,M} - \mathbf{u}_{1,M} - \mathbf{u}_{0,M-1} = h^2/2\mathbf{f}_{0,M} + 2h\mathbf{g}_{0,M}.$$

We can now collect the stencil coefficients for $\mathbf{u}_0$ into a matrix:

$$\mathbf{D}_{N,1} = \begin{bmatrix} 2 & -1 & & & \\ -1 & 4 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 4 & -1 \\ & & & -1 & 2 \end{bmatrix} \in \mathbb{R}^{(M+1) \times (M+1)},$$

and the corresponding right-hand side:

$$\mathbf{f}_0 = \begin{bmatrix} h^2/2\mathbf{f}_{0,0} + 2h\mathbf{g}_{0,0} \\ h^2\mathbf{f}_{0,1} + 2h\mathbf{g}_{0,1} \\ \vdots \\ h^2\mathbf{f}_{0,M-1} + 2h\mathbf{g}_{0,M-1} \\ h^2/2\mathbf{f}_{0,M} + 2h\mathbf{g}_{0,M} \end{bmatrix}.$$

The coupling to the neighboring column $\mathbf{u}_1$ comes from:

$$\mathbf{D}_{N,2} = \text{diag}(-1, -2, \dots, -2, -1) \in \mathbb{R}^{(M+1) \times (M+1)}.$$

So, the boundary system at $i = 0$ becomes:

$$\mathbf{D}_{N,2}\mathbf{u}_1 + \mathbf{D}_{N,1}\mathbf{u}_0 = \mathbf{f}_0.$$

The analogous structure holds at the right boundary:

$$\mathbf{D}_{N,1}\mathbf{u}_M + \mathbf{D}_{N,2}\mathbf{u}_{M-1} = \mathbf{f}_M.$$

Now, comes the inner grid points, i.e., $i = 1, 2, \dots, M-1$. For the grid point $(i, 0)$ and (i, M) we need to use *Inner Grid Points* method. Here the equation would look like

$$4\mathbf{u}_{i,0} - 2\mathbf{u}_{i,1} - \mathbf{u}_{i-1,0} - \mathbf{u}_{i+1,0} = h^2\mathbf{f}_{i,0} + 2h\mathbf{g}_{i,0},$$

and similarly for $j = M$

$$4\mathbf{u}_{i,M} - 2\mathbf{u}_{i,M-1} - \mathbf{u}_{i-1,M} - \mathbf{u}_{i+1,M} = h^2\mathbf{f}_{i,M} - 2h\mathbf{g}_{i,M}.$$

For $j = 1, 2, \dots, M-1$ we use the five-point stencil and hence for $\mathbf{u}_i$ we get the system of equations as

$$-\mathbb{I}\mathbf{u}_{i-1} + \mathbf{D}_{N,3}\mathbf{u}_i - \mathbb{I}\mathbf{u}_{i+1} = \mathbf{f}_i,$$

where $\mathbb{I}$ is a $(M+1) \times (M+1)$ identity matrix and

$$\mathbf{D}_{N,3} = \begin{bmatrix} 4 & -2 & & & \\ -1 & 4 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 4 & -1 \\ & & & -2 & 4 \end{bmatrix} \in \mathbb{R}^{(M+1) \times (M+1)}.$$

The full block structure of the matrix $\mathbf{A}$ is:

$$\mathbf{A} = \begin{bmatrix} \mathbf{D}_{N,1} & \mathbf{D}_{N,2} & & & \\ -\mathbb{I} & \mathbf{D}_{N,3} & -\mathbb{I} & & \\ & \ddots & \ddots & \ddots & \\ & & -\mathbb{I} & \mathbf{D}_{N,3} & -\mathbb{I} \\ & & & \mathbf{D}_{N,2} & \mathbf{D}_{N,1} \end{bmatrix}.$$

Let us now express $\mathbf{A}$ using Kronecker products:

$$\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2 + \mathbf{A}_3 + \mathbf{A}_4,$$

where: $\mathbf{A}_1 = \mathbf{A}_{1,1} \otimes \mathbf{D}_{N,3}$, $\mathbf{A}_2 = \mathbf{A}_{2,2} \otimes \mathbb{I}$, $\mathbf{A}_3 = \mathbf{A}_{3,3} \otimes \mathbf{D}_{N,1}$, and $\mathbf{A}_4 = \mathbf{A}_{4,4} \otimes \mathbf{D}_{N,2}$.

The matrix blocks $\mathbf{A}_{i,i}$ are defined as follows (all of size $(M+1) \times (M+1)$):

$$\mathbf{A}_{1,1} = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix}, \quad \mathbf{A}_{2,2} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ -1 & 0 & -1 & \cdots & 0 \\ 0 & -1 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 0 \end{bmatrix}, \quad \mathbf{A}_{3,3} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix},$$

and

$$\mathbf{A}_{4,4} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$$

Example 2.2. Consider Eq. (2.10) with

$$f = 8\pi^2 \cos(2\pi x) \cos(2\pi y)$$

on $\Omega = (0, 1)^2$, together with the full Neumann boundary condition

$$\frac{\partial u}{\partial \mathbf{n}} = 0 \quad \text{on } \Gamma.$$

The analytical solution is

$$u(x, y) = \cos(2\pi x) \cos(2\pi y).$$

Since this is a full Neumann problem, the solution is determined only up to an additive constant. To obtain a unique numerical solution, we impose the normalization

$$\mathbf{u}_{0,0} = 1.$$

Moreover, the compatibility condition is satisfied since

$$\int_{\Omega} f d\mathbf{x} = 8\pi^2 \int_0^1 \cos(2\pi x) dx \int_0^1 \cos(2\pi y) dy = 0,$$

while $g = 0$ on $\partial\Omega$.

We compute the numerical solution using the five-point stencil with a second-order approximation of the Neumann boundary conditions and report the results in Table 2.2. The computed E.O.C. approaches the theoretical prediction of second-order accuracy.

h	$\ u - \mathbf{u}\ _{h,\infty}$	E.O.C.
0.125000	0.106059	-
0.062500	0.025901	2.033754
0.031250	0.006438	2.008367
0.015625	0.001607	2.002087
0.007812	0.000402	2.000522

Table 2.2: Example 2.2: Experimental order of convergence for the full Neumann problem.

Compatibility condition. Since the Poisson problem with pure Neumann boundary conditions is solvable only if the right-hand side satisfies a compatibility condition, the discrete system must fulfill

$$\sum_{i=0}^M \sum_{j=0}^M h^2 \mathbf{f}_{i,j} = \sum_{\text{boundary nodes}} (2h \mathbf{g}_{i,j}),$$

where at the corners both boundary flux contributions are included. Equivalently, $\mathbf{A}$ is singular with a one-dimensional nullspace (the constant vector), and one must either enforce a mean-zero condition or fix one degree of freedom to obtain a unique solution.

2.2.2 Non-Rectangular Domains

Until now we have dealt with “good” domains—those having a nice geometrical structure such as a square. The ideas can, of course, be extended to rectangular domains as well. However, in practice we may encounter more complicated domains (see Fig. 2.9). In such cases, the main difficulty arises in computing the values at grid points located near the boundary.

Consider the Dirichlet problem, Eq. (1.4), defined on a simply connected domain¹ Ω .

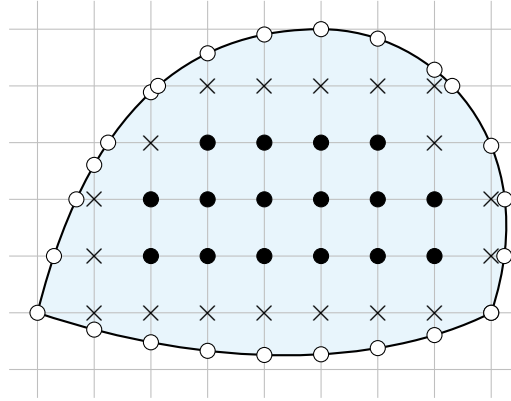


Figure 2.9: Non-square domain. Boundary nodes are denoted by $\circ$, nodes adjacent to boundary nodes are denoted by $\times$, and inner nodes by $\bullet$.

If we cover the domain with a grid of mesh size h , then the solution is known at the Dirichlet nodes from the boundary condition $g(x, y)$. We denote these boundary nodes by white dots ($\circ$).

The grid points marked with a cross ($\times$) are those for which at least one neighboring point belongs to $\partial\Omega_h$. Clearly, we cannot have the case when all neighbors belong to $\partial\Omega_h$. We denote this set by $\Omega_h^\times$.

The remaining interior grid points are denoted by black dots ($\bullet$), with the corresponding set denoted $\Omega_h^\bullet$. Thus, we have the decomposition

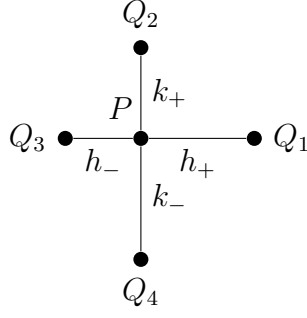
$$\bar{\Omega}_h = \Omega_h^\bullet \cup \Omega_h^\times \cup \partial\Omega_h.$$

The solution at $\partial\Omega_h$ is known from the Dirichlet condition; the solution at $\Omega_h^\bullet$ can be approximated using the standard five-point stencil. Hence, the only remaining task is to compute the solution at $\Omega_h^\times$.

Let us take a general point $P \in \Omega_h^\times$, and let $\{Q_i\}_{i=1}^4$ be its neighboring nodes, labeled counterclockwise, with Q_1 being the right neighbor. The distances from P to its neighbors are denoted h_+, h_-, k_+, k_- (see Fig. 2.10). Here, $h_\pm$ correspond to x -direction distances and $k_\pm$ to y -direction distances.

To approximate the Laplacian Δu at P , we approximate u_{xx} and u_{yy} separately.

¹Informally, a simply connected domain is an open connected set with “no holes.”

Figure 2.10: Point P with uneven spacing to its neighbors.

Approximation of u_{xx} : Using Taylor expansion about P in the x -direction:

$$\begin{aligned} u(x + h_+, y) &= u(x, y)|_P + h_+ u_x|_P + \frac{h_+^2}{2} u_{xx}|_P + \mathcal{O}(h^3), \\ u(x - h_-, y) &= u(x, y)|_P - h_- u_x|_P + \frac{h_-^2}{2} u_{xx}|_P + \mathcal{O}(h^3), \end{aligned}$$

where h is the maximum grid spacing in the x -direction.

Multiplying the first equation by h_- , the second by h_+ , and adding eliminates u_x :

$$\begin{aligned} h_- u|_{Q_1} &= h_- u|_P + h_- h_+ u_x|_P + \frac{h_- h_+^2}{2} u_{xx}|_P + \mathcal{O}(h^4), \\ h_+ u|_{Q_3} &= h_+ u|_P - h_- h_+ u_x|_P + \frac{h_-^2 h_+}{2} u_{xx}|_P + \mathcal{O}(h^4). \end{aligned}$$

Adding and simplifying gives:

$$u_{xx}|_P = \frac{2}{h_- + h_+} \left(\frac{u|_{Q_1} - u|_P}{h_+} + \frac{u|_{Q_3} - u|_P}{h_-} \right) + \mathcal{O}(h), \quad (2.29)$$

with a formal truncation error of $\mathcal{O}(h)$.

Approximation of u_{yy} : By the same reasoning in the y -direction:

$$u_{yy}|_P = \frac{2}{k_- + k_+} \left(\frac{u|_{Q_2} - u|_P}{k_+} + \frac{u|_{Q_4} - u|_P}{k_-} \right) + \mathcal{O}(k), \quad (2.30)$$

where k is the maximum grid spacing in the y -direction.

Discrete PDE at P : Combining (2.29) and (2.30), the PDE can be approximated at P by:

$$-\frac{2}{h_- + h_+} \left(\frac{u|_{Q_1} - u|_P}{h_+} + \frac{u|_{Q_3} - u|_P}{h_-} \right) - \frac{2}{k_- + k_+} \left(\frac{u|_{Q_2} - u|_P}{k_+} + \frac{u|_{Q_4} - u|_P}{k_-} \right) = f|_P. \quad (2.31)$$

If $P \in \Omega_h^\bullet$ and $h_- = h_+ = k_- = k_+ = h$, then we recover the standard second-order finite difference method. For Neumann boundary conditions, similar approximations can be derived using the ideas presented in Sec. 2.2.1.

Now, we turn to the convergence of the method. For this, we require the notions of *stability* and *consistency*.

For stability, we require that the finite difference operator $\mathcal{L}_h$ is stable. Stability of $\mathcal{L}_h$ means that it must be: Linear, Inverse-monotone, and satisfy the comparison principle. Linearity is straightforward to verify. We therefore move on to *inverse-monotonicity*. To discuss this property, we first recall the concept of operators of *general positive type*.

Definition 2.8 (General Positive Type Operators). Suppose that P is a grid point and $\{Q_i\}_{i=1}^\nu$ are its neighboring grid points. A finite difference operator of the form

$$\mathcal{L}_h u|_P = \alpha_0 u|_P - \sum_{j=1}^\nu \alpha_j u|_{Q_j}, \quad (2.32)$$

is said to be of *general positive type* if the coefficients satisfy:

$$\alpha_j \geq 0, \quad j = 0, 1, \dots, \nu, \quad \text{and} \quad \alpha_0 \geq \sum_{j=1}^\nu \alpha_j. \quad (2.33)$$

Theorem 2.16 (Inverse Monotonicity). Suppose that the difference operator $\mathcal{L}_h$ is defined by

$$\mathcal{L}_h \mathbf{u}_{i,j} = \begin{cases} \mathbf{g}_{i,j}, & \text{for } i = 0 \text{ or } M, \text{ or } j = 0 \text{ or } M, \\ \mathfrak{L}_h \mathbf{u}_{i,j}, & \text{otherwise,} \end{cases}$$

where $\mathfrak{L}_h$ is of positive type and $\alpha_j > 0$ for $j = 0, 1, \dots, \nu$. Then $\mathcal{L}_h$ is inverse monotone.

Proof. We proceed by contradiction. Suppose there exists $\mathbf{u}$ such that $\mathcal{L}_h \mathbf{u} \geq 0$ but $\mathbf{u} \not\geq 0$. Then there exists a grid point (i_0, j_0) such that $\mathbf{u}_{i_0, j_0} < 0$.

Since $\mathcal{L}_h \mathbf{u} \geq 0$ and the Dirichlet boundary conditions are applied directly (implying $\mathbf{u}_{i,j} \geq 0$ for $i, j \in \{0, M\}$), the point (i_0, j_0) must be in the interior. Let $\{Q_j\}_{j=1}^{\nu_0}$ be the neighboring nodes of (i_0, j_0) . By minimality, we have

$$\mathbf{u}_{i_0, j_0} \leq \mathbf{u}|_{Q_j}, \quad j = 1, 2, \dots, \nu_0.$$

From the definition of $\mathfrak{L}_h$,

$$\begin{aligned} \mathfrak{L}_h \mathbf{u}_{i_0, j_0} &= \alpha_0 \mathbf{u}_{i_0, j_0} - \sum_{j=1}^{\nu_0} \alpha_j \mathbf{u}|_{Q_j} \\ &\leq \alpha_0 \mathbf{u}_{i_0, j_0} - \sum_{j=1}^{\nu_0} \alpha_j \mathbf{u}_{i_0, j_0} \\ &= \mathbf{u}_{i_0, j_0} \left(\alpha_0 - \sum_{j=1}^{\nu_0} \alpha_j \right). \end{aligned}$$

From (2.33), $\alpha_0 \geq \sum_{j=1}^{\nu_0} \alpha_j$, and since $\mathbf{u}_{i_0, j_0} < 0$, it follows that

$$\mathfrak{L}_h \mathbf{u}_{i_0, j_0} \leq 0.$$

But we also have $\mathfrak{L}_h \mathbf{u}_{i_0, j_0} \geq 0$ by assumption, hence

$$\mathfrak{L}_h \mathbf{u}_{i_0, j_0} = 0.$$

Equality can hold only if $\alpha_0 = \sum_{j=1}^{\nu_0} \alpha_j$. Substituting into the definition of $\mathfrak{L}_h$ gives

$$\sum_{j=1}^{\nu_0} \alpha_j (\mathbf{u}_{i_0, j_0} - \mathbf{u}|_{Q_j}) = 0.$$

Since $\alpha_j > 0$ and $\mathbf{u}_{i_0, j_0} - \mathbf{u}|_{Q_j} \leq 0$ for all j , the above can hold only if

$$\mathbf{u}_{i_0, j_0} = \mathbf{u}|_{Q_j}, \quad j = 1, 2, \dots, \nu_0.$$

Thus, the same strict minimum value is attained at all neighboring nodes. Repeating this argument propagates the minimum to the boundary, where $\mathbf{u} \geq 0$, leading to a contradiction. Therefore, $\mathcal{L}_h$ is inverse monotone. $\square$

Now, we check whether (2.31) can be written in the form (2.32). Let $\mathbf{u}|_P = \mathbf{u}_0$ and $\mathbf{u}|_{Q_i} = \mathbf{u}_i$ for $i = 1, 2, 3, 4$. Then

$$\begin{aligned} \mathfrak{L}_h \mathbf{u}_0 &= -\frac{2}{h_- + h_+} \left(\frac{\mathbf{u}_1 - \mathbf{u}_0}{h_+} + \frac{\mathbf{u}_3 - \mathbf{u}_0}{h_-} \right) - \frac{2}{k_- + k_+} \left(\frac{\mathbf{u}_2 - \mathbf{u}_0}{k_+} + \frac{\mathbf{u}_4 - \mathbf{u}_0}{k_-} \right) \\ &= \left[\frac{2}{h_+ + h_-} \left(\frac{1}{h_+} + \frac{1}{h_-} \right) + \frac{2}{k_+ + k_-} \left(\frac{1}{k_+} + \frac{1}{k_-} \right) \right] \mathbf{u}_0 \\ &\quad - \frac{2}{h_+(h_+ + h_-)} \mathbf{u}_1 - \frac{2}{h_-(h_+ + h_-)} \mathbf{u}_3 - \frac{2}{k_+(k_+ + k_-)} \mathbf{u}_2 - \frac{2}{k_-(k_+ + k_-)} \mathbf{u}_4. \end{aligned}$$

The coefficients clearly satisfy the conditions for general positive type operators. Hence, $\mathcal{L}_h$ is inverse monotone. It is then stable by the comparison principle, as shown next.

Corollary 2.17. *The operator $\mathcal{L}_h$ comprising $\mathfrak{L}_h$ at points of $\Omega_h^\bullet \cup \Omega_h^\times$ together with the Dirichlet operator is stable.*

Proof. Without loss of generality, suppose $\Omega \subset \{(x, y) \mid 0 \leq x \leq a\}$. Then the comparison function

$$\Phi(x, y) = 1 + \frac{1}{2}x(a - x)$$

is non-negative on Ω and satisfies $\mathcal{L}_h \Phi \geq 1$. Hence, by the comparison principle, $\mathcal{L}_h$ is stable. $\square$

Now, the only thing left to show is the second-order convergence of the method. We have already seen that it is consistent with $\mathcal{O}(h^2)$ in $\Omega_h^\bullet$ and $\mathcal{O}(h)$ in $\Omega_h^\times$.

Corollary 2.18. *The five-point approximation (2.31) to the Poisson equation with Dirichlet boundary conditions in a general, simply connected domain Ω is second-order convergent, provided the fourth derivatives of the exact solution are bounded in Ω .*

Proof. We split the global error $\mathbf{e}$ into two parts:

$$\mathbf{e} = \mathbf{e}^\bullet + \mathbf{e}^\times,$$

where $\mathbf{e}^\bullet$ is the error on $\Omega_h^\bullet$ and $\mathbf{e}^\times$ is the error on $\Omega_h^\times$. They satisfy the discrete problems

$$\mathcal{L}_h \mathbf{e}^\bullet = \begin{cases} 0, & \text{on } \partial\Omega_h, \\ 0, & \text{on } \Omega_h^\times, \\ \mathfrak{R}_h^\bullet, & \text{on } \Omega_h^\bullet, \end{cases} \quad \mathcal{L}_h \mathbf{e}^\times = \begin{cases} 0, & \text{on } \partial\Omega_h, \\ \mathfrak{R}_h^\times, & \text{on } \Omega_h^\times, \\ 0, & \text{on } \Omega_h^\bullet, \end{cases} \quad (2.34)$$

where $\mathfrak{R}_h^\bullet = \mathfrak{L}_h u - f$ on $\Omega_h^\bullet$ is $\mathcal{O}(h^2)$ and $\mathfrak{R}_h^\times = \mathfrak{L}_h u - f$ on $\Omega_h^\times$ is $\mathcal{O}(h)$.

To control $\mathbf{e}^\times$ we introduce a non-negative discrete function Ψ defined by

$$\Psi|_P = \begin{cases} 0, & \text{on } \partial\Omega_h, \\ h^2, & \text{on } \Omega_h, \end{cases}$$

so that $\mathfrak{L}_h \Psi|_P = 0$ for $P \in \Omega_h^\bullet$ (since Ψ is constant there).

If $P \in \Omega_h^\times$, then at least one of its neighbors lies on $\partial\Omega_h$. Let Q_1 be such a boundary neighbor, so that $\Psi|_{Q_1} = 0$. Using (2.33) we have

$$\begin{aligned} \mathfrak{L}_h \Psi|_P &= \alpha_0 \Psi|_P - \sum_{j=1}^4 \alpha_j \Psi|_{Q_j} \\ &\geq h^2 \left(\alpha_0 - \sum_{j=2}^4 \alpha_j \right) \\ &= h^2 \left(\alpha_0 - \sum_{j=1}^4 \alpha_j \right) + \alpha_1 h^2 \\ &\geq \alpha_1 h^2. \end{aligned}$$

Since $\alpha_1 = \frac{2}{h_+(h_++h_-)}$ and $h_+, h_- \leq h$, it follows that $\alpha_1 h^2 \geq 1$. If P has more than one boundary neighbor, this lower bound is even larger. Thus $\mathfrak{L}_h \Psi|_P \geq 1$ for all $P \in \Omega_h^\times$.

From (2.34) and the above, we compute

$$\mathcal{L}_h (\mathbf{e}^\times - \|\mathfrak{R}_h^\times\|_{h,\infty} \Psi) \leq 0,$$

because on $\Omega_h^\times$ we have

$$\mathfrak{L}_h \mathbf{e}^\times - \|\mathfrak{R}_h^\times\|_{h,\infty} \mathfrak{L}_h \Psi \leq \mathfrak{R}_h^\times - \|\mathfrak{R}_h^\times\|_{h,\infty} \leq 0.$$

By inverse monotonicity of $\mathcal{L}_h$,

$$\mathbf{e}^\times \leq \|\mathfrak{R}_h^\times\|_{h,\infty} \Psi.$$

A similar argument applied to $\mathbf{e}^\times$ yields

$$|\mathbf{e}^\times| \leq \|\mathfrak{R}_h^\times\|_{h,\infty} h^2.$$

Since $\|\mathfrak{R}_h^\times\|_{h,\infty} = \mathcal{O}(h)$, this gives $\mathbf{e}^\times = \mathcal{O}(h^3)$ in the maximum norm.

Finally, $\mathbf{e}^\bullet$ is governed by a residual $\mathcal{O}(h^2)$ and the stable operator $\mathcal{L}_h$, hence $\mathbf{e}^\bullet = \mathcal{O}(h^2)$. Since $\mathbf{e} = \mathbf{e}^\bullet + \mathbf{e}^\times$ and $\mathbf{e}^\bullet$ dominates, we conclude

$$\mathbf{e} = \mathcal{O}(h^2),$$

establishing second-order convergence. □

Remark 2.19. It is tempting to repeat the above barrier argument in an attempt to obtain a higher-order estimate for the two components of the error. However, the argument does not extend to yield an $\mathcal{O}(h^4)$ estimate, and the obstruction is different for $\mathbf{e}^\times$ and $\mathbf{e}^\bullet$.

For $\mathbf{e}^\times$, since $\|\mathfrak{R}_h^\times\|_{h,\infty} = \mathcal{O}(h)$, an $\mathcal{O}(h^4)$ estimate would require a barrier of size $\mathcal{O}(h^3)$. If $\Psi = \mathcal{O}(h^3)$ were used in place of the present $\mathcal{O}(h^2)$ barrier, then at a point $P \in \Omega_h^\times$ the crucial estimate would become

$$\mathfrak{L}_h \Psi|_P \gtrsim \alpha_1 h^3 = \mathcal{O}(h),$$

since $\alpha_1 = \mathcal{O}(h^{-2})$. Thus $\mathfrak{L}_h \Psi$ would no longer be bounded below by a positive constant independent of h , and the comparison argument would fail precisely at the step $\alpha_1 h^2 \geq 1$.

For $\mathbf{e}^\bullet$, on the other hand, the residual is already $\mathcal{O}(h^2)$, so the existing $\mathcal{O}(h^2)$ barrier would have the appropriate magnitude to formally produce an $\mathcal{O}(h^4)$ bound. However, the same barrier cannot be used for this purpose because it is constant on $\Omega_h^\bullet$, and consequently

$$\mathfrak{L}_h \Psi = 0 \quad \text{on } \Omega_h^\bullet,$$

which is precisely the region where $\mathfrak{R}_h^\bullet$ is nonzero. Hence the key comparison inequality needed to control $\mathbf{e}^\bullet$ cannot be established using this barrier. In fact, the interior five-point discretization has a truncation error of order $\mathcal{O}(h^2)$ in general, so an $\mathcal{O}(h^4)$ estimate for $\mathbf{e}^\bullet$ should not be expected without additional structure or cancellation.

Thus, the above barrier argument is sufficient to establish the $\mathcal{O}(h^3)$ contribution from the irregular region, but it cannot be iterated to obtain fourth-order estimates for either component.

2.2.3 Higher-Order Discretisation

In all the examples we have studied until now, all of them are second-order convergent. But can we achieve a higher-order discretisation, say of order four?

To answer this question, we start with the Taylor expansion formula in two dimensions.

Theorem 2.20. Suppose that $u(x, y)$ and all its partial derivatives of order less than or equal to $n+1$ are continuous on $D = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$ and let $(x_0, y_0) \in D$. For every $(x, y) \in D$, there exist ξ between x and x_0 and μ between y and y_0 such that

$$u(x, y) = P_n(x, y) + R_n(x, y),$$

where

$$\begin{aligned} P_n(x, y) = & u(x_0, y_0) + \left[(x - x_0) \frac{\partial u}{\partial x}(x_0, y_0) + (y - y_0) \frac{\partial u}{\partial y}(x_0, y_0) \right] \\ & + \left[\frac{(x - x_0)^2}{2} \frac{\partial^2 u}{\partial x^2}(x_0, y_0) + (x - x_0)(y - y_0) \frac{\partial^2 u}{\partial x \partial y}(x_0, y_0) \right. \\ & \left. + \frac{(y - y_0)^2}{2} \frac{\partial^2 u}{\partial y^2}(x_0, y_0) \right] \\ & + \cdots + \frac{1}{n!} \sum_{j=0}^n \binom{n}{j} (x - x_0)^{n-j} (y - y_0)^j \frac{\partial^n u}{\partial x^{n-j} \partial y^j}(x_0, y_0), \end{aligned}$$

and

$$R_n(x, y) = \frac{1}{(n+1)!} \sum_{j=0}^{n+1} \binom{n+1}{j} (x - x_0)^{n+1-j} (y - y_0)^j \frac{\partial^{n+1} u}{\partial x^{n+1-j} \partial y^j}(\xi, \mu).$$

The function $P_n(x, y)$ is called the n^{th} Taylor polynomial in two variables for the function u about (x_0, y_0) , and $R_n(x, y)$ is the remainder term associated with $P_n(x, y)$.

Now, consider the four diagonal points for (i, j) as shown in Fig. 2.11, labelled R_1, R_2, R_3 , and R_4 .

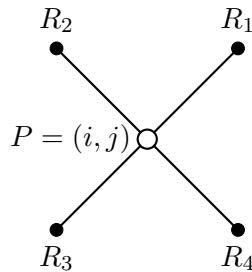


Figure 2.11: The white point represents an interior grid node (i, j) , and the black points are its four nearest diagonal neighbors used in the five-point stencil.

Using the Taylor expansion at these four points gives

$$\begin{aligned}
u|_{R_1} &= u(x+h, y+h) = u_{i,j} + hu_x|_{i,j} + hu_y|_{i,j} + \frac{h^2}{2}(u_{xx} + 2u_{xy} + u_{yy})|_{i,j} + \mathcal{O}(h^3), \\
u|_{R_2} &= u(x-h, y+h) = u_{i,j} - hu_x|_{i,j} + hu_y|_{i,j} + \frac{h^2}{2}(u_{xx} - 2u_{xy} + u_{yy})|_{i,j} + \mathcal{O}(h^3), \\
u|_{R_3} &= u(x-h, y-h) = u_{i,j} - hu_x|_{i,j} - hu_y|_{i,j} + \frac{h^2}{2}(u_{xx} + 2u_{xy} + u_{yy})|_{i,j} + \mathcal{O}(h^3), \\
u|_{R_4} &= u(x+h, y-h) = u_{i,j} + hu_x|_{i,j} - hu_y|_{i,j} + \frac{h^2}{2}(u_{xx} - 2u_{xy} + u_{yy})|_{i,j} + \mathcal{O}(h^3).
\end{aligned}$$

Adding the above four equations and cancelling the third-order $\mathcal{O}(h^3)$ terms, we get

$$u_{i+1,j+1} + u_{i+1,j-1} + u_{i-1,j-1} + u_{i-1,j+1} = 4u_{i,j} + 2h^2(u_{xx} + u_{yy})|_{(i,j)} + \mathcal{O}(h^4).$$

Now the Laplacian $-\Delta u$ can be approximated by

$$-\Delta u = \frac{1}{2h^2} [4u_{i,j} - u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j-1} - u_{i-1,j+1}] + \mathcal{O}(h^2).$$

Let us denote the new difference operator by $\mathfrak{L}_h^\times$, i.e.,

$$\mathfrak{L}_h^\times \mathbf{u}_{i,j} = \frac{1}{2h^2} [4\mathbf{u}_{i,j} - \mathbf{u}_{i+1,j+1} - \mathbf{u}_{i+1,j-1} - \mathbf{u}_{i-1,j-1} - \mathbf{u}_{i-1,j+1}].$$

Notice that this is a second-order consistent approximation. In the case of the Laplace equation, we notice that this approximation also satisfies the mean-value property. Now, to obtain a higher-order approximation we combine the newly developed five-point stencil with the one developed in Sec. 2.1.

Let us denote the five-point stencil developed in the previous section by $\mathfrak{L}_h^+$, i.e.,

$$\mathfrak{L}_h^+ = \frac{\delta_x^2 + \delta_y^2}{h^2}.$$

To obtain a fourth-order approximation, consider $\lambda \mathfrak{L}_h^\times + (1-\lambda) \mathfrak{L}_h^+$ for a generic λ . The local truncation error for this method is given by

$$\begin{aligned}
\lambda \mathfrak{L}_h^\times + (1-\lambda) \mathfrak{L}_h^+ &= \lambda \left[-\Delta u - \frac{h^2}{12} (\partial_x^4 u + 6\partial_x^2 \partial_y^2 u + \partial_y^4 u) + \mathcal{O}(h^4) \right] \\
&\quad + (1-\lambda) \left[-\Delta u - \frac{h^2}{12} (\partial_x^4 u + \partial_y^4 u) + \mathcal{O}(h^4) \right] \\
&= -\Delta u - \frac{h^2}{12} (\partial_x^4 u + \partial_y^4 u) - \frac{6\lambda h^2}{12} \partial_x^2 \partial_y^2 u + \mathcal{O}(h^4).
\end{aligned}$$

Now, for $\lambda = 1/3$ we get

$$\lambda \mathfrak{L}_h^\times + (1-\lambda) \mathfrak{L}_h^+ = -\Delta u - \frac{h^2}{12} \Delta^2 u + \mathcal{O}(h^4),$$

where $\Delta^2 u = \partial_x^4 u + 2\partial_x^2 \partial_y^2 u + \partial_y^4 u$. Since $-\Delta u = f$, we have $-\Delta^2 u = \Delta f$. Therefore, we have an order-4 consistency term.

Remark 2.21. We have not talked about the exact form of the local truncation error. An interested reader can use Taylor expansion to see that the $\mathcal{O}(h^4)$ term remains the leading term. However, for higher-order accuracy we require $u \in \mathcal{C}^6(\Omega)$, which demands quite high smoothness.

Nine-Point Stencil: Let $u \in \mathcal{C}^6(\Omega)$. Then the nine-point stencil formula for the Poisson equation, Eq. (2.2), is given by

$$\mathcal{L}_h^9 \mathbf{u}_{i,j} = \mathbf{f}_{i,j}^\times,$$

where

$$\mathcal{L}_h^9 \mathbf{u}_{i,j} = \begin{cases} \mathbf{u}_{i,j}, & \text{if } i, j = 0 \text{ or } M, \\ \mathfrak{L}_h^9 \mathbf{u}_{i,j}, & \text{if } 1 \leq i, j \leq M-1, \end{cases} \quad \text{and} \quad \mathbf{f}_{i,j}^\times = \mathbf{f}_{i,j} + \frac{h^2}{12} \Delta \mathbf{f}_{i,j},$$

and

$$\begin{aligned} \mathfrak{L}_h^9 \mathbf{u}_{i,j} = & \frac{1}{3h^2} \left[10\mathbf{u}_{i,j} - \frac{1}{2} (\mathbf{u}_{i+1,j+1} + \mathbf{u}_{i+1,j-1} + \mathbf{u}_{i-1,j+1} + \mathbf{u}_{i-1,j-1}) \right. \\ & \left. - 2(\mathbf{u}_{i,j+1} + \mathbf{u}_{i,j-1} + \mathbf{u}_{i+1,j} + \mathbf{u}_{i-1,j}) \right]. \end{aligned}$$

Consistency of the method is already obvious. The stability of the method is also straightforward to see, as the operator is of general positive type (see Def. 2.2.2) and hence $\mathfrak{L}_h^9$ is inverse-monotone. The next step is to show the satisfaction of the comparison lemma; for this, we can choose the same $\Phi(x, y)$ as in Fig. 2.5 to get the result. Hence, we have a convergent method of order four.

However, this method has drawbacks. The matrix obtained will not be a penta-diagonal matrix; instead, we get a nine-diagonal matrix (see Fig. 2.12). Furthermore, solving it will be more challenging as the number of non-zero entries increases.

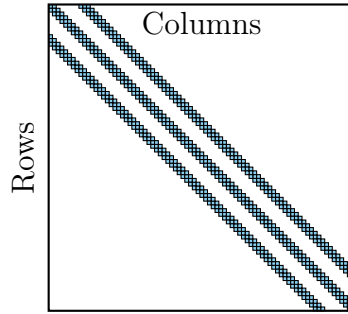


Figure 2.12: Sparsity pattern for matrix $\mathbf{A}$ developed using the nine-point stencil for $M = 10$.

Regarding the implementation of the method, the matrix $\mathbf{A}$ will have the structure

$$-\mathbf{D}_1 \mathbf{u}_{i-1} + \mathbf{D}_2 \mathbf{u}_i - \mathbf{D}_1 \mathbf{u}_{i+1}, \quad \text{for } i = 1, 2, \dots, M-1,$$

where

$$\mathbf{D}_1 = \begin{bmatrix} 4 & 1 & 0 & \cdots & 0 \\ 1 & 4 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 4 \end{bmatrix}, \quad \mathbf{D}_2 = \begin{bmatrix} 20 & -4 & 0 & \cdots & 0 \\ -4 & 20 & -4 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 20 \end{bmatrix}.$$

The right-hand side $\mathbf{f}$ will have an extra term to incorporate $\Delta \mathbf{f}_{i,j}$. The boundary conditions can be applied in the same way as before. For Neumann boundary conditions or irregular domains, the ideas developed in the previous examples remain applicable.

Example 2.3. Consider Eq. (2.2) with

$$f = -(12x^2y^5 + 20x^4y^3 + 17\sin(xy)(x^2 + y^2)), \quad g = 0.$$

The analytical solution is

$$u(x, y) = x^4y^5 - 17\sin(xy).$$

We compute the numerical solution using the nine-point stencil and report the results in Table 2.3. We observe that the computed E.O.C. approaches the theoretical prediction of fourth-order accuracy. We also plot the log-log plot in Fig. 2.13.

h	$\ u - \mathbf{u}\ _{h,\infty}$	E.O.C.
0.250000	0.0010709	-
0.125000	0.0000717	3.90039
0.062500	0.0000045	3.98456
0.031250	0.0000003	3.99164
0.015625	0.0000000	3.99900

Table 2.3: Example 2.3: Experimental order of convergence for the nine-point stencil.

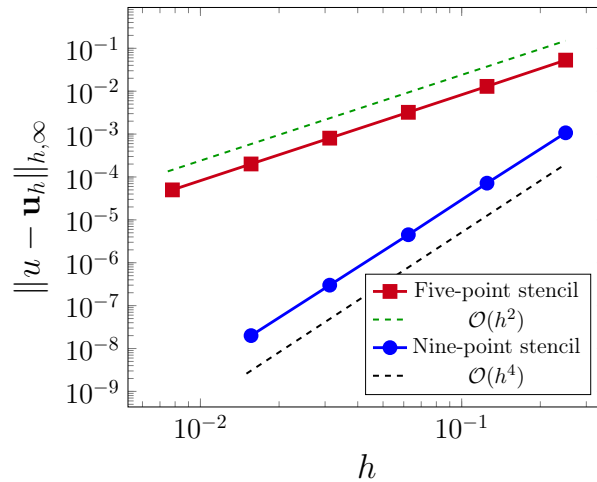


Figure 2.13: Log-log plot of the maximum-norm error versus mesh size h for the five-point and nine-point finite difference stencils, together with the reference slopes $\mathcal{O}(h^2)$ and $\mathcal{O}(h^4)$, respectively.

2.3 Summary

In this chapter, we examined the finite difference method (FDM) for second-order elliptic equations in two dimensions. The ideas developed here extend naturally to three dimensions. The

convergence theory remains unchanged, and the approach can also be generalized to Neumann boundary conditions, irregular domains, and higher-order discretizations.

If, instead of the Poisson equation, we consider a more general elliptic operator such as the convection–diffusion equation

$$-\varepsilon \Delta u + \mathbf{b} \cdot \nabla u = f \quad \text{in } \Omega, \quad (2.35)$$

with appropriate boundary conditions, the implementation of the FDM follows the same principles. In this case, in addition to approximating the Laplacian Δ , we must also approximate the gradient ∇u . This introduces an additional challenge. An excellent survey on this topic is given by Stynes [17].

Despite the development of modern numerical methods, the FDM remains popular in the scientific community because of its simplicity. However, it has certain limitations. One major drawback is the regularity assumption on the solution. The numerical analysis of the FDM relies on Taylor expansions of u , and achieving the expected order of accuracy typically requires $u \in \mathcal{C}^4(\Omega)$. In practice, solutions may lack such high smoothness, making it necessary to use methods that relax this assumption.

Another difficulty arises when incorporating Neumann boundary conditions or handling irregular domains. These cases require special treatment, often involving modifications to either the system matrix $\mathbf{A}$ or the right-hand side vector $\mathbf{f}$.

Also, suppose we want to find the solution at a point between the grid points, say $(i+0.5, j+0.7)$, then in this case, one either needs to do re-grid or use interpolation techniques, leading to further difficulties.

This naturally leads to the question: *do we have better methods?* In the next part of the course, we will explore approaches designed to overcome these limitations.

Chapter 3

Introduction to Functional Analysis

In the last chapter we noted that the analysis of the Finite Difference Method relies heavily on Taylor expansions, which require high differentiability of the solution, also referred to as smoothness. Let us look at some examples to see what smoothness actually means. Define

$$f_0(x) = \begin{cases} -1 & \text{if } -1 \leq x < 0, \\ 1 & \text{if } 0 \leq x \leq 1, \end{cases} \quad f_1(x) = |x|, \quad f_2(x) = x|x|, \quad f_3(x) = x^2|x|,$$

on the interval $[-1, 1]$. We now examine the behaviour of the functions $\{f_i(x)\}_{i=0}^3$ at the point $x = 0$:

1. $f_0(x)$ is not continuous at $x = 0$.
2. $f_1(x)$ is continuous at $x = 0$, but is not differentiable there.
3. $f_2(x)$ is continuous and differentiable at $x = 0$, but its second derivative does not exist at that point.
4. $f_3(x)$ is twice differentiable at $x = 0$, but not three times differentiable.

If we look at Fig. 3.1 at $(0, 0)$, we see what is meant by smoothness. As i increases, the graphs of the functions look progressively *smoother*, near the point $x = 0$.

But why do we need smoothness, or even continuity, in the first place? In real-world applications such as stock pricing, the price of a share may exhibit very sharp changes. For example, the value of a stock can drop from \$100 to \$0 and then rise back to \$100 within a short period of time. Like many other phenomena, this too can be modeled mathematically. The evaluation of an option price is described by the *Black–Scholes equation*, named after the American economists Fischer Sheffey Black and Myron Scholes. It is given by

$$\frac{\partial V}{\partial t} + \frac{\sigma^2}{2} S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0,$$

where $V(t, S)$ is the price of the option as a function of stock price S and time t , r is the risk-free interest rate, and σ is the volatility of the stock. As mentioned, the solution $V(t, S)$ will, in general, not be a smooth function, and hence we need to study a more general class of functions together with their properties.

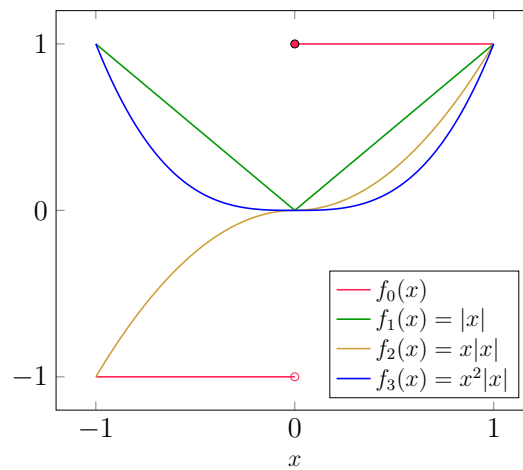


Figure 3.1: Smoothness of the functions $\{f_i(x)\}_{i=0}^3$ around $x = 0$.



Figure 3.2: Fischer Black (11 January 1938 – 30 August 1995, left) and Myron Scholes (1 July 1941–, right).

This chapter is intended as an introduction to Functional Analysis. The main topics to be covered include measurable functions, function spaces, weak derivatives, and their properties. It will present the most important concepts required for the study of PDEs, but it is by no means a complete text. For a deeper understanding of Functional Analysis, I recommend the references [13, 12].

3.1 Function Spaces

3.1.1 Banach Spaces

Until now we have looked at some very particular classes of functions such as $\mathcal{C}(\Omega)$ and $\mathcal{C}^4(\Omega)$, but now we move towards a more general notion. Let us denote such a space by X . We start with the most fundamental structure, referred to as a *vector space*.

Definition 3.1. (Vector Space) A *vector space* or a *linear space* over $\mathbb{R}$ (whose elements are called *scalars*) is a set X , whose elements are called *vectors*, equipped with two operations: *addition* and *scalar multiplication*, such that the following properties hold:

- **Addition:** $(x, y) \in X \times X \mapsto x + y \in X$ such that for all $x, y, z \in X$:
 - $x + y = y + x$ (Commutativity).
 - $x + (y + z) = (x + y) + z$ (Associativity).
 - There exists a unique vector $0 \in X$, called the zero vector, such that $x + 0 = x$ for all $x \in X$.
 - For all $x \in X$ there exists $-x \in X$ such that $x + (-x) = 0$.
- **Scalar Multiplication:** $(\lambda, x) \in \mathbb{R} \times X \mapsto \lambda x \in X$ such that for all $\lambda, \beta \in \mathbb{R}$ and $x, y \in X$:
 - For all $x \in X$, $1 \cdot x = x$, where 1 is the multiplicative identity of $\mathbb{R}$.
 - $\lambda(\beta x) = (\lambda\beta)x$.
 - $(\lambda + \beta)x = \lambda x + \beta x$.
 - $\lambda(x + y) = \lambda x + \lambda y$.

Example 3.1. 1. The set of real numbers $\mathbb{R}$ is a vector space over $\mathbb{R}$.
 2. $\mathbb{R}^d$ is a vector space over $\mathbb{R}$.
 3. The set of integers $\mathbb{Z}$ with scalars in $\mathbb{R}$ is *not* a vector space, since for $\lambda = \sqrt{3}$ and any $x \in \mathbb{Z}$, we have $\lambda x \notin \mathbb{Z}$.

After defining vector spaces, we would like to measure distances in such spaces, which leads to the notion of *normed spaces*. We have already seen the notation of a norm (see $\|\cdot\|_{\infty, h}$), but we revisit it here.

Definition 3.2. (Normed Space) Let X be a real vector space. A mapping $\|\cdot\| : X \rightarrow \mathbb{R}$ is called a *norm* on X if

1. $\|x\| \geq 0$ for all $x \in X$ (Non-negativity).
 2. $\|x\| = 0 \Leftrightarrow x = 0$ (Definiteness).
 3. $\|\lambda x\| = |\lambda| \|x\|$ for all $x \in X$ and $\lambda \in \mathbb{R}$ (Homogeneity).
 4. $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$ (Triangle Inequality).
- The pair $(X, \|\cdot\|)$ is called a *normed space*.

Example 3.2. 1. The set of real numbers $\mathbb{R}$ with the absolute value $|\cdot|$ is a normed space.
 2. The space $\mathcal{C}[a, b]$ is a normed space with $\|x\| = \max_{t \in [a, b]} |x(t)|$ for $x \in \mathcal{C}[a, b]$.
 3. If $\|\cdot\|$ is defined by $\|x\| = 0$ for all $x \in X$, then $(X, \|\cdot\|)$ is *not* a normed space.

We saw the importance of norms in the previous chapter, as they allow us to quantify errors. In Numerical Analysis, approximations naturally give rise to sequences of numbers, and we hope that these sequences converge to the true solution.

In Calculus, we primarily dealt with sequences of real numbers. Later, if one studies complex analysis, this notion extends naturally to sequences of complex numbers. In more

advanced analysis, we also encounter sequences of functions. The concepts that we develop from now on apply more generally to sequences of mathematical objects. These objects may be real or complex numbers, functions, or even more general objects such as distributions.

Before studying the convergence of sequences, however, we introduce a more general concept: the notion of a *Cauchy sequence*, named after the French mathematician Augustin-Louis Cauchy.



Figure 3.3: Augustin-Louis Cauchy (21 August 1789 – 23 May 1857).

Definition 3.3. (Cauchy Sequence) A sequence $\{x_n\}_{n \in \mathbb{N}} \subset X$ is called a *Cauchy sequence* if for all $\varepsilon > 0$ there exists $n_0(\varepsilon) \in \mathbb{N}$ such that

$$\|x_m - x_n\| < \varepsilon \quad \forall m, n > n_0(\varepsilon).$$

In other words, a sequence is said to be Cauchy if for any $\varepsilon > 0$ we can find an index n_0 (depending on ε) such that all terms of the sequence beyond this index are within ε of each other.

Definition 3.4. (Convergent Sequence) A sequence $\{x_n\}_{n \in \mathbb{N}} \subset X$ is said to *converge* to $x \in X$ if for all $\varepsilon > 0$ there exists $n_0(\varepsilon) \in \mathbb{N}$ such that

$$\|x_n - x\| < \varepsilon \quad \forall n > n_0(\varepsilon).$$

In this case we write $x_n \rightarrow x$ as $n \rightarrow \infty$.

Example 3.3. In Real Analysis a classic result tells us that every convergent sequence is a Cauchy sequence, but the converse is not true in general [15, Theorem 3.11]. For instance, consider $X = \mathbb{P}[0, 1]$, the space of all polynomials on $[0, 1]$, equipped with the supremum norm

$$\|p\|_\infty = \max_{x \in [0, 1]} |p(x)|.$$

Define

$$p_n(x) = \sum_{k=0}^n \frac{x^k}{k!}.$$

Then $\{p_n\}$ is a Cauchy sequence in $(\mathcal{C}[0, 1], \|\cdot\|_\infty)$, since $p_n \rightarrow e^x$ uniformly on $[0, 1]$. However, e^x is not a polynomial, and hence $\{p_n\}$ does not converge in $X = \mathbb{P}[0, 1]$.

This illustrates that Cauchy sequences are more general than convergent sequences: a Cauchy sequence may fail to converge if the space is not *complete*. If we enlarge the space to $\mathcal{C}[0, 2]$, then the same sequence $\{p_n\}$ does converge (to e^x). This observation motivates the notion of completeness.

Definition 3.5. (Complete Space) A normed space $(X, \|\cdot\|)$ is called *complete* if every Cauchy sequence in X converges to an element of X .

The notion of a complete space was introduced by Stefan Banach, and hence giving rise to Banach Spaces.



Figure 3.4: Stefan Banach (30 March 1892 – 31 August 1945).

Definition 3.6. (Banach Space) A complete normed space is called a *Banach space*.

Example 3.4. The Euclidean space $\mathbb{R}^d$ is a Banach space with the Euclidean norm

$$\|x\|_2 = \sqrt{\sum_{i=1}^d |x_i|^2}.$$

This follows since $\mathbb{R}^d$ is finite-dimensional and every finite-dimensional normed space is complete.

let us look if the space of continuous functions is complete or not with the maximum norm that we have seen.

Lemma 3.1. *The space of continuous functions $\mathcal{C}[a, b]$ is a Banach space with*

$$\|x\|_\infty = \max_{t \in [a, b]} |x(t)|.$$

Proof. Let $\{x_n\}_{n \in \mathbb{N}}$ be a Cauchy sequence in $\mathcal{C}[a, b]$. By definition, for every $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that

$$\|x_m - x_n\|_\infty < \varepsilon \quad \forall m, n \geq n_0.$$

This means

$$|x_m(t) - x_n(t)| < \varepsilon \quad \forall t \in [a, b], \quad \forall m, n \geq n_0.$$

Fix $t_0 \in [a, b]$. Then $\{x_n(t_0)\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$, hence convergent (since $\mathbb{R}$ is complete). Define

$$x(t) = \lim_{n \rightarrow \infty} x_n(t), \quad t \in [a, b].$$

We now claim that $x_n \rightarrow x$ uniformly.

Indeed, given $\varepsilon > 0$, choose n_0 as above. Then for all $m \geq n_0$ and all $t \in [a, b]$,

$$|x_m(t) - x(t)| \leq \lim_{n \rightarrow \infty} |x_m(t) - x_n(t)| \leq \varepsilon.$$

Taking maximum over t , we get

$$\|x_m - x\|_\infty \leq \varepsilon.$$

Thus $x_n \rightarrow x$ in the $\|\cdot\|_\infty$ norm.

Finally, since x_n converges uniformly to x and each x_n is continuous, the limit x is continuous (Uniform limit theorem, [15, Theorem 7.12]). Therefore $x \in \mathcal{C}[a, b]$.

Hence $\mathcal{C}[a, b]$ is complete under $\|\cdot\|_\infty$. □

Now, we look at the completeness of the same space but with different norm.

Lemma 3.2. *The space of continuous functions $\mathcal{C}[0, 2]$ is not a Banach space with*

$$\|x\|_1 = \int_0^1 |x(t)| dt.$$

Proof. We construct a Cauchy sequence in $\mathcal{C}[0, 2]$ that does not converge in $\mathcal{C}[0, 2]$ under $\|\cdot\|_1$.

Define

$$x_n(t) = \begin{cases} 0 & t \in [0, 0.5], \\ \frac{t - 0.5}{a_n - 0.5} & t \in [0.5, a_n], \\ 1 & t \in [a_n, 2], \end{cases} \quad a_n = 0.5 + \frac{1}{n}.$$

Graphically, x_n is a continuous function that rises linearly from 0 at $t = 0.5$ to 1 at $t = a_n$, and stays constant outside this interval (see Fig. 3.5).

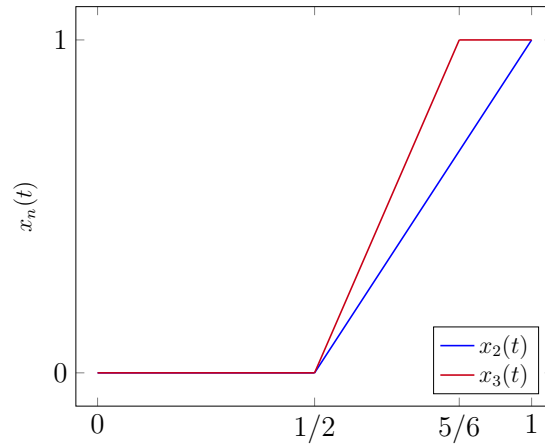


Figure 3.5: Cauchy sequence of continuous functions in $(\mathcal{C}[0, 2], \|\cdot\|_1)$.

Now consider

$$\|x_m - x_n\|_1 = \int_0^1 |x_m(t) - x_n(t)| dt.$$

The difference is nonzero only in the interval $[0.5, \max\{a_m, a_n\}]$. A careful computation shows

$$\|x_m - x_n\|_1 = \frac{1}{2} \left| \frac{1}{m} - \frac{1}{n} \right|.$$

Hence, given $\varepsilon > 0$, if we choose $n_0 > 1/2\varepsilon$, then for $m, n \geq n_0$, we get

$$\|x_m - x_n\|_1 < \varepsilon,$$

so $\{x_n\}$ is a Cauchy sequence in $(\mathcal{C}[0, 2], \|\cdot\|_1)$.

Non-convergence: Suppose $x_n \rightarrow x$ in $\|\cdot\|_1$ for some $x \in \mathcal{C}[0, 2]$. But pointwise, we see

$$x_n(t) \rightarrow \begin{cases} 0 & t \in [0, 0.5), \\ 1 & t \in (0.5, 1], \end{cases}$$

which is a discontinuous step function.

Since x would have to equal this pointwise limit almost everywhere, the candidate limit is not continuous. Hence $x \notin \mathcal{C}[0, 2]$.

Therefore $\{x_n\}$ does not converge in $\mathcal{C}[0, 2]$ under $\|\cdot\|_1$, and $\mathcal{C}[0, 2]$ is not complete with this norm. $\square$

In the last example, if we take the norm

$$\|x\|_2 = \sqrt{\int_0^1 |x(t)|^2 dt},$$

then the same argument will show that $\mathcal{C}[a, b]$ is not a Banach space.

From the examples that we have seen, we have noted that we have certain spaces which form a Banach space with respect to a norm but not with others. As noted we like complete spaces as they give us the limit and hence now the question is can we create a “bigger” space which is a *completion* of $\mathcal{C}[a, b]$.

3.1.2 Space of Measurable Functions

Definition 3.7. (L^p Measurable Functions) Let $\Omega \subset \mathbb{R}^d$ be an open and bounded domain. We denote by L^p , $1 \leq p < \infty$, the set of *measurable functions* $f : \Omega \rightarrow \mathbb{R}$ for which

$$\int_{\Omega} |f(x)|^p dx < \infty.$$

Similarly, the set of measurable functions $f : \Omega \rightarrow \mathbb{R}$ satisfying

$$\text{ess sup } \{|f(x)| : x \in \Omega\} < \infty,$$

is denoted by $L^\infty(\Omega)$. Then $L^p(\Omega)$, $p \in [1, \infty]$, is a real vector space with the norm

$$\|f\|_p = \left(\int_{\Omega} |f(x)|^p dx \right)^{1/p}, \text{ for } 1 \leq p < \infty, \quad \text{and} \quad \|f\|_\infty = \text{ess sup } \{|f(x)| : x \in \Omega\},$$

respectively.

Now, to check if $L^p(\Omega)$ is a Banach space, we first need to verify whether $\|\cdot\|_p$ is a norm. The conditions of non-negativity and homogeneity are straightforward. For definiteness, we identify functions that are equal up to a set of *measure zero*. This identification is important, since if $M \subset \Omega$ has measure zero, then

$$\int_{\Omega} |f(x)|^p dx = 0 \quad \Rightarrow \quad f(x) = 0 \text{ for } x \in \Omega \setminus M.$$

The last property is the triangle inequality, which in the case of L^p spaces is known as the Minkowski inequality. Its proof follows from another important result, namely Hölder's inequality.

Fun Fact: Hölder's inequality was actually proven by the British mathematician Leonard James Rogers but Hölder cites Rogers work and reproves it.



Figure 3.6: Otto Hölder (22 December 1859 – 29 August 1937, left) and Hermann Minkowski (22 June 1864–, 12 January 1909, right).

Lemma 3.3. (Hölder's Inequality)[12, Proposition 6.1.1] *Let $1 \leq p < \infty$ and let q be the conjugate exponent of p , i.e.,*

$$\frac{1}{p} + \frac{1}{q} = 1,$$

with $q = \infty$ when $p = 1$. If $f \in L^p(\Omega)$ and $g \in L^q(\Omega)$, then

$$\|fg\|_1 = \int_{\Omega} |f(x)g(x)| d\mu \leq \|f\|_p \|g\|_q.$$

Remark 3.4. For $p = q = 2$, we recover the well-known Cauchy–Schwarz inequality.

Lemma 3.5. (Minkowski Inequality)[12, Proposition 6.1.2] *Let $1 \leq p \leq \infty$ and $f, g \in L^p(\Omega)$. Then $f + g \in L^p(\Omega)$ and*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

One of the important properties of these L^p spaces is that they are contained in one another, a property known as embedding.

Lemma 3.6. (Embedding Lemma) *Let $1 \leq p < q \leq \infty$. Then*

$$L^q(\Omega) \subset L^p(\Omega),$$

and

$$\|f\|_p \leq \mu(\Omega)^{\frac{1}{p} - \frac{1}{q}} \|f\|_q.$$

Proof. Let $f \in L^q(\Omega)$. First suppose that $q < \infty$. We write $|f|^p = |f|^p \cdot 1$, and apply Hölder's inequality with

$$r = \frac{q}{p}, \quad r' = \frac{q}{q-p}.$$

Since $1/r + 1/r' = 1$, we obtain

$$\int_{\Omega} |f|^p dx \leq \left(\int_{\Omega} |f|^{pr} dx \right)^{1/r} \left(\int_{\Omega} 1^{r'} dx \right)^{1/r'}.$$

Since $pr = q$, this gives

$$\|f\|_{L^p(\Omega)}^p \leq \left(\int_{\Omega} |f|^q dx \right)^{p/q} \mu(\Omega)^{(q-p)/q}.$$

Taking the p^{th} root, we obtain

$$\|f\|_{L^p(\Omega)} \leq \mu(\Omega)^{\frac{1}{p} - \frac{1}{q}} \|f\|_{L^q(\Omega)}.$$

For $q = \infty$, we use the L^∞ bound:

$$|f(x)| \leq \|f\|_{L^\infty(\Omega)} \quad \text{a.e. in } \Omega.$$

Hence

$$\|f\|_{L^p(\Omega)}^p = \int_{\Omega} |f|^p dx \leq \mu(\Omega) \|f\|_{L^\infty(\Omega)}^p.$$

Taking the p^{th} root gives

$$\|f\|_{L^p(\Omega)} \leq \mu(\Omega)^{1/p} \|f\|_{L^\infty(\Omega)}.$$

Thus, whenever $\mu(\Omega) < \infty$,

$$L^q(\Omega) \subseteq L^p(\Omega), \quad 1 \leq p < q \leq \infty.$$

□

Remark 3.7. The above embedding lemma states that if $p < q$, then $L^q(\Omega) \subset L^p(\Omega)$. In particular, if a function is L^q -integrable, then it is also L^p -integrable.

Example 3.5. In the above lemma if Ω is not bounded, then the result does not hold. Consider $f(x) = \frac{1}{1+|x|}$, $x \in \mathbb{R}$. Then

$$\|f\|_1 = \int_{\mathbb{R}} \frac{1}{1+|x|} dx = 2 \int_0^\infty \frac{1}{1+x} dx = 2 [\log(1+x)]_0^\infty = \infty.$$

Hence $f \notin L^1(\mathbb{R})$. However,

$$\|f\|_2^2 = \int_{\mathbb{R}} \frac{1}{(1+|x|)^2} dx = 2 \int_0^\infty \frac{1}{(1+x)^2} dx = -2 \left[\frac{1}{1+x} \right]_0^\infty = 2 < \infty.$$

Thus, $f \in L^2(\mathbb{R})$.

Introduction to Measure Theory

In the above definition of a measurable function, we introduced the notion of *measure*.

Before delving deeper into function spaces, we take a brief detour to understand what we mean by a measure and a measurable function. This section is intended only to provide a basic understanding of the technical terminology from measure theory that you may encounter in other references. If you are comfortable with thinking of a measure simply as a generalization of the notion of “measuring” the size of a set, you may skip this section and return to it when needed.

Remark 3.8. In mathematics, the concept of measure is a generalization of geometrical measures (length, area, volume). Since the idea of area is closely related to that of integration, measure plays a fundamental role in integration theory.

Before defining a measure, we first want to characterize the subsets of a set X that can be regarded as measurable. This leads to the notion of a σ -algebra, introduced by Émile Borel in 1898.



Figure 3.7: Émile Borel (7 January 1871 – 3 February 1956).

Definition 3.8. (σ -algebra) Let X be a set. A σ -algebra is a collection $\mathcal{S}$ of subsets of X such that:

1. $X \in \mathcal{S}$,
2. If $E \in \mathcal{S}$, then $E^C \in \mathcal{S}$,
3. If $E_i \in \mathcal{S}$ for $i \in \mathbb{N}$, then

$$\bigcup_{i=1}^{\infty} E_i \in \mathcal{S}.$$

The pair $(X, \mathcal{S})$ is called a *measurable space*, and the members of $\mathcal{S}$ are called *measurable sets*.

Example 3.6.

1. On any set X , $\mathcal{S} = \{\emptyset, X\}$ is the trivial σ -algebra.
2. The power set of X is the largest possible σ -algebra if X is discrete.
3. The Borel σ -algebra on $\mathbb{R}$ is the smallest σ -algebra containing all open intervals of $\mathbb{R}$.

Now that we have the notion of measurable sets, we can introduce the concept of a measure.

Definition 3.9. (Measure) Let $(X, \mathcal{S})$ be a measurable space. A measure on X (denoted by μ) is a function

$$\mu : \mathcal{S} \rightarrow [0, \infty],$$

satisfying:

1. $\mu(\emptyset) = 0$,
2. $\mu(E) \geq 0$ for all $E \in \mathcal{S}$,
3. (Countable additivity) For any sequence $\{E_k\}_{k=1}^{\infty}$ of pairwise disjoint sets in $\mathcal{S}$,

$$\mu \left(\bigcup_{k=1}^{\infty} E_k \right) = \sum_{k=1}^{\infty} \mu(E_k).$$

The triple $(X, \mathcal{S}, \mu)$ is called a *measure space*.

Example 3.7. 1. **Counting Measure:** Let X be a non-empty set and let $\mathcal{S}$ be the collection of all subsets of X (the power set, which is a σ -algebra). For $E \subset X$, define

$$\mu(E) = \begin{cases} \text{number of elements of } E, & \text{if } E \text{ is finite,} \\ \infty, & \text{if } E \text{ is infinite.} \end{cases}$$

This defines a measure called the *counting measure*.

2. **Dirac Measure:** Let $(X, \mathcal{S})$ be as above and let $x_0 \in X$ be a fixed point. For $E \subset X$, define

$$\mu(E) = \begin{cases} 1, & \text{if } x_0 \in E, \\ 0, & \text{otherwise.} \end{cases}$$

This defines a measure on X called the *Dirac measure*. Thus, the same measurable space can carry different measures.

After these examples of measures, we now turn to measurable functions.

Definition 3.10. (Measurable Function) Let $f : X \rightarrow \mathbb{R}$ be a given function. It is said to be a *measurable function* if for all $\alpha \in \mathbb{R}$ we have

$$f^{-1}((\alpha, \infty)) = \{x \in X : f(x) > \alpha\}$$

is a measurable set.

Let us look at examples of some measurable functions.

Example 3.8. 1. **Indicator Function:** Let $(X, \mathcal{S})$ be a measure space and $E \in \mathcal{S}$. The indicator function is given by

$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E, \\ 0 & \text{if } x \notin E. \end{cases}$$

Here, for $\alpha \in \mathbb{R}$ we have $\{x \in X : \chi_E(x) > \alpha\}$ equal to E , $\emptyset$, or X .

2. **Simple Functions:** Finite linear combinations of indicator functions are called *simple functions*:

$$\varphi(x) = \sum_{i=1}^n a_i \chi_{E_i}(x), \quad E_i \in \mathcal{S}, \quad a_i \in \mathbb{R}.$$

We will discuss them later in detail.

3. **Continuous Functions:** Continuous functions are measurable since the inverse image of an open set is open, and hence belongs to the Borel σ -algebra.

One of the most important classes of measures is the *Lebesgue measure*, named after the French mathematician Henri Léon Lebesgue. This is the generalization we mentioned at the beginning of this section.



Figure 3.8: Henri Léon Lebesgue (28 June 1875 – 26 July 1941).

Let $I = [a, b)$ be an interval whose length is denoted by $\ell(I)$. For any subset $E \subset \mathbb{R}$, the *Lebesgue outer measure*, denoted by $\mu^*(E)$, is defined as

$$\mu^*(E) = \inf \sum_{n=1}^{\infty} \ell(I_n),$$

where the infimum is taken over all countable collections of intervals with $E \subset \bigcup_{n=1}^{\infty} I_n$.

A set E is said to be *Lebesgue measurable* if it satisfies the **Carathéodory criterion**, i.e.,

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c),$$

for every set $A \subset \mathbb{R}$. In this case, the outer measure μ^* restricted to measurable sets is called the *Lebesgue measure* μ .

In the case $E \subset \mathbb{R}^d$, the outer measure is defined as

$$\mu^*(E) = \inf \left\{ \sum_{B \in C} \text{vol}(B) \right\},$$

where C is a countable collection of boxes covering E , and $\text{vol}(B)$ denotes the volume of the box. If E satisfies the Carathéodory criterion, then μ^* defines the Lebesgue measure on $\mathbb{R}^d$.

Remark 3.9. A set $E \subset \mathbb{R}^d$ is said to have *measure zero* if for every $\varepsilon > 0$ there exists a countable collection of d -dimensional rectangles $\{I_k\}$ such that

1. $E \subset \bigcup_{k=1}^{\infty} I_k$,
2. $\sum_{k=1}^{\infty} \mu(I_k) < \varepsilon$.

That is, for every $\varepsilon > 0$ we can cover E with a collection of intervals whose total measure is smaller than ε .

Let us recap what we have done so far: we introduced the notion of a σ -algebra, then defined measure, and more importantly, the Lebesgue measure. The reason we studied all this is to generalize the usual notion of integration.

Definition 3.11. (Simple Function) Let $(X, \mathcal{S}, \mu)$ be a measure space and let $E \in \mathcal{S}$. The characteristic function of E , denoted by χ_E , is defined by

$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E, \\ 0 & \text{if } x \notin E. \end{cases}$$

A *simple function* is a function $s : X \rightarrow \mathbb{R}$ of the form

$$s(x) = \sum_{i=1}^m \alpha_i \chi_{A_i}(x),$$

where $\alpha_i \in \mathbb{R}$ and $A_i \in \mathcal{S}$ for $1 \leq i \leq m$.

We define the integral of a non-negative simple function $s(x)$ by

$$\int_{\mu} s \, d\mu := \sum_{i=1}^m \alpha_i \mu(A_i).$$

Lemma 3.10. [12, Proposition 1.3.2] Let $(X, \mathcal{S})$ be a measurable space and let $f : X \rightarrow \mathbb{R}$ be a non-negative measurable function. Then there exists a sequence $\{s_n\}$ of non-negative simple functions such that

$$0 \leq s_n \leq s_{n+1} \leq f, \quad \forall n,$$

and

$$\lim_{n \rightarrow \infty} s_n(x) = f(x) \quad \forall x \in X.$$

This means that every non-negative measurable function can be approximated by non-negative simple functions. Hence, we define the integral as

$$\int_X f \, d\mu := \sup \left\{ \int_X s \, d\mu : s \text{ simple, } 0 \leq s \leq f \right\}.$$

We are now in a position to define the integrability of a general measurable function f .

Definition 3.12. (Integrable Function) Let $(X, \mathcal{S}, \mu)$ be a measure space and let $f : X \rightarrow \mathbb{R}$ be a measurable function. Then f is said to be *integrable* if

$$\int_X |f| \, d\mu < \infty,$$

where

$$|f| = f^+ + f^-, \quad f^+ = \max\{f, 0\}, \quad f^- = -\min\{f, 0\}.$$

In this case, we define

$$\int_X f \, d\mu := \int_X f^+ \, d\mu - \int_X f^- \, d\mu.$$

Example 3.9. 1. If μ is the counting measure, then any real-valued function f on $\mathbb{N}$ is measurable and can be identified with a sequence $\{a_n\}$, where $f(n) = a_n$. In this case,

$$\int_{\mathbb{N}} f d\mu = \sum_{n=1}^{\infty} a_n.$$

2. If μ is the Dirac measure concentrated at $x_0 \in X$, then for any function $f : X \rightarrow \mathbb{R}$,

$$\int_X f d\mu = f(x_0).$$

Remark 3.11. If E is a set of measure zero, then

$$\int_E |f| d\mu = 0.$$

Examples of sets of measure zero include any finite set of points, or any countable set such as $\mathbb{Q} \cap [0, 1]$. This observation is important: suppose we have an integrable function defined on a domain Ω . If we replace Ω with $\Omega^* := \Omega \setminus E$, where E has measure zero, then the integral remains unchanged.

In particular, the limit obtained in Lemma 3.2 is a measurable and integrable function, since the discontinuity occurs only at finitely many points (a set of measure zero).

Let us think about this in a simple way. Suppose we have a square sheet of paper with side length a . Its area is a^2 . Now, suppose we mark a single point on the sheet with a sharp pin. Has the area of the sheet changed? No—it is still a^2 . If we mark two points, the area is still a^2 . In fact, we can mark any finite number of points, and the area remains unchanged.

This gives us the basic intuition behind a set of *measure zero*: a set can contain points, or even infinitely many points, yet be so “small” that it does not contribute anything to the measure (area, in this case).

Finally, we recall the notion of the essential supremum, which is used in the definition of the $\|\cdot\|_{\infty}$ norm.

Definition 3.13. (Essential Supremum) Let f be a measurable function. The *essential supremum* of f , denoted by $\text{ess sup}(f)$, is defined as

$$\text{ess sup}(f) = \inf \{ \alpha \in \mathbb{R} : f \leq \alpha \text{ a.e.} \}.$$

The last thing we want to mention is the completeness of $L^p(\Omega)$ spaces.

Theorem 3.12. [12, Theorem 6.1.1] Let $(X, \mathcal{S}, \mu)$ be a measure space. Let $1 \leq p \leq \infty$. Then L^p is a Banach space.

We have now completed a brief introduction to measure theory, focusing only on the results essential for our purposes. The key points included: the notion of a measure, measurable

functions, and most importantly Lebesgue integration. A good introduction to Measure Theory can be found in [18].

3.1.3 Dual Space

We have defined what is a normed space, a Banach space, and even looked at a very important class of Banach spaces. Now we look at another space, a space of functions (more precisely, functionals) that act on the space X .

Definition 3.14. (Linear Functional) Let $(X, \|\cdot\|_X)$ be a normed space. A mapping $g : X \rightarrow \mathbb{R}$ is called *linear* if

$$g(\alpha x + \beta y) = \alpha g(x) + \beta g(y) \quad \forall \alpha, \beta \in \mathbb{R}, \quad x, y \in X.$$

A linear mapping $g : X \rightarrow \mathbb{R}$ is continuous if there exists a constant $C > 0$ such that

$$|g(x)| \leq C\|x\|_X \quad \forall x \in X.$$

Such a mapping g is referred to as a *linear functional*.

The definition of continuity might appear similar to the definition of boundedness. In fact, in Functional Analysis they are equivalent.

Remark 3.13. On finite-dimensional spaces all linear functionals are continuous, but in infinite-dimensional spaces this is not necessarily true.

Definition 3.15. (Dual Space) Let us define the sum of two continuous linear functionals g_1 and g_2 , and the multiplication of a continuous linear functional g with a real number α , by

$$(g_1 + g_2)(x) = g_1(x) + g_2(x) \quad \text{and} \quad (\alpha g)(x) = \alpha g(x), \quad \alpha \in \mathbb{R}, \quad x \in X.$$

Then the set of all continuous linear functionals forms a linear space, called the *dual space* X^* . If $g \in X^*$, we denote $g(x)$ by $\langle g, x \rangle$ for $x \in X$.

Lemma 3.14. *The set X^* of continuous linear functionals $x \mapsto \langle g, x \rangle$ on X is a Banach space with respect to the norm*

$$\|g\|_{X^*} := \sup_{0 \neq x \in X} \frac{|\langle g, x \rangle|}{\|x\|_X}.$$

Proof. We need to show two things: (1) $\|\cdot\|_{X^*}$ is a norm, and (2) $(X^*, \|\cdot\|_{X^*})$ is complete.

Step 1: $\|\cdot\|_{X^*}$ is a norm. Clearly $\|g\|_{X^*} \geq 0$ for $g \in X^*$. Also,

$$\|g\|_{X^*} = 0 \Leftrightarrow \langle g, x \rangle = 0 \quad \forall x \in X \Leftrightarrow g = 0.$$

For homogeneity, let $\lambda \in \mathbb{R}$:

$$\begin{aligned}
 \|\lambda g\|_{X^*} &= \sup_{0 \neq x \in X} \frac{|\langle \lambda g, x \rangle|}{\|x\|_X} \\
 &= \sup_{0 \neq x \in X} \frac{|\lambda \langle g, x \rangle|}{\|x\|_X} \\
 &= |\lambda| \sup_{0 \neq x \in X} \frac{|\langle g, x \rangle|}{\|x\|_X} \\
 &= |\lambda| \|g\|_{X^*}.
 \end{aligned}$$

For the triangle inequality:

$$\begin{aligned}
 \|g_1 + g_2\|_{X^*} &= \sup_{0 \neq x \in X} \frac{|\langle g_1 + g_2, x \rangle|}{\|x\|_X} \\
 &= \sup_{0 \neq x \in X} \frac{|\langle g_1, x \rangle + \langle g_2, x \rangle|}{\|x\|_X} \\
 &\leq \sup_{0 \neq x \in X} \frac{|\langle g_1, x \rangle|}{\|x\|_X} + \sup_{0 \neq x \in X} \frac{|\langle g_2, x \rangle|}{\|x\|_X} \\
 &= \|g_1\|_{X^*} + \|g_2\|_{X^*}.
 \end{aligned}$$

Hence $\|\cdot\|_{X^*}$ is a norm.

Step 2: X^* is complete. Let $\{g_n\}_{n \in \mathbb{N}}$ be a Cauchy sequence in X^* . Then for all $\varepsilon > 0$, there exists $n_0(\varepsilon) \in \mathbb{N}$ such that

$$\|g_m - g_n\|_{X^*} < \varepsilon \quad \text{for all } m, n > n_0(\varepsilon).$$

Fix $x \in X$. Then by definition of the norm,

$$\begin{aligned}
 |\langle g_n, x \rangle - \langle g_m, x \rangle| &= |\langle g_n - g_m, x \rangle| \\
 &\leq \|g_n - g_m\|_{X^*} \|x\|_X \\
 &< \varepsilon \|x\|_X.
 \end{aligned}$$

Thus $\{\langle g_n, x \rangle\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$, hence convergent. Define

$$g(x) := \lim_{n \rightarrow \infty} \langle g_n, x \rangle \quad \forall x \in X.$$

Linearity: For $\alpha, \beta \in \mathbb{R}$ and $x, y \in X$,

$$\begin{aligned}
 g(\alpha x + \beta y) &= \lim_{n \rightarrow \infty} \langle g_n, \alpha x + \beta y \rangle \\
 &= \alpha \lim_{n \rightarrow \infty} \langle g_n, x \rangle + \beta \lim_{n \rightarrow \infty} \langle g_n, y \rangle \\
 &= \alpha g(x) + \beta g(y).
 \end{aligned}$$

Boundedness: Since every Cauchy sequence in a normed space is bounded,¹ there exists $C > 0$ such that $\|g_n\|_{X^*} \leq C$ for all n . Then

$$|\langle g_n, x \rangle| \leq \|g_n\|_{X^*} \|x\|_X \leq C \|x\|_X.$$

Passing to the limit $n \rightarrow \infty$, we obtain

$$|g(x)| \leq C \|x\|_X, \quad \forall x \in X,$$

which shows that g is continuous.

Therefore, $g \in X^*$ and $g_n \rightarrow g$ in X^* . Hence X^* is complete, i.e., a Banach space. $\square$

Example 3.10. For $1 < p < \infty$, the dual of $L^p(\Omega)$ is $L^q(\Omega)$, where q is the conjugate exponent of p , i.e.,

$$\frac{1}{p} + \frac{1}{q} = 1.$$

For $p = 1$, the dual of $L^1(\Omega)$ is $L^\infty(\Omega)$, but for $p = \infty$ the dual space is the space of bounded finitely additive signed measures (strictly larger than $L^1(\Omega)$).

At first glance, it may appear strange that $L^q(\Omega)$ is the dual of $L^p(\Omega)$, since L^q is the space of q -integrable functions, whereas $(L^p)^*$ is the space of continuous linear functionals. In fact, there exists an *isometric isomorphism*^a between $(L^p)^*$ and L^q : every $g \in L^q$ induces a functional $T_g \in (L^p)^*$ defined by

$$T_g(f) = \int_{\Omega} f(x)g(x) dx \quad \text{for } f \in L^p(\Omega),$$

and this correspondence preserves the norm. Hence we may identify elements of $(L^p)^*$ with elements of L^q .

^aSuppose X and Y are normed vector spaces. A linear map $T : X \rightarrow Y$ is an *isometric isomorphism* if T is bijective and

$$\|Tx\|_Y = \|x\|_X, \quad \forall x \in X,$$

i.e., T preserves the norm. For example, consider the map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(x, y) = (-y, x).$$

With the Euclidean norm, we have

$$\|T(x, y)\|_2 = \sqrt{(-y)^2 + x^2} = \sqrt{x^2 + y^2} = \|(x, y)\|_2,$$

and T is bijective. Hence, T is an isometric isomorphism.

¹Every Cauchy sequence (x_n) in a normed space $(X, \|\cdot\|)$ is bounded. Indeed, taking $\varepsilon = 1$ in the Cauchy condition, there exists $n_0(1) \in \mathbb{N}$ such that $\|x_n - x_{n_0(1)}\| < 1$ for all $n \geq n_0(1)$. Hence, by the triangle inequality,

$$\|x_n\| \leq \|x_n - x_{n_0(1)}\| + \|x_{n_0(1)}\| < 1 + \|x_{n_0(1)}\|, \quad n \geq n_0(1).$$

The finitely many terms $x_1, \dots, x_{n_0(1)-1}$ are also bounded. Therefore, $\{x_n\}$ is bounded.

3.1.4 Hilbert Space

Until now we have discussed vector spaces and how to measure distances in them. Now, we want to see how the elements of X interact with one another. For this, the notion of an inner product comes.

Definition 3.16. (Inner Product and Hilbert Space) Let X be a linear space. A mapping $(\cdot, \cdot) : X \times X \rightarrow \mathbb{R}$ is called an inner product on X if

1. $(x, y) = (y, x)$ for all $x, y \in X$ (symmetry),
2. $(\alpha x + \beta y, z) = \alpha(x, z) + \beta(y, z)$ for all $x, y, z \in X$ and $\alpha, \beta \in \mathbb{R}$ (linearity in the first argument),
3. $(x, x) > 0$ for $x \neq 0$ (positive-definiteness).

The space $(X, (\cdot, \cdot))$ is referred to as an *inner product space*. An inner product also induces a norm on X denoted by

$$\|x\|_X = \sqrt{(x, x)}.$$

If $(X, \|\cdot\|_X)$ is complete, then it is referred to as a *Hilbert space*, named after the German mathematician David Hilbert.

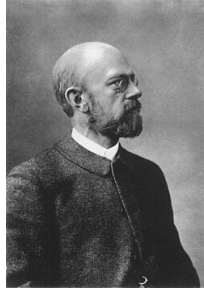


Figure 3.9: David Hilbert (23 January 1862 – 14 February 1943).

Example 3.11. 1. $X = \mathbb{R}^n$ is an inner product space with $(\cdot, \cdot)$ being the dot product, i.e.,

$$(x, y) = \sum_{i=1}^n x_i y_i \quad \text{for } x, y \in \mathbb{R}^n.$$

2. $X = L^2(\Omega)$. For $f, g \in X$, the inner product defined by

$$(f, g) = \int_{\Omega} f(x)g(x) dx,$$

makes $L^2(\Omega)$ into a Hilbert space.

3. $X = \mathcal{C}(\Omega)$ with the inner product

$$(f, g) = \int_{\Omega} f(x)g(x) dx$$

is an inner product space, but it is not a Hilbert space as it is not complete under the induced norm.

4. $X = L^p(\Omega)$, $p \neq 2$, is not an inner product space but merely a normed space.

Remark 3.15. One way to check if a normed space is induced by an inner product is to verify the *parallelogram law*, i.e.,

$$\|f + g\|^2 + \|f - g\|^2 = 2\|f\|^2 + 2\|g\|^2,$$

for all $f, g \in X$. For $(L^p, \|\cdot\|_p)$ with $p \neq 2$, this law fails and hence it is not an inner product space.

For instance, consider

$$f(x) = \begin{cases} 1 & 0 \leq x < 0.5, \\ 0 & 0.5 \leq x \leq 1, \end{cases} \quad g(x) = \begin{cases} 0 & 0 \leq x < 0.5, \\ 1 & 0.5 \leq x \leq 1. \end{cases}$$

Then $\|f\|_1^2 = \|g\|_1^2 = 0.25$, but $\|f + g\|_1^2 = 1$ and $\|f - g\|_1^2 = 1$. Hence, $(L^1, \|\cdot\|_1)$ is not a Hilbert space.

Remark 3.16. So far, we have looked at normed spaces, Banach spaces, inner product spaces, and Hilbert spaces.

- Normed spaces are the most general framework.
- Banach spaces are normed spaces that are complete.
- Inner product spaces are spaces equipped with an inner product, which induces a norm. Hence, every inner product space is a normed space.
- Hilbert spaces are special classes of inner product spaces that are complete in the norm induced by the inner product.

Remark 3.17. Given a normed space, if its norm comes from an inner product, then this inner product is unique (determined by the norm) via the polarization identity².

Now, what is so nice about inner product spaces? We introduced the notion of functionals in the last section. If we have a linear functional $g \in X^*$, then we can characterise its entire action on X using only an element of X and the inner product. This is called the *Riesz Representation Theorem*, named after the Hungarian mathematician Frigyes Riesz.

Theorem 3.18. (Riesz Representation Theorem)[2, Theorem 2.4.2] *Let X be a Hilbert space. Then for every $g \in X^*$, there exists a unique $u_g \in X$ such that*

$$\langle g, x \rangle = (x, u_g), \quad \forall x \in X,$$

and

$$\|u_g\|_X = \|g\|_{X^*}.$$

²**Polarization Identity.** Let X be a real inner product space, and let $\|x\|^2 = (x, x)$. Then

$$(x, y) = \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2), \quad \forall x, y \in X.$$



Figure 3.10: Frigyes Riesz (22 January 1880 – 28 February 1956).

Proof. We divide the proof into two cases.

1. **Case 1:** Let $g = 0$. Take $u_g = 0$. Then

$$\langle g, x \rangle = 0 = (x, 0) = (x, u_g), \quad \forall x \in X,$$

and

$$\|u_g\|_X = 0 = \|g\|_{X^*}.$$

2. **Case 2:** Let $g \neq 0$. Define

$$\text{Ker}(g) = \{x \in X : \langle g, x \rangle = 0\}.$$

Since g is continuous,

$$\text{Ker}(g) = g^{-1}(\{0\})$$

is a closed subspace of X^3 . Since $g \neq 0$, we have $\text{Ker}(g) \neq X$.

As X is a Hilbert space and $\text{Ker}(g)$ is a closed subspace, we have the orthogonal decomposition⁴

$$X = \text{Ker}(g) \oplus \text{Ker}(g)^\perp.$$

Since $g \neq 0$, $\text{Ker}(g)$ has codimension one⁵. Hence,

$$\dim(\text{Ker}(g)^\perp) = 1.$$

Choose $e \in \text{Ker}(g)^\perp$ such that $\|e\|_X = 1$. For any $x \in X$, there exist $n \in \text{Ker}(g)$ and $\alpha \in \mathbb{R}$ such that $x = n + \alpha e$.

Therefore,

$$\langle g, x \rangle = \langle g, n \rangle + \alpha \langle g, e \rangle = \alpha \langle g, e \rangle,$$

³ $g : X \rightarrow \mathbb{R}$ is continuous and 0 is a closed subset of $\mathbb{R}$. Therefore, $g^{-1}(0)$ is closed.

⁴If M is a closed subspace of a Hilbert space X , then $X = M \oplus M^\perp$.

⁵To see why $\text{Ker}(g)$ has codimension one, note that $g \neq 0$ implies that there exists $e \in X$ such that $g(e) \neq 0$. If $z \in \text{Ker}(g)^\perp$, then

$$g(e)z - g(z)e \in \text{Ker}(g),$$

while $g(e)z - g(z)e \in \text{Ker}(g)^\perp$. Hence $g(e)z - g(z)e = 0$, and therefore

$$z = \frac{g(z)}{g(e)}e.$$

Thus $\text{Ker}(g)^\perp = \text{span } e$, so $\dim \text{Ker}(g)^\perp = 1$, and consequently $\text{Ker}(g)$ has codimension one.

since $n \in \text{Ker}(g)$.

Moreover, $n \perp e$, and hence

$$(x, e) = (n + \alpha e, e) = (n, e) + \alpha(e, e) = \alpha,$$

because $\|e\|_X = 1$. Thus,

$$\langle g, x \rangle = (x, e)\langle g, e \rangle.$$

Define $u_g = \langle g, e \rangle e$. Then

$$\langle g, x \rangle = (x, u_g), \quad \forall x \in X.$$

Uniqueness: Suppose $u_1, u_2 \in X$ both represent g . Then

$$(x, u_1) = (x, u_2), \quad \forall x \in X.$$

Hence,

$$(x, u_1 - u_2) = 0, \quad \forall x \in X.$$

Taking $x = u_1 - u_2$, we obtain $\|u_1 - u_2\|_X^2 = 0$, and therefore $u_1 = u_2$.

Equality of norms: By the Cauchy–Schwarz inequality⁶

$$|\langle g, x \rangle| = |(x, u_g)| \leq \|x\|_X \|u_g\|_X.$$

Therefore,

$$\|g\|_{X^*} = \sup_{x \neq 0} \frac{|\langle g, x \rangle|}{\|x\|_X} \leq \|u_g\|_X.$$

Conversely, taking $x = u_g$ gives

$$\|g\|_{X^*} \geq \frac{|\langle g, u_g \rangle|}{\|u_g\|_X}.$$

Since $\langle g, u_g \rangle = (u_g, u_g) = \|u_g\|_X^2$, we obtain

$$\|g\|_{X^*} \geq \|u_g\|_X.$$

Hence,

$$\|g\|_{X^*} = \|u_g\|_X.$$

□

Hence, we can identify the elements of the dual space X^* with the elements of the Hilbert space X itself.

We have proved some theorems here. In the following sections, if required, we will use these results and prove them as needed.

⁶The Cauchy–Schwarz inequality holds in every inner product space: $|(x, y)| \leq \|x\| \|y\|$. For a real inner product space, if $y \neq 0$, then for any scalar t , $0 \leq \|x - ty\|^2$. Expanding the right-hand side and taking $t = (x, y)/\|y\|^2$ gives the result. If $y = 0$, the inequality is immediate.

3.2 Sobolev Spaces

Until now we have discussed integrable functions, but as we are working in a PDE setting, we are interested in differentiable functions.

The classical notion of the derivative of a function $u(x)$ that we are familiar with is

$$u'(x) = \lim_{h \rightarrow 0} \frac{u(x+h) - u(x)}{h}.$$

The above definition is “local” in the sense that it describes the derivative around the point x . However, in PDE analysis we are interested in a more global view: we want to interpret derivatives in such a way that they belong to L^2 (or more generally L^p) spaces.

To generalize the notion of derivatives, we need some preliminary notations.

Definition 3.17. (Compact Support) Let $f : \Omega \rightarrow \mathbb{R}$ be a real-valued function defined on a domain $\Omega \subseteq \mathbb{R}^d$. The *support* of f , written as $\text{supp}(f)$, is the closure of the set of points in Ω where f is non-zero:

$$\text{supp}(f) = \overline{\{x \in \Omega : f(x) \neq 0\}}.$$

If $\text{supp}(f)$ is compact,^a then we say that f has *compact support*.

^aIn $\mathbb{R}^d$ a compact set is one which is closed and bounded. For example, $[0, 1]$ is compact, but $[0, \infty)$ is not compact (closed but not bounded).

Outside the support of f one can naturally extend the function by defining it to be zero. If Ω is bounded, then saying that u has compact support in Ω is equivalent to saying that u vanishes in a neighborhood of $\partial\Omega$.

Example 3.12. The function $f : [-1, 1] \rightarrow \mathbb{R}$ defined by $f(x) = 1 - x^2$ has $\text{supp}(f) = [-1, 1]$, which is compact.

Definition 3.18. Let $\Omega \subset \mathbb{R}^d$. We denote by $\mathcal{D}(\Omega)$ or $\mathcal{C}_0^\infty(\Omega)$ the set of infinitely differentiable functions with compact support in Ω .

The space $\mathcal{D}(\Omega)$ will be our “test function” space, which we use to generalize the notion of derivatives.

Definition 3.19. (Locally Integrable Functions) Given a domain $\Omega \subseteq \mathbb{R}^d$, the set of *locally integrable* functions, denoted by $L_{\text{loc}}^1(\Omega)$, is defined as

$$L_{\text{loc}}^1(\Omega) = \{f : f \in L^1(K) \text{ for all compact } K \subset \Omega\}.$$

Remark 3.19. Given $f \in L^1_{\text{loc}}(\Omega)$, one can always define a linear functional on $\mathcal{D}(\Omega)$ by

$$T_f(\varphi) = \int_{\Omega} f(x)\varphi(x) dx, \quad \varphi \in \mathcal{D}(\Omega).$$

Such functionals are called *distributions*.

Example 3.13. 1. Any continuous function f defined on Ω belongs to $L^1_{\text{loc}}(\Omega)$, since continuous functions are bounded on compact sets, and compact subsets of Ω have finite measure.

2. Let $\mathbf{x} \in \mathbb{R}^d$ and define

$$\delta_{\mathbf{x}}(\varphi) = \varphi(\mathbf{x}) \quad \forall \varphi \in \mathcal{D}(\mathbb{R}^d).$$

This is a linear functional on $\mathcal{D}(\mathbb{R}^d)$. If $\mathbf{x} = \mathbf{0}$, then this is the *Dirac distribution* at the origin.

This functional cannot be represented by a function in L^1_{loc} , and hence it is not itself a function.

Proof of non-representability: Assume that $\delta_{\mathbf{0}}$ can be represented by a function $f \in L^1_{\text{loc}}$. Then, for every $\varepsilon > 0$, let $\varphi_{\varepsilon} \in \mathcal{D}(\mathbb{R}^d)$ be such that - $\text{supp}(\varphi_{\varepsilon}) \subset B(\mathbf{0}; \varepsilon)$, - $0 \leq \varphi_{\varepsilon} \leq 1$, - $\varphi_{\varepsilon} = 1$ on $B[\mathbf{0}; \varepsilon/2]$.

Then

$$\delta_{\mathbf{0}}(\varphi_{\varepsilon}) = \varphi_{\varepsilon}(\mathbf{0}) = 1,$$

while, by the assumption,

$$\delta_{\mathbf{0}}(\varphi_{\varepsilon}) = \int_{\mathbb{R}^d} f(\mathbf{x})\varphi_{\varepsilon}(\mathbf{x}) d\mathbf{x} = \int_{B(\mathbf{0}; \varepsilon)} f(\mathbf{x})\varphi_{\varepsilon}(\mathbf{x}) d\mathbf{x} \leq \int_{B(\mathbf{0}; \varepsilon)} |f(\mathbf{x})| d\mathbf{x}.$$

As $\varepsilon \rightarrow 0$, the last integral tends to zero by local integrability of f , leading to a contradiction. Therefore, $\delta_{\mathbf{0}}$ cannot be represented by an L^1_{loc} function.

Definition 3.20. (Weak Derivatives) A function $f \in L^1_{\text{loc}}(\Omega)$ has a *weak derivative* $D^{\alpha}f$ provided there exists a function $g \in L^1_{\text{loc}}(\Omega)$ such that

$$\int_{\Omega} g(x)\varphi(x) dx = (-1)^{|\alpha|} \int_{\Omega} f(x) \varphi^{(\alpha)}(x) dx \quad \forall \varphi \in \mathcal{D}(\Omega).$$

If such a g exists, we define $D^{\alpha}f := g$.

The notion of weak derivative may appear abstract at first, but let us see how it naturally arises. Suppose $f \in C^1(\Omega)$ with $\Omega \subset \mathbb{R}$. Then, by integration by parts,

$$\int_{\Omega} f(x)\varphi'(x) dx = f(x)\varphi(x) \Big|_{\partial\Omega} - \int_{\Omega} f'(x)\varphi(x) dx.$$

Since $\varphi \in \mathcal{D}(\Omega)$ vanishes on $\partial\Omega$, the boundary term drops out:

$$\int_{\Omega} f(x)\varphi'(x) dx = - \int_{\Omega} f'(x)\varphi(x) dx.$$

Thus $f'(x)$ appears in the identity above.

If $f'(x)$ does not exist in the classical sense, but some $g \in L^1_{\text{loc}}(\Omega)$ satisfies the same identity for all test functions $\varphi \in \mathcal{D}(\Omega)$, then we call g the *weak derivative* of f .

The reason this works is that $\mathcal{D}(\Omega)$ is dense⁷ in $L^p(\Omega)$ for $1 \leq p < \infty$, so such g (if it exists) is uniquely determined.

Example 3.14. Let $d = 1$, $\Omega = [-1, 1]$, and $f(x) = 1 - |x|$.

First weak derivative. We claim the weak derivative of f is

$$g(x) = \begin{cases} 1, & x < 0, \\ -1, & x > 0. \end{cases}$$

(its value at $x = 0$ is irrelevant). Indeed, for $\varphi \in \mathcal{D}(\Omega)$,

$$\begin{aligned} \int_{-1}^1 f(x) \varphi'(x) dx &= \int_{-1}^0 (1+x) \varphi'(x) dx + \int_0^1 (1-x) \varphi'(x) dx \\ &= (1+x) \varphi(x) \Big|_{-1}^0 - \int_{-1}^0 \varphi(x) dx \\ &\quad + (1-x) \varphi(x) \Big|_0^1 + \int_0^1 \varphi(x) dx. \end{aligned}$$

Since $\varphi(\pm 1) = 0$ and φ is continuous at 0, this becomes

$$- \int_{-1}^0 (1) \cdot \varphi(x) dx - \int_0^1 (-1) \cdot \varphi(x) dx = - \int_{-1}^1 g(x) \varphi(x) dx.$$

Hence g is the weak derivative of f .

Second weak derivative. Now let h be the weak derivative of g . Then for $\varphi \in \mathcal{D}([-1, 1])$,

$$\begin{aligned} \int_{-1}^1 h(x) \varphi(x) dx &= - \int_{-1}^1 g(x) \varphi'(x) dx \\ &= - \left(\int_{-1}^0 1 \cdot \varphi'(x) dx + \int_0^1 (-1) \cdot \varphi'(x) dx \right) \\ &= -(\varphi(0) - \varphi(-1) - (\varphi(1) - \varphi(0))). \end{aligned}$$

Since $\varphi(\pm 1) = 0$, this reduces to

$$2\varphi(0) = \int_{-1}^1 2\delta_0(x) \varphi(x) dx.$$

Thus the weak derivative is $h(x) = 2\delta_0(x)$. However, $\delta_0 \notin L^1_{\text{loc}}(\Omega)$, so the second weak derivative (and higher ones) do not exist as functions in L^1_{loc} .

⁷A subset A is *dense* in X if $\overline{A} = X$.

But in the second differentiation example we obtained an object that is not a weak derivative in L^1_{loc} , but still makes sense as a *distribution*. This motivates a more general concept, the *distributional derivative*. Before that, we recall the notion of distributions.

Definition 3.21. (Distribution) A continuous linear functional on $\mathcal{D}(\Omega)$ is called a *distribution* on Ω . The space of all distributions is denoted by $\mathcal{D}'(\Omega)$, i.e., the (topological) dual of $\mathcal{D}(\Omega)$.

Example 3.15. 1. Let $f \in L^1_{\text{loc}}(\Omega)$. Define $T_f : \mathcal{D}(\Omega) \rightarrow \mathbb{R}$ by

$$T_f(\varphi) = \int_{\Omega} f(x)\varphi(x) dx.$$

Then T_f is a distribution.

2. Let $\mathbf{x} \in \mathbb{R}^d$. Define the Dirac distribution by

$$\delta_{\mathbf{x}}(\varphi) = \varphi(\mathbf{x}), \quad \varphi \in \mathcal{D}(\mathbb{R}^d).$$

Physicist Paul Dirac introduced the notion of the Dirac δ “function,” but it initially lacked rigorous mathematical meaning. Laurent Schwartz later provided the correct framework in the theory of distributions.



Figure 3.11: Paul Dirac (8 August 1902 – 20 October 1984, left) and Laurent Schwartz (5 March 1915 – 4 July 2002, right).

Definition 3.22. (Distributional Derivative) Let $\Omega \subset \mathbb{R}^d$ be an open set and let $T \in \mathcal{D}'(\Omega)$. Given a multi-index α , the *distributional derivative* of T is defined by

$$D^{\alpha}T(\varphi) = (-1)^{|\alpha|}T(D^{\alpha}\varphi), \quad \forall \varphi \in \mathcal{D}(\Omega).$$

- Example 3.16.** 1. The distribution $2\delta_0$ obtained earlier is the distributional derivative of $g(x) = \text{sign}(x)$.
 2. Consider the Heaviside function on $\mathbb{R}$:

$$H(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Clearly $H \in L^1_{\text{loc}}(\mathbb{R})$, hence it defines a distribution. For $\varphi \in \mathcal{D}(\mathbb{R})$,

$$T'_H(\varphi) = (-1)T_H(\varphi') = - \int_0^\infty \varphi'(x) dx = \varphi(0) = \delta_0(\varphi).$$

Thus, δ_0 is the distributional derivative of H .

If a function f is sufficiently smooth, then the classical, weak, and distributional derivatives all coincide. If $f \in L^1_{\text{loc}}$ but is not classically differentiable, its weak derivative may still exist; if not, its distributional derivative always exists.

Having introduced weak derivatives, we can now generalize Lebesgue norms and spaces to include derivatives.

Definition 3.23. (Sobolev Spaces) Let $k \in \mathbb{N}_0$ and let $1 \leq p \leq \infty$. Suppose $f \in L^1_{\text{loc}}(\Omega)$ and all weak derivatives $D^\alpha f$ exist for $|\alpha| \leq k$. Then we define the Sobolev norm

$$\|f\|_{W^{k,p}(\Omega)} := \begin{cases} \left(\sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^p(\Omega)}^p \right)^{1/p}, & 1 \leq p < \infty, \\ \max_{|\alpha| \leq k} \|D^\alpha f\|_{L^\infty(\Omega)}, & p = \infty. \end{cases}$$

The *Sobolev space* $W^{k,p}(\Omega)$ is then defined as

$$W^{k,p}(\Omega) := \{f \in L^1_{\text{loc}}(\Omega) : \|f\|_{W^{k,p}(\Omega)} < \infty\}.$$

Remark 3.20. The reason we define weak derivatives starting with $L^1_{\text{loc}}(\Omega)$ rather than $L^p_{\text{loc}}(\Omega)$ is that $L^1_{\text{loc}}(\Omega)$ is the natural minimal class of functions that can be identified with distributions via integration. Thus, for $u \in L^1_{\text{loc}}(\Omega)$, we can define its distributional derivatives $D^\alpha u$. We then impose the additional requirement $D^\alpha u \in L^p(\Omega)$ to define the Sobolev space $W^{k,p}(\Omega)$.

The name Sobolev space comes from the Russian mathematician Sergei Sobolev.



Figure 3.12: Sergei Sobolev (6 October 1908 – 3 January 1989).

It is straightforward to check that $\|\cdot\|_{W^{k,p}(\Omega)}$ defines a norm, and the corresponding Sobolev space is in fact a Banach space.

When we defined a norm in Sec. 3.1.1 we forgot to mention another important concept called a *semi-norm*. A semi-norm is one which satisfies properties 1, 3, and 4 of a norm, but may have $\|x\| = 0$ even though $x \neq 0$.

Example 3.17. Define $p : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $p(\mathbf{x}) = x_1$. Then this is a semi-norm since, for $\mathbf{x} = (0, 2)$, we have $p(\mathbf{x}) = 0$ but $\mathbf{x} \neq \mathbf{0}$.

Definition 3.24. For k a non-negative integer and $f \in W^{k,p}(\Omega)$ we define the *Sobolev semi-norm* by

$$|f|_{W^{k,p}(\Omega)} = \left(\sum_{|\alpha|=k} \|D_{\mathbf{w}}^{\alpha} f\|_{L^p(\Omega)}^p \right)^{1/p},$$

for $1 \leq p < \infty$, and in the case $p = \infty$

$$|f|_{W^{k,\infty}(\Omega)} = \max_{|\alpha|=k} \|D_{\mathbf{w}}^{\alpha} f\|_{L^{\infty}(\Omega)}.$$

Now let us look at some Sobolev spaces and how these norms look in practice.

Example 3.18. 1. For $p = 1$, $k = 1$,

$$W^{1,1}(\Omega) = \{f \in L^1_{\text{loc}}(\Omega) : \|f\|_{W^{1,1}(\Omega)} < \infty\},$$

where

$$\|f\|_{W^{1,1}(\Omega)} = \sum_{|\alpha| \leq 1} \|D_{\mathbf{w}}^{\alpha} f\|_{L^1(\Omega)}.$$

For $\Omega \subset \mathbb{R}^2$, we have $\alpha = (0, 0), (1, 0), (0, 1)$. Hence

$$\|f\|_{W^{1,1}(\Omega)} = \|f\|_{L^1(\Omega)} + \left\| \frac{\partial f}{\partial x} \right\|_{L^1(\Omega)} + \left\| \frac{\partial f}{\partial y} \right\|_{L^1(\Omega)}.$$

2. For $p = 2$, $k = 1$,

$$W^{1,2}(\Omega) = \{f \in L^1_{\text{loc}}(\Omega) : \|f\|_{W^{1,2}(\Omega)} < \infty\},$$

where

$$\|f\|_{W^{1,2}(\Omega)}^2 = \sum_{|\alpha| \leq 1} \|D_{\mathbf{w}}^{\alpha} f\|_{L^2(\Omega)}^2 = \|f\|_{L^2(\Omega)}^2 + \left\| \frac{\partial f}{\partial x} \right\|_{L^2(\Omega)}^2 + \left\| \frac{\partial f}{\partial y} \right\|_{L^2(\Omega)}^2.$$

For the special case $p = 2$, the space $L^2(\Omega)$ is a Hilbert space. Similarly, the Sobolev

space $W^{k,2}(\Omega)$ is a Hilbert space with the inner product

$$(u, v)_{W^{k,2}(\Omega)} := \sum_{|\alpha| \leq k} (D_w^\alpha u, D_w^\alpha v)_{L^2(\Omega)} = \sum_{|\alpha| \leq k} \int_{\Omega} D_w^\alpha u(\mathbf{x}) D_w^\alpha v(\mathbf{x}) d\mathbf{x}, \quad \forall u, v \in W^{k,2}(\Omega).$$

This space is so important that it has another notation:

$$W^{k,2}(\Omega) = H^k(\Omega).$$

From now on we shorten some notation. If the domain is clear we write $\|\cdot\|_{W^{k,p}(\Omega)}$ simply as $\|\cdot\|_{k,p}$ (and similarly for the semi-norms). For $H^k(\Omega)$ we write $\|\cdot\|_{H^k(\Omega)}$ as $\|\cdot\|_k$.

We also use D^α to denote D_w^α , and write ∇f for the gradient, i.e.,

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_d} \right).$$

Definition 3.25 ($W_0^{k,p}(\Omega)$ Space). We denote by $W_0^{k,p}(\Omega)$ the closure of $\mathcal{C}_0^\infty(\Omega)$ in the norm $\|\cdot\|_{k,p}$. Similarly, $H_0^k(\Omega)$ is the closure of $\mathcal{C}_0^\infty(\Omega)$ in $H^k(\Omega)$.

Remark 3.21. What do we mean by the closure in the $\|\cdot\|_{k,p}$ norm? The closure consists of all functions that can be approximated by a sequence of functions in $\mathcal{C}_0^\infty(\Omega)$ with respect to the $\|\cdot\|_{k,p}$ norm. In other words, $u \in W_0^{k,p}(\Omega)$ if and only if there exists a sequence $\{u_n\} \subset \mathcal{C}_0^\infty(\Omega)$ such that

$$\|u_n - u\|_{k,p} \rightarrow 0.$$

Thus, $W_0^{k,p}(\Omega)$ consists precisely of the limits of $\|\cdot\|_{k,p}$ -convergent sequences of functions from $\mathcal{C}_0^\infty(\Omega)$.

We now mention an important inequality that relates the Sobolev semi-norm and the norm: the Poincaré inequality named after the French mathematician, Henri Poincaré, who has a Millenium problem named after him, *Poincaré conjecture*.



Figure 3.13: Henri Poincaré (29 April 1854 – 17 July 1912).

Lemma 3.22 (Poincaré Inequality). [8, Lemma 1.7] *There exists a positive constant C_P (depending only on Ω) such that*

$$\|v\|_{0,p} \leq C_P |v|_{1,p}, \quad \forall v \in W_0^{1,p}(\Omega),$$

where $\|v\|_{0,p} = \|v\|_{L^p(\Omega)}$.

Proof. We split the proof into two steps.

Step 1: Reduction to smooth compactly supported functions. Assume the Poincaré inequality holds for every $v \in \mathcal{C}_0^\infty(\Omega)$, i.e. there is a constant C_P such that

$$\|w\|_{L^p(\Omega)} \leq C_P |w|_{1,p}, \quad \forall w \in \mathcal{C}_0^\infty(\Omega).$$

Let $v \in W_0^{1,p}(\Omega)$. By definition of $W_0^{1,p}(\Omega)$ there exists a sequence $\{v_n\} \subset \mathcal{C}_0^\infty(\Omega)$ with $v_n \rightarrow v$ in the $\|\cdot\|_{1,p}$ -norm. Using the triangle inequality and the assumed inequality for v_n we get, for every n ,

$$\begin{aligned} \|v\|_{L^p(\Omega)} &\leq \|v - v_n\|_{L^p(\Omega)} + \|v_n\|_{L^p(\Omega)} \\ &\leq \|v - v_n\|_{L^p(\Omega)} + C_P |v_n|_{1,p} \\ &\leq \|v - v_n\|_{L^p(\Omega)} + C_P (|v - v_n|_{1,p} + |v|_{1,p}) \\ &\leq (1 + C_P) \|v - v_n\|_{1,p} + C_P |v|_{1,p}. \end{aligned}$$

Letting $n \rightarrow \infty$ and using $v_n \rightarrow v$ in $\|\cdot\|_{1,p}$ gives $\|v\|_{L^p(\Omega)} \leq C_P |v|_{1,p}$. Thus it suffices to prove the inequality for $v \in \mathcal{C}_0^\infty(\Omega)$.

Step 2: Proof for $v \in \mathcal{C}_0^\infty(\Omega)$. Extend v by zero outside Ω . Choose $a > 0$ such that $\Omega \subset [-a, a] \times \mathbb{R}^{d-1}$. For $\mathbf{x} = (x_1, \dots, x_d)$ we have

$$v(\mathbf{x}) = \int_{-a}^{x_1} \frac{\partial v}{\partial x_1}(s, x_2, \dots, x_d) ds,$$

because $v(-a, x_2, \dots, x_d) = 0$ (extension by zero and compact support).

Let q be the conjugate exponent of p (so $1/p + 1/q = 1$). By Hölder's inequality,

$$\begin{aligned} |v(\mathbf{x})| &\leq \left(\int_{-a}^{x_1} \left| \frac{\partial v}{\partial x_1}(s, x_2, \dots, x_d) \right|^p ds \right)^{1/p} \left(\int_{-a}^{x_1} 1^q ds \right)^{1/q} \\ &\leq (2a)^{1/q} \left(\int_{-a}^a \left| \frac{\partial v}{\partial x_1}(s, x_2, \dots, x_d) \right|^p ds \right)^{1/p}. \end{aligned}$$

Raising to the p -th power gives

$$|v(\mathbf{x})|^p \leq (2a)^{p/q} \int_{-a}^a \left| \frac{\partial v}{\partial x_1}(s, x_2, \dots, x_d) \right|^p ds.$$

Integrate this inequality over all $\mathbf{x} \in \Omega$. For fixed $x_2, \dots, x_d$ integrate in $x_1 \in [-a, a]$ and then integrate over the remaining variables; by Fubini's theorem we obtain

$$\begin{aligned} \int_{\Omega} |v(\mathbf{x})|^p d\mathbf{x} &\leq (2a)^{p/q} \int_{\Omega} \int_{-a}^a \left| \frac{\partial v}{\partial x_1}(s, x_2, \dots, x_d) \right|^p ds d\mathbf{x} \\ &= (2a)^{p/q} \int_{-a}^a \int_{\{(x_2, \dots, x_d) : (s, x_2, \dots, x_d) \in \Omega\}} \left| \frac{\partial v}{\partial x_1}(s, x_2, \dots, x_d) \right|^p d(x_2 \cdots x_d) ds \\ &\leq (2a)^{p/q} \cdot (2a) \int_{\Omega} \left| \frac{\partial v}{\partial x_1}(\mathbf{x}) \right|^p d\mathbf{x} \\ &= (2a)^{1+p/q} \int_{\Omega} \left| \frac{\partial v}{\partial x_1}(\mathbf{x}) \right|^p d\mathbf{x}. \end{aligned}$$

Since $1 + p/q = p$ (because $1/q = 1 - 1/p$), we get

$$\|v\|_{L^p(\Omega)}^p \leq (2a)^p \|\partial_{x_1} v\|_{L^p(\Omega)}^p,$$

hence

$$\|v\|_{L^p(\Omega)} \leq 2a \|\partial_{x_1} v\|_{L^p(\Omega)}.$$

Finally, recall the definition of the Sobolev seminorm:

$$|v|_{1,p} = \left(\sum_{i=1}^d \|\partial_{x_i} v\|_{L^p(\Omega)}^p \right)^{1/p}.$$

Since each term in the sum is nonnegative, in particular $\|\partial_{x_1} v\|_{L^p(\Omega)} \leq |v|_{1,p}$. Combining with the previous inequality yields

$$\|v\|_{L^p(\Omega)} \leq 2a |v|_{1,p}.$$

Thus the Poincaré inequality holds for all $v \in \mathcal{C}_0^\infty(\Omega)$ with $C_P = 2a$, and by Step 1 it holds for all $v \in W_0^{1,p}(\Omega)$. $\square$

By successive application of the Poincaré inequality shows that the semi-norm $|\cdot|_{k,p}$ is equivalent to the norm $\|\cdot\|_{k,p}$ on $W_0^{k,p}(\Omega)$, i.e.,

$$|v|_{k,p} \leq \|v\|_{k,p} \leq C |v|_{k,p} \quad \forall v \in W_0^{k,p}(\Omega).$$

Why are semi-norms important? Recall the Poisson equation with pure Neumann boundary conditions: here, the choice of norm is crucial. If we take the L^2 -norm or the full H^1 -norm, constants still remain an issue since these norms do not annihilate them, and hence the bilinear form is not coercive. In this case, the H^1 semi-norm makes sense as it vanishes on constants. Moreover, the H^1 semi-norm quantifies the gradient, which is useful for measuring the irregularity of solutions.

3.2.1 Sobolev Embedding Theorem

We know that certain L^p spaces are contained within others. This also holds for Sobolev spaces. Some inclusions are straightforward:

$$W^{m,p}(\Omega) \subset W^{k,p}(\Omega) \quad \text{for } m \geq k,$$

with k, m non-negative integers. Another one is

$$W^{k,q}(\Omega) \subset W^{k,p}(\Omega) \quad \text{for } 1 \leq p \leq q \leq \infty,$$

since $L^q \subset L^p$ when $q \geq p$.

But there are more subtle inclusions. Before stating them we define what a *continuous embedding* is.

Definition 3.26. (Continuous Embedding) Let X and Y be two normed spaces with norms $\|\cdot\|_X$ and $\|\cdot\|_Y$ respectively. We say X is *continuously embedded* into Y if $X \subset Y$ and there exists a constant $C > 0$ such that

$$\|x\|_Y \leq C\|x\|_X \quad \forall x \in X. \quad (3.1)$$

We denote the embedding by $X \hookrightarrow Y$.

Theorem 3.23. (Sobolev Embedding Theorem) [8, Theorem 1.5] *Let $\Omega \subset \mathbb{R}^d$ be a domain with Lipschitz boundary. Then, for $k \geq 0$ and $1 \leq p \leq \infty$, the following continuous embeddings hold:*

$$\begin{aligned} W^{k,p}(\Omega) &\hookrightarrow L^{p^*}(\Omega) \quad \text{with } \frac{1}{p^*} = \frac{1}{p} - \frac{k}{d}, \quad \text{if } k < \frac{d}{p}, \\ W^{k,p}(\Omega) &\hookrightarrow L^q(\Omega) \quad \forall q < \infty, \quad \text{if } k = \frac{d}{p}, \\ W^{k,p}(\Omega) &\hookrightarrow C^{0,k-d/p}(\overline{\Omega}) \quad \text{if } \frac{d}{p} < k < \frac{d}{p} + 1, \\ W^{k,p}(\Omega) &\hookrightarrow C^{0,\alpha}(\overline{\Omega}) \quad \forall \alpha \in (0,1), \quad \text{if } k = \frac{d}{p} + 1, \\ W^{k,p}(\Omega) &\hookrightarrow C^{0,1}(\overline{\Omega}) \quad \text{if } k > \frac{d}{p} + 1. \end{aligned}$$

The last three embeddings are into spaces of Hölder continuous functions (see Definition 1.3). Note that L^p is a much larger space than $C(\overline{\Omega})$, so in the last three cases one modifies a function $v \in W^{k,p}(\Omega)$ on a set of measure zero to obtain a continuous representative $\tilde{v}$. In practice, we do not distinguish between v and $\tilde{v}$.

Why are these embeddings important? They tell us about the additional regularity that functions in Sobolev spaces possess. For example, let $d = 1$, $p = 2$, and $k = 1$. Then $d/p = 1/2$, so $k = 1 > d/p$ and hence

$$H^1(\Omega) \hookrightarrow C^{0,1/2}(\overline{\Omega}).$$

Thus one-dimensional H^1 functions are continuous. If $d = 2$, then $d/p = 1$ and $k = 1 = d/p$, so

$$H^1(\Omega) \hookrightarrow L^q(\Omega), \quad \forall q < \infty,$$

but not into continuous functions. This shows the critical nature of Sobolev embeddings.

Example 3.19. Let $\Omega = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1/4\}$ and set, with $r = \sqrt{x_1^2 + x_2^2}$,

$$f(r) = \log(-\log r), \quad 0 < r < \frac{1}{2}.$$

Since $-\log r \rightarrow +\infty$ as $r \rightarrow 0^+$, f is unbounded on Ω and therefore admits no bounded, let alone continuous, representative.

Nevertheless $f \in H^1(\Omega)$. The function is smooth on $\Omega \setminus \{0\}$ with $f'(r) = (r \log r)^{-1}$, and by radial symmetry $|\nabla f| = |f'(r)|$. Passing to polar coordinates and substituting $t = \log r$,

$$\|f\|_{1,2}^2 = \int_0^{2\pi} \int_0^{1/2} \frac{r dr d\varphi}{r^2 (\log r)^2} = 2\pi \int_{-\infty}^{-\log 2} \frac{dt}{t^2} = \frac{2\pi}{\log 2} < \infty,$$

while $\|f\|_{0,2}^2 = 2\pi \int_0^{1/2} (\log(-\log r))^2 r dr < \infty$. Thus $H^1(\Omega) \not\hookrightarrow L^\infty(\Omega)$ for $d = 2$: the Sobolev embedding is borderline and fails to give continuity.

Remark 3.24. The critical nature of Sobolev embeddings in three dimensions is also reflected in the Millennium problem for the Navier–Stokes equations. In 3D, the embedding $H^1(\mathbb{R}^3) \hookrightarrow L^6(\mathbb{R}^3)$ is just sufficient to control certain nonlinear terms, but not strong enough to guarantee global existence and uniqueness of solutions. This borderline embedding is one of the reasons the global regularity problem remained open, [14].

3.2.2 Trace

The reason we are studying Sobolev spaces is to solve Poisson boundary value problems. Now the boundary of $\Omega \subset \mathbb{R}^d$ is in $\mathbb{R}^{d-1}$ and hence its d -measure is zero (for example, the area of a line is zero). But one still needs a way to represent the boundary values by a function defined on Ω .

Let us take an example. Say $\Omega = (0, 1)^2$ and $u(x_1, x_2) = x_1^{-\alpha/2}$ with $0 < \alpha < 1$. Now,

$$\begin{aligned} \|u\|_2^2 &= \int_{\Omega} (x_1)^{-\alpha} d\mathbf{x} \\ &= \int_0^1 \int_0^1 x_1^{-\alpha} dx_2 dx_1 \\ &= \frac{1}{1-\alpha} < \infty. \end{aligned}$$

Hence $u \in L^2(\Omega)$ but $u|_{x_1=0} = \infty$. Therefore we need a way of defining the boundary values.

Theorem 3.25. (Trace Theorem)[5, Theorem B.52],[1, Theorem 5.36] *Let $1 \leq p < \infty$ and $\Omega \subset \mathbb{R}^d$ be a bounded domain.*

- *If Ω has Lipschitz boundary Γ , then the trace operator*

$$\text{tr} : W^{1,p}(\Omega) \rightarrow W^{1-1/p,p}(\Gamma)$$

is bounded and surjective. Moreover,

$$W_0^{1,p}(\Omega) = \{v \in W^{1,p}(\Omega) : \text{tr}(v) = 0\}.$$

- *More generally, if Ω has $\mathcal{C}^{m,\alpha}$ boundary, then for $m \geq 1$,*

$$\text{tr} : W^{m,p}(\Omega) \rightarrow W^{m-1/p,p}(\Gamma)$$

is bounded and surjective.

Now for $p = 2$ the trace operator is

$$\text{tr} : H^1(\Omega) \rightarrow H^{1/2}(\Gamma),$$

and as it is surjective every function in $H^{1/2}(\Gamma)$ is the trace of a function in $H^1(\Omega)$. This is useful in the case we seek solutions to PDEs that satisfy given values on Γ .

Corollary 3.26. [8, Corollary 1.1] *Let $1 \leq p < \infty$ and $\Omega \subset \mathbb{R}^d$ be a bounded domain with Lipschitz continuous boundary Γ . Then there exists a positive constant C such that for all $g \in W^{1-1/p,p}(\Gamma)$ there exists $u_g \in W^{1,p}(\Omega)$ satisfying*

$$\text{tr}(u_g) = g \quad \text{and} \quad \|u_g\|_{1,p,\Omega} \leq C\|g\|_{1-1/p,p,\Gamma}.$$

The function u_g is called the lifting operator of g in $W^{1,p}(\Omega)$.

Remark 3.27. The target space $W^{m-1/p,p}(\Gamma)$ we have used is called the Sobolev–Slobodeckij space. If we drop the smoothness condition of $\mathcal{C}^{m,\alpha}$ boundary to just Lipschitz boundary then the natural target becomes a larger space called a *Besov space*.

Now, in the last result we have introduced half spaces like $H^{1/2}(\Gamma)$. But what is half differentiability? Fractional spaces are quite an advanced topic and are clearly out of scope of this work. For completeness we mention them, but the only important thing to remember is that the solution loses $1/p$ regularity on the boundary.

For $1 \leq p < \infty$ and $0 < \sigma < 1$ we define

$$W^{\sigma,p}(\Omega) = \{u \in L^p(\Omega) : [u]_{W^{\sigma,p}(\Omega)} < \infty\},$$

where

$$[u]_{W^{\sigma,p}(\Omega)}^p = \int_{\Omega} \int_{\Omega} \frac{|u(\mathbf{x}) - u(\mathbf{y})|^p}{|\mathbf{x} - \mathbf{y}|^{d+\sigma p}} d\mathbf{x}d\mathbf{y},$$

and we define the norm on this space by

$$\|u\|_{W^{\sigma,p}(\Omega)}^p = \|u\|_{L^p(\Omega)}^p + [u]_{W^{\sigma,p}(\Omega)}^p.$$

In the case of a general exponent $s \in \mathbb{R}^+$, we write $s = m + \sigma$ with integer m and $0 < \sigma < 1$, and define the space

$$W^{s,p}(\Omega) = \{u \in W^{m,p}(\Omega) : D^\alpha u \in W^{\sigma,p}(\Omega) \ \forall |\alpha| = m\},$$

with norm

$$\|u\|_{W^{s,p}(\Omega)}^p = \|u\|_{W^{m,p}(\Omega)}^p + \sum_{|\alpha|=m} \|D^\alpha u\|_{W^{\sigma,p}(\Omega)}^p.$$

For an in-depth study we suggest the paper [4].

Remark 3.28. The last thing we want to mention about Sobolev spaces is their dual. Let $1 \leq p < \infty$ and q be the conjugate exponent of p . Let $\Omega \subset \mathbb{R}^d$. Then the dual of $W_0^{k,p}(\Omega)$ is denoted by $W^{-k,q}(\Omega)$. In particular, if $p = 2$ then $H^{-k}(\Omega)$ is the dual of $H_0^k(\Omega)$. Negative Sobolev spaces are often defined using Fourier transforms and in practice measure how non-smooth the functions are. We will revisit this topic later. The important thing to note is the hierarchy of spaces

$$H_0^1(\Omega) \hookrightarrow L^2(\Omega) \hookrightarrow H^{-1}(\Omega).$$

3.2.3 Gauss and Green Formulas

One key aspect that we will use further is the generalization of the integration by parts formula. Here the regularity of the domain and the solution becomes very important.

Theorem 3.29. (Gauss Theorem) *Let $\Omega \subset \mathbb{R}^d$ for $d \geq 2$ be a bounded domain with Lipschitz boundary Γ . Then for $u \in W^{1,1}(\Omega)$,*

$$\int_{\Omega} \partial_i u(\mathbf{x}) \, d\mathbf{x} = \int_{\Gamma} u(s) \, \mathbf{n}_i(s) \, ds,$$

where $\mathbf{n} = (\mathbf{n}_1, \dots, \mathbf{n}_d)$ is the unit outward normal vector to Γ .

Corollary 3.30. *Let the conditions of Theorem 3.29 on Ω be satisfied. Consider $u \in W^{1,p}(\Omega)$ and $v \in W^{1,q}(\Omega)$ with $p \in (1, \infty)$ and $1/p + 1/q = 1$. Then*

$$\int_{\Omega} \partial_i u(\mathbf{x}) v(\mathbf{x}) \, d\mathbf{x} = \int_{\Gamma} u(s) v(s) \, \mathbf{n}_i(s) \, ds - \int_{\Omega} u(\mathbf{x}) \partial_i v(\mathbf{x}) \, d\mathbf{x}.$$

Proof. Since $u \in W^{1,p}(\Omega)$ and $v \in W^{1,q}(\Omega)$, we have $uv \in W^{1,1}(\Omega)$ (by Hölder's inequality). Applying the product rule and Theorem 3.29 to uv yields the result. $\square$

From the above corollary we obtain Green's formula, first given by the British mathematical physicist George Green in 1828.

Corollary 3.31. (Green's Formula)[5, Corollary B.59] *Let the conditions of Theorem 3.29 on Ω be satisfied. Then*

$$\int_{\Omega} \nabla u(\mathbf{x}) \cdot \nabla v(\mathbf{x}) d\mathbf{x} = \int_{\Gamma} \frac{\partial u}{\partial \mathbf{n}}(s) v(s) ds - \int_{\Omega} \Delta u(\mathbf{x}) v(\mathbf{x}) d\mathbf{x},$$

for all $u \in H^2(\Omega)$ and $v \in H^1(\Omega)$.

Proof. The proof follows by summing over $i = 1, \dots, d$ in the previous corollary. $\square$



Figure 3.14: Carl Friedrich Gauss (30 April 1777 – 23 February 1855, left) and George Green (14 July 1793 – 31 May 1841, right).

3.2.4 Domains

We have seen certain types of domains that play an important role in defining results for PDEs. Now, we want to go a little deeper and give a mathematically rigorous definition of them.

Definition 3.27. (Lipschitz Domain) Let Ω be a bounded domain in $\mathbb{R}^d$. Then Ω is called a *Lipschitz domain* if for every $\mathbf{x} \in \Gamma$ there exists a neighbourhood U of $\mathbf{x}$ in $\mathbb{R}^d$ and new orthogonal coordinates $(y_1, \dots, y_d)$ such that

1. U is a hypercube in the new coordinates, i.e.,

$$U = \{(y_1, \dots, y_d) : -a_i < y_i < a_i, \quad i = 1, \dots, d\}.$$

2. There exists a Lipschitz continuous function ϕ defined on

$$U' = \{(y_1, \dots, y_{d-1}) : -a_i < y_i < a_i, \quad i = 1, \dots, d-1\},$$

such that

$$|\phi(\mathbf{y}')| \leq a_d \quad \text{for every } \mathbf{y}' = (y_1, \dots, y_{d-1}) \in U',$$

$$\Omega \cap U = \{\mathbf{y} = (\mathbf{y}', y_d) \in U : y_d < \phi(\mathbf{y}')\},$$

$$\Gamma \cap U = \{\mathbf{y} = (\mathbf{y}', y_d) \in U : y_d = \phi(\mathbf{y}')\}.$$

Graphically, this means that for every $\mathbf{x} \in \Gamma$ there exists a local coordinate system and an open hypercube U (an interval in 1D, an open square in 2D, an open cube in 3D, etc.) such that the boundary can be represented as the graph of a Lipschitz continuous function ϕ over U' , and the domain Ω locally lies on one side of this graph. In other words, $\Gamma \cap U$ corresponds

to the points on the graph $y_d = \phi(\mathbf{y}')$, while $\Omega \cap U$ corresponds to the points lying below it (i.e., $y_d < \phi(\mathbf{y}')$). This is illustrated in Fig. 3.15.

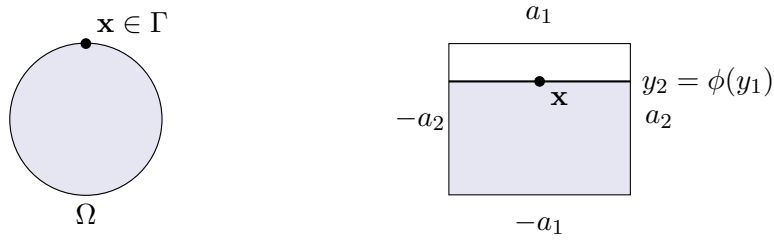


Figure 3.15: Localization of a boundary point. Left: global domain Ω and a boundary point $\mathbf{x} \in \Gamma$. Right: the zoomed neighbourhood U where the boundary is represented locally as the graph $y_2 = \phi(y_1)$ and $\Omega \cap U$ lies below the graph.

Example 3.20. 1. Domains such as balls and polygons in 2D are Lipschitz.
2. Consider the domain

$$\Omega = \{(x, y) : x^2 + y^2 < 1\} \setminus \{(x, y) : x \geq 0, y = 0\},$$

i.e., a domain with slit. Then this domain is not Lipschitz as for any hypercube along the slit will have domain on both the sides.

3. In three dimensions, a polyhedral domain need not be Lipschitz. A simple example is shown in Fig. 3.16, where two rectangular blocks meet at a right angle. Near a point on the common edge, the boundary is made up of two perpendicular surface patches. This means that for the same base point $y' \in \mathbb{R}^2$, there are two different possible boundary heights y_3 . Thus the boundary cannot be represented locally as the graph of a single Lipschitz function $y_3 = \varphi(y')$. For this reason, the Lipschitz condition fails precisely along such non-smooth edges.

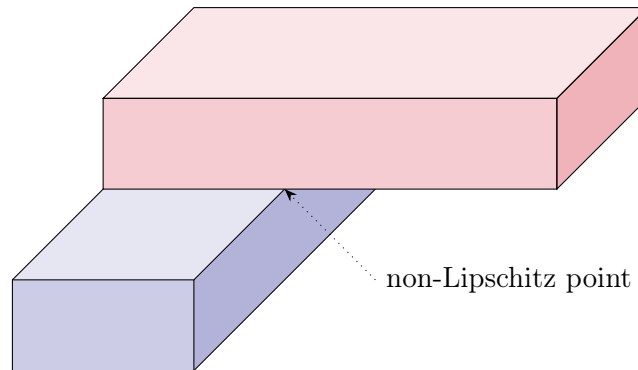


Figure 3.16: A polyhedral domain in 3D that is not Lipschitz: two blocks meet at a right angle, producing a non-Lipschitz point.

If you recall from Sec. 1.3, we defined Hölder continuous functions. The hierarchy of spaces is

$$\mathcal{C}^1(\Omega) \subset \mathcal{C}^{0,1}(\Omega) \subset \mathcal{C}^{0,\alpha}(\Omega) \subset \mathcal{C}^0(\Omega),$$

where $0 < \alpha < 1$. Theorem 1.3 requires a $\mathcal{C}^{2,\alpha}$ boundary, which means the function ϕ in Definition 3.27 must be $\mathcal{C}^{2,\alpha}$ as a function of $\mathbb{R}^{d-1}$. This excludes many important domains —

the square being the most obvious example. Therefore, in applications we usually restrict our analysis to Lipschitz domains.

Remark 3.32. In the above inclusion chain, the space $\mathcal{C}^{0,\alpha}(\Omega)$ may appear “larger” than $\mathcal{C}^{0,1}(\Omega)$, but *larger is not always better*. A typical example is $|x|^\alpha$ for $0 < \alpha < 1$. Although this function is $\mathcal{C}^{0,\alpha}$, it is not Lipschitz: near $x = 0$ its slope blows up, producing a cusp. At such points normals and tangents are not well-defined, so many geometric results (e.g. the Gauss theorem or trace theorems) fail.

3.3 Fixed Points

The final idea that we want to mention is the idea of *fixed points*, which plays a critical role in showing the existence and uniqueness of solutions.

Let us consider the following problem in a Banach space X with an operator $P : X \rightarrow X$: Find $x \in X$ such that

$$x = Px. \quad (3.2)$$

The solution $x \in X$ of the above problem is called a *fixed point* of the operator P .

Definition 3.28. (Contraction) The mapping $P : X \rightarrow X$ is called a *contraction* if there exists a constant $\rho < 1$ such that

$$\|Px_1 - Px_2\|_X \leq \rho \|x_1 - x_2\|_X \quad \forall x_1, x_2 \in X.$$

Now, we state one of the most important results regarding the existence and uniqueness of fixed points.

Theorem 3.33. (Banach Fixed Point Theorem) Let X be a Banach space and $P : X \rightarrow X$ be a contraction. Then:

1. There exists a unique fixed point $x^* \in X$ solving Eq. (3.2).
2. The sequence $x_n = Px_{n-1}$, $n \geq 1$, converges to the fixed point $x^* \in X$ for any initial guess $x_0 \in X$.
3. The following error estimate holds:

$$\|x_n - x^*\|_X \leq \frac{\rho^n}{1 - \rho} \|Px_0 - x_0\|_X \quad \forall n \in \mathbb{N}.$$

Proof. We start by showing that the sequence $\{x_n\}$ is Cauchy. Observe that

$$\begin{aligned} \|x_i - x_{i-1}\|_X &= \|Px_{i-1} - Px_{i-2}\|_X \\ &\leq \rho \|x_{i-1} - x_{i-2}\|_X \\ &\leq \rho^2 \|x_{i-2} - x_{i-3}\|_X \\ &\vdots \\ &\leq \rho^{i-1} \|Px_0 - x_0\|_X \quad \forall i \in \mathbb{N}. \end{aligned}$$

Hence, for $m > n \geq 1$,

$$\begin{aligned}
\|x_m - x_n\|_X &= \left\| \sum_{i=n+1}^m (x_i - x_{i-1}) \right\|_X \\
&\leq \sum_{i=n+1}^m \|x_i - x_{i-1}\|_X \\
&\leq \sum_{i=n+1}^m \rho^{i-1} \|Px_0 - x_0\|_X \\
&= \|Px_0 - x_0\|_X \sum_{i=n+1}^m \rho^{i-1} \\
&\leq \rho^n \frac{1 - \rho^{m-n}}{1 - \rho} \|Px_0 - x_0\|_X \\
&\leq \frac{\rho^n}{1 - \rho} \|Px_0 - x_0\|_X \longrightarrow 0,
\end{aligned}$$

as $n \rightarrow \infty$. Hence $\{x_n\}$ is a Cauchy sequence in X , and since X is complete, it converges to some $x^* \in X$.

We claim that this x^* is a fixed point of P . Indeed,

$$\begin{aligned}
\|x^* - Px^*\|_X &\leq \|x^* - x_n\|_X + \|x_n - Px^*\|_X \\
&= \|x_n - x^*\|_X + \|Px_{n-1} - Px^*\|_X \\
&\leq \|x_n - x^*\|_X + \rho \|x_{n-1} - x^*\|_X \longrightarrow 0,
\end{aligned}$$

as $n \rightarrow \infty$. Hence $x^* = Px^*$.

Next, we show uniqueness. Suppose x_1^*, x_2^* are fixed points. Then

$$\|x_1^* - x_2^*\|_X = \|Px_1^* - Px_2^*\|_X \leq \rho \|x_1^* - x_2^*\|_X.$$

Since $\rho < 1$, this implies $\|x_1^* - x_2^*\|_X = 0$, i.e., $x_1^* = x_2^*$.

Finally, for the error estimate, let $m > n$:

$$\begin{aligned}
\|x_n - x^*\|_X &\leq \|x_n - x_m\|_X + \|x_m - x^*\|_X \\
&\leq \frac{\rho^n}{1 - \rho} \|Px_0 - x_0\|_X + \|x_m - x^*\|_X.
\end{aligned}$$

Letting $m \rightarrow \infty$ gives the desired bound. □

This concludes the discussion of the functional analytic concepts that are required for the numerical analysis of PDEs. Of course, the theory of partial differential equations is a vast subject, and this chapter by no means provides a complete coverage. For a more in-depth treatment, we refer the reader to the books [6, 3].

List of Algorithms

1 Finite Difference Method for 2D Poisson Equation with Dirichlet BCs 26

Index

- ℓ_∞ Stability, 28
- L^p Measurable Functions, 60
- σ -algebra, 63

- Backward Difference Operator, 36
- Banach Fixed Point Theorem, 90
- Banach Space, 57
- Black–Scholes Equation, 53

- Cauchy Sequence, 56
- Compact Support, 75
- Comparison Function, 29
- Complete Space, 57
- Conservation Law, 8
- Consistency, 27
- Continuous Embedding, 84
- Contraction, 90
- Convergence, 28
- Convergent Sequence, 56

- Dirac Measure, 64
- Dirichlet Boundary Condition, 10
- Discrete Maximum Principle, 34
- Distribution, 78
- Distributional Derivative, 78
- Dual Space, 68

- Elliptic PDE, 15
- Embedding Lemma, 61
- Experimental Order of Convergence, 32

- Five Point Stencil, 21
- Fixed Point, 90
- Forward Difference Operator, 36

- Gauss Theorem, 87
- General Positive Type Operators, 44
- Grid Function, 20

- Hölder Inequality, 61
- Heaviside Function, 79
- Hyperbolic PDE, 15

- Hölder Continuous Functions, 16

- Inverse Monotone, 29
- Isometric Isomorphism, 70

- Kronecker Product, 25

- Lifting Operator, 86
- Linear Functional, 68
- Lipschitz Domain, 88
- Local Truncation Error, 27
- Locally Integrable Functions, 75

- Maximum Principles, 33
- Measurable Function, 64
- Measurable Sets, 63
- Measurable Space, 63
- Measure, 62
- Measure Space, 63
- Minkowski Inequality, 61
- Multi-Index, 16

- Neumann Boundary Condition, 11
- Normed Space, 55

- Order of a PDE, 14

- Parabolic PDE, 15
- Poincaré Inequality, 81
- Poisson Equation, 11
- Polarization Identity, 72

- Riesz Representation Theorem, 72
- Robin Boundary Condition, 11

- Semi-norm, 80
- Simple Function, 66
- Sobolev Embedding Theorem, 84
- Sobolev Spaces, 79
- Sparse Matrix, 24
- Support, 75

- Trace Theorem, 86

Vector Space, 55

Weak Derivatives, 76

Bibliography

- [1] R. A. Adams. *Sobolev spaces*. Pure and Applied Mathematics, Vol. 65. Academic Press [A subsidiary of Harcourt Brace Jovanovich, Publishers], New York-London, 1975, pp. xviii+268.
- [2] S. C. Brenner and L. R. Scott. *The mathematical theory of finite element methods*. Third. Vol. 15. Texts in Applied Mathematics. Springer, New York, 2008, pp. xviii+397. ISBN: 978-0-387-75933-3. DOI: 10.1007/978-0-387-75934-0.
- [3] H. Brezis. *Functional analysis, Sobolev spaces and partial differential equations*. Universitext. Springer, New York, 2011, pp. xiv+599. ISBN: 978-0-387-70913-0.
- [4] E. Di Nezza, G. Palatucci, and E. Valdinoci. “Hitchhikers guide to the fractional Sobolev spaces”. In: *Bulletin des Sciences Mathématiques* 136.5 (July 2012), pp. 521–573. ISSN: 0007-4497. DOI: 10.1016/j.bulsci.2011.12.004.
- [5] A. Ern and J.-L. Guermond. *Theory and practice of finite elements*. Vol. 159. Applied Mathematical Sciences. Springer-Verlag, New York, 2004, pp. xiv+524. ISBN: 0-387-20574-8. DOI: 10.1007/978-1-4757-4355-5.
- [6] L. C. Evans. *Partial differential equations*. Second. Vol. 19. Graduate Studies in Mathematics. American Mathematical Society, Providence, RI, 2010, pp. xxii+749. ISBN: 978-0-8218-4974-3. DOI: 10.1090/gsm/019.
- [7] J. D. Faires and R. Burden. *Numerical methods*. Second. With 1 IBM-PC floppy disk (3.5 inch; HD). Brooks/Cole Publishing Co., Pacific Grove, CA, 1998, pp. xii+594. ISBN: 0-534-35187-5.
- [8] S. Ganesan and L. Tobiska. *Finite Elements: Theory and Algorithms*. 1st. New York, NY, USA: Cambridge University Press, 2017. ISBN: 1108415709, 9781108415705.
- [9] D. Gilbarg and N. S. Trudinger. *Elliptic partial differential equations of second order*. Second. Vol. 224. Grundlehren der mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]. Springer-Verlag, Berlin, 1983, pp. xiii+513. ISBN: 3-540-13025-X. DOI: 10.1007/978-3-642-61798-0.
- [10] D. F. Griffiths, J. W. Dold, and D. J. Silvester. *Essential Partial Differential Equations: Analytical and Computational Aspects*. Springer International Publishing, 2015. ISBN: 9783319225692. DOI: 10.1007/978-3-319-22569-2.
- [11] Pierre Grisvard. *Elliptic problems in nonsmooth domains*. Vol. 69. Classics in Applied Mathematics. Reprint of the 1985 original [MR0775683], With a foreword by Susanne C. Brenner. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2011, pp. xx+410. ISBN: 978-1-611972-02-3. DOI: 10.1137/1.9781611972030.ch1. URL: <https://doi.org/10.1137/1.9781611972030.ch1>.
- [12] S. Kesavan. *Functional analysis*. Second. Vol. 52. Texts and Readings in Mathematics. Springer, Singapore; Hindustan Book Agency, New Delhi, [2023] ©2023, pp. xv+268. ISBN: 978-981-19-7633-9; 978-819-51-9613-5. DOI: 10.1007/978-981-19-7633-9.

- [13] E. Kreyszig. *Introductory functional analysis with applications*. John Wiley & Sons, New York-London-Sydney, 1978, pp. xiv+688. ISBN: 0-471-50731-8.
- [14] OpenAI. *On the Navier–Stokes Millennium Prize Problem*. <https://openai.com/index/navier-stokes-solution/>. Accessed September 14, 2026. Sept. 2026.
- [15] W. Rudin. *Principles of mathematical analysis*. Third. International Series in Pure and Applied Mathematics. McGraw-Hill Book Co., New York-Auckland-Düsseldorf, 1976, pp. x+342.
- [16] G. Strang. *Linear algebra and its applications*. Academic Press [Harcourt Brace Jovanovich, Publishers], New York-London, 1976, pp. xi+374.
- [17] M. Stynes. “Steady-state convection-diffusion problems”. In: *Acta Numer.* 14 (2005), pp. 445–508. ISSN: 0962-4929. DOI: 10.1017/S0962492904000261.
- [18] T. Tao. *An Introduction to Measure Theory*. Graduate Studies in Mathematics. American Mathematical Society, 2021. ISBN: 9781470466404. URL: <https://books.google.co.in/books?id=k0lDEAAAQBAJ>.